Adding 200 Yeast Datasets

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The served graph, KG 4.0 (release 2026.10.17-4295f7ca, paired with torchcell v1.8.3), holds 70 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets. Nineteen of them, from sixteen sources, were added since the previous ranking (*The Final Fifty for Datasets 51–100*), and thirteen of those were rows of that ranking. Those rows have left the list, and the 154 candidates that remain are ranked here for the next 200 additions, from 70 served to 270. The pool is 46 rows short of that number; the shortfall is open discovery, named as such, not a reserve.

The ordering changes in one way. The first band is now **state across genomes**: a transcriptome or a proteome measured on strains whose genomes differ by sequence, which is where the served graph is thinnest relative to what a model of cell state needs (Sec. 1). Eight rows qualify and fill ranks 1 to 8. Behind them the rule of the previous pass holds: the rows a yeast Perturb-seq is designed against, then the rows that put two molecular layers on one joinable axis, then scale; inside a band, tier, then measurements (Sec. 2).

Table 4 is the recommended first fifty in rank order and ten extra, so a row that proves unreachable has a named replacement. Table 5 gives each row's citation, link, data location and the reason for its position; Table 6 names the joins each row opens against a served dataset or another row; Table 7 records what was considered and dropped; Sec. 7 closes with the summary statistics, and Sec. 8 prints the whole pool with a source for every row.

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1 State across genomes ✗

The **state** of a cell, for the purposes of this list, is its transcriptome or its proteome: the abundance of every transcript or every protein at once. A **genome** is a strain's total genomic content, written down as sequence. The first band of this ranking is the set of rows that measure a state on more than one genome, where the genomes differ by sequence rather than by one designed edit: natural isolates, the segregants of a cross, a recombinant panel, or an interspecies hybrid read allele by allele. The band comes first because a model that predicts state from genome has to be fitted on pairs of the two, and the served graph holds far more of each half than of the pair.

1.1 What the served graph holds on that axis ✗

Table 1 is every served dataset whose genotype axis is a set of sequenced genomes rather than the S288C reference with one cataloged edit. Three of the ten measure a state. Caudal 2024 is 943 isolate transcriptomes on the 1,011-isolate panel; Muenzner 2024 is 795 proteomes on the same panel and 55 turnover profiles; Boocock 2025 and N'Guessan 2025 are single-cell transcriptomes over 82,467 segregant genotypes of four crosses. Everything else on a sequenced-genome axis is growth, a life-cycle trait, or a guide-by-background fitness, which fixes the genotype side of the pair without supplying a state.

Table 1: Served datasets whose genotype axis is a set of sequenced genomes. Records are the served counts of release 2026.10.17-4295f7ca. *State* marks a transcriptome or proteome readout.

Dataset	Genomes	Readout	Records	State
Caudal 2024	943 natural isolates	RNA-seq TPM per gene	943	yes
Muenzner 2024	795 natural isolates	protein abundance per protein	795	yes
Muenzner 2024	55 natural isolates	protein degradation rate	55	yes
Boocock 2025	79,516 segregants, 3 crosses	single-cell UMI counts	79,516	yes
N'Guessan 2025	2,951 BY x RM segregants	single-cell UMI counts, pseudobulk	2,951	yes
Hale 2024	169 segregants x 8,046 guides	CRISPRi fitness	8,692,635	
Bloom 2019	13,950 segregants, 16 crosses	growth over 38 traits	530,100	
Peter 2018	971 natural isolates	growth over 35 conditions	33,985	
De Chiara 2022	1,011-isolate panel	doubling time, cell yield, life cycle	76,620	
Puddu 2019	4,800 deletion strains, sequenced	genome architecture	8,835	

How different those genomes are from the reference is a measured quantity, not a figure of speech. Against S288C, the 943 isolates of the served Caudal 2024 store carry a median of 78 accessory open reading frames the reference lacks and lack a median of 126 the reference has (ranges 34 to 352 and 43 to 453), so a typical isolate is about 50 ORFs smaller than BY4741 and differs from it at about 200 ORFs in presence. The larger difference is inside the shared genes: a median of about 5,000 sequence variants and roughly 360 natural loss-of-function alleles per isolate. Across the panel, 1,614 distinct non-reference ORFs appear in at least one isolate and 1,118 reference ORFs are absent in at least one. The population catalog these sit in is Peter 2018's pangenome of 7,796 ORFs, 4,940 core and 2,856 variable.

1.2 What the eight rows add ✗

The eight rows of the band, ranks 1 to 8 of Table 4, each put a state on a genome axis the graph does not yet pair it with.

Albert 2018 and Jakobson 2025, the segregant pair. Albert 2018 is bulk RNA-seq of 1,012 sequenced BY x RM segregants, the largest genotype-indexed transcriptome in yeast and the bulk partner of the served Boocock 2025, whose pooled panel holds 393 of these same segregants read per cell. Jakobson 2025 measures protein abundance on the same sequenced segregant class, 800 F6 segregants with more than 6,400 protein quantitative trait loci. Together they are transcript and protein on one recombinant panel, which is the natural-variation counterpart of the served Kemmeren 2014 and Messner 2023 pair on deletions.

Teyssonniere 2024 and Grossbach 2022, within-strain pairing. Teyssonniere 2024 is proteomes for 942 of the 1,011 isolates, so against the served Caudal 2024 the transcript-to-protein pairing is one to one by strain at species scale. It shares authors and a panel with the served Muenzner 2024, and whether the two proteomes are independent acquisitions is the first thing to settle, since building both without settling it risks serving one measurement twice. Grossbach 2022 is the one row where transcript, protein and phosphosite are all observed within a genotype, on 112 sequenced BY x RM strains, which turns the RNA-to-protein question from a population correlation into a within-strain residual.

Smith 2008, the condition axis. Smith 2008 profiles 109 BY x RM segregants in glucose and in ethanol. It is the only segregant transcriptome with an environment axis, so genotype by environment on expression is a contrast inside one panel rather than a join across two.

Skelly 2013, McManus 2014 and Foss 2007, the smaller rows. Skelly 2013 pairs transcriptome and proteome on the same 22 sequenced isolates. McManus 2014 reads ribosome occupancy beside mRNA abundance in *S. cerevisiae*, *S. paradoxus* and their F1 hybrid, so translation is measured allele by allele across two genomes; the *S. paradoxus* arm is out of species and is stored apart or dropped. Foss 2007 is the first segregant proteome on the BY x RM cross, fifteen years and two platforms before Grossbach 2022, and its figures are the least confirmed of the eight: neither a segregant count nor a proteome depth is stated in the abstract.

1.3 What is still missing after the eight ✗

Three things no row in the pool supplies. There is no proteome on a condition axis across isolates; Teyssonniere 2024 and Muenzner 2024 are one medium each, and the only proteome with a condition axis anywhere in the list, Lahtvee 2017, is wild type. There is no translation layer on natural isolates beyond McManus 2014's three genomes. And there is no single-cell proteome on any genome axis at all; the per-cell rows in the perturb-seq band are transcriptomes. These are the shape of the 46-row shortfall, and discovery for the next pass should start there.

2 The selection rule ✗

2.1 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down. Every row in Table 4 therefore names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The eleven bases in use are

- **S288C-KO** – the reference genome minus one cataloged ORF (BY4741, BY4742, BY4743). Thirteen of the sixty in Table 4 sit here, second only to reference-only.
- **S288C-KO/het** – a heterozygous diploid deletion, so a dosage edit rather than a null.
- **S288C+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level.
- **S288C+designed-edit** – a designed single-nucleotide or small-indel library. Designed, and not verified per strain.
- **S288C+tag** – the reference plus a designed tag, degron or promoter cassette.
- **S288C+ORF-plasmid** – the reference plus a barcoded ORF on a known plasmid.
- **S288C+reporter-locus** – the reference plus a designed regulatory sequence driving a reporter, which is what every massively parallel reporter assay perturbs.
- **isolate-WGS** – a published assembly or variant call set per isolate.
- **segregant-WGS** – genotype calls or whole-genome sequence per progeny.
- **engineered-chassis** – a named production strain plus heterologous cassettes.
- **reference-only** – wild type, where the environment carries the perturbation.

What fails the gate is a strain whose genome is unknown even in principle. Genome shuffling and adaptive laboratory evolution without resequencing both land there: the selected progeny differ from their parents at unmapped positions, so there is no genotype to map a phenotype from. Those are dropped in Table 7 rather than ranked low, since no amount of phenotypic value substitutes for a missing genotype.

The **S288C+designed-edit** and **S288C+guide** bases carry a weaker claim than the others and the distinction is worth holding. A designed variant library states what each member was built to be, not what it turned out to be, so its genotypes are designed rather than realized. Ingesting one commits to storing the design and a verification status, not a verified sequence.

2.2 Band comes before tier, and why ✗

Scale is the right default ordering and it is the wrong first question when a specific campaign is being planned. The largest row in this table, de Boer 2020 at 10^8 promoter sequences, tells a model of state across genomes nothing; 112 sequenced strains with a transcriptome, a proteome and a phosphoproteome each tell it what the pairing looks like. Each row therefore carries a **band**, applied before tier and before scale:

- **state across genomes**, 8 rows. A transcriptome or proteome measured on strains whose genomes differ by sequence: isolates, segregants, a recombinant panel or an interspecies hybrid. The owner's focus for this pass, set on 2026-10-19, and the axis Sec. 1 shows the served graph is thinnest on. It replaces the previous pass's *next build* band, whose three rows (Muenzner 2024, Peter 2018, De Chiara 2022) are served.
- **perturb-seq**, 23 rows. A CRISPR interference or activation library whose per-guide matrix is released, a barcoded genotype set read out per cell, a designed single-gene perturbation crossed with a transcriptome, or the strain and noise measurements a pooled per-cell campaign depends on. Seven rows of the previous pass's thirty are served (Boocock 2025, Hale 2024, Puddu 2019, Hackett 2020, Hu 2007, N'Guessan 2025, Jackson 2020).
- **molecular layers**, 14 rows. Two or more of transcript, translation, protein, phosphosite, metabolite, flux and turnover, joinable on one axis, either inside one study or against a served dataset. The join is per *strain* when both layers are measured on the same genotypes, and per *gene* when one layer is a single genome-wide coefficient such as a half-life. The eight state rows were in this band last pass and are promoted out of it.
- **scale**, 109 rows. Everything else, ordered as before.

Within a band the order is tier, then measurements. The cost is visible and is not hidden: the three bands total 45 rows, so they fill ranks 1 to 45 and the first five scale rows complete the fifty: de Boer 2020, Lee 2014, Nguyen Ba 2022, Galardini 2019 and Dutta 2026. The ten extra are all scale. de Boer 2020 and Lee 2014 are the largest things in the ranking by instance count and sit at 46 and 47, which is the deliberate consequence of ordering by what a row combines with rather than by how many numbers it holds. A reader who wants scale back reads rows 46 to 60 first.

2.3 Tier, and what rows are ranked on ✗

Tier is assigned from stated bars rather than judgment:

- **Tier 1** clears all three of $\geq 10^3$ genotypes, $\geq 10^4$ instances, and a clean sequence basis, and lands in one of the nine classes.
- **Tier 2** clears two of the three, or is the only dataset covering a requested class, or directly de-risks the Perturb-seq proposal.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. Turco 2023 has the widest environment axis in the pool at 7,536 and is tier 2, because it is a meta-aggregation: its instance count re-serves primary screens, several of them already served, and the net-new fraction is unknown until it is de-duplicated against the rest of this table.

Within a tier, rows sort by **measurements**, meaning instances times phenotype dimensionality. Instances alone is the wrong statistic and gets the order visibly wrong: Albert 2018 is 1,012 instances, which ranks it below a mid-sized fitness screen, but each instance is a transcriptome, so it is roughly 5.8×10^6 numbers. This is the same normalization the supported-datasets table already applies through its gzip-signal column.

2.4 Where the lines fall, and what they cost ✗

The long-run cut is 200 additions, $270 - 70$, and the pool holds 154 rows, so every row in the pool is inside the cut and there is no reserve. The 46-row shortfall is not padded; Sec. 1.3 names where the next discovery pass should look.

The two nearer lines are what a build week is planned against. Wave 1 is rows 1–50 and wave 2 rows 51–70; a wave-2 row is promoted the moment a wave-1 row turns out to have no recoverable per-strain data, which has already happened once to Lee 2014. Wave 1 is the eight state rows, 23 Perturb-seq rows, 14 molecular-layer rows and five scale rows; wave 2 is scale throughout, so the two waves differ in kind and not only in rank. Table 2 gives the composition by class and by band.

Table 2: Candidates by class and by band, split at the two wave lines. *Genotypes* and *Instances* sum the per-row axes over all waves; rows with no count contribute nothing, so both totals are lower bounds.

	Wave 1	Wave 2	Reserve	Genotypes	Instances
Natural variation	14	4	14	1,225,745	3.1×10^6
Tolerance / robustness	3	4	15	122,906	5.3×10^7
CRISPR library screen	8	0	3	328,794	3.9×10^5
Expression / single cell	15	1	3	529	1.0×10^3
Metabolite / precursor	5	2	7	40,631	1.3×10^5
Modality / backbone	4	4	29	785,185	1.3×10^6
Regulatory DNA	1	1	5	100,509,217	1.0×10^8
Deep mutational scan	0	4	5	811,600	8.2×10^5
Combinatorial genome	0	0	3	400	600
The same rows, by band					
state across genomes	8	0	0	3,000	3.1×10^3
perturb-seq	23	0	0	327,436	4.0×10^5
molecular layers	14	0	0	5,500	5.7×10^3
scale	5	20	84	103,489,071	1.6×10^8
Total	50	20	84	103,825,007	1.6×10^8

A dataset named explicitly in the scoping request is pinned into the recommended set even if it ranks below the cut, and the pin is reported rather than folded into the score, so the set stays auditable. With the whole pool inside the cut no pin can bind, so none is in force; the two requested rows, Blank 2005 and Trikka 2015, rank on their own figures.

2.5 What moved since the previous pass ✗

Table 3 lists every position change against the committed ranking of the previous document, bounded to rows inside either pass's working set. Thirteen rows left the list because KG 4.0 serves them, each with the dataset class it is served as. The eight state rows climbed from ranks 34 to 112 into the first eight. Everything else in the table moved because rows above it left or were promoted, and says so.

Table 3: Rank changes between the previous pass and this one, for every row in either pass's first 70, every row added, and every row that left. *Was* is recomputed under the previous rule, tier then measurements, over the rows that pass held. A dash means the row is new or has gone.

Dataset	Was	Now	Reason
Albert 2018 (eQTL in 1,012 segregants)	34	1	State across genomes, first: bulk RNA-seq of 1,012 sequenced BY x RM segregants, the largest genotype-indexed transcriptome in yeast, and the bulk partner of the served Boocock 2025, whose pooled panel stores 393 of these same segregants.
Jakobson 2025 (genome-to-proteome map)	35	2	State across genomes: protein abundance across the same sequenced segregant panel Albert 2018 profiles by transcriptome, 800 F6 segregants, so protein and transcript loci are measurable on one genotype set.
Teyssonniere 2024 (species-wide proteome against transcriptome)	37	3	State across genomes: proteomes for 942 of the 1,011 isolates whose transcriptomes the served Caudal 2024 holds and whose proteomes the served Muenzner 2024 holds for 796, so the de-duplication against Muenzner is the first task and the pairing with Caudal is one to one.
Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	38	4	State across genomes: transcript, protein and phosphosite on one set of 112 sequenced BY x RM strains, the only row where all three layers are observed within a genotype.
Smith 2008 (gene-environment interaction eQTL)	112	5	State across genomes: mRNA abundance of 109 BY x RM segregants in glucose and in ethanol, the one segregant transcriptome with a condition axis, so genotype by environment is a within-panel contrast.
Skelly 2013 (expression variation across isolates)	46	6	State across genomes: transcriptome and proteome on the same 22 sequenced isolates, the cheap and the expensive layer paired within one experiment on natural genomes.
McManus 2014 (ribosome profiling, allele-specific)	48	7	State across genomes: ribosome occupancy beside mRNA abundance for S. cerevisiae, S. paradoxus and their F1 hybrid, so translation is read allele by allele across two genomes; the paradoxus arm is out of species and is stored apart or dropped.

Dataset	Was	Now	Reason
Foss 2007 (BY x RM segregant proteome)	53	8	State across genomes: the first segregant proteome on the BY x RM cross and the independent replicate of Grossbach 2022 fifteen years and two platforms earlier.
Dong 2021 (MAGIC + SAM biosensor)	11	9	A genome-wide CRISPR activation, interference and deletion library sorted on a metabolite biosensor, so the label is product concentration rather than fitness. The pattern a sorted Perturb-seq would reuse.
Momen-Roknabadi 2020 (inducible CRISPRi library)	12	10	A genome-scale inducible CRISPR interference library with a released per-guide matrix. Induction control is what lets a knockdown be applied after a cell is captured rather than during outgrowth.
McGlincy 2021 (genome-scale CRISPRi library)	13	11	The second independently designed genome-scale yeast CRISPR interference library with a released per-guide matrix, so guide design and gene effect can be separated.
Bao 2018 (CHAnGE single-nucleotide library)	14	12	A guide-indexed genome-wide library whose edits sit below the open reading frame, so the perturbation is an allele rather than a null.
Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	15	13	A dose-graded knockdown axis. Graded rather than binary perturbation is what turns a per-cell readout into a dose-response curve.
Roy 2018 (multiplexed precision editing)	16	14	Multiplexed designed edits per cell, which is combinatorial input in the sense the Perturb-seq specification requires.
Newman 2006 (protein noise, GFP library)	17	15	Per-cell protein abundance distributions across a genome-scale tagged collection. Expression noise per gene is what sets how large a perturbation effect has to be before a per-cell readout can resolve it.
Mukherjee 2021 (CRISPRi essential genes x acetic acid)	18	16	Knockdown reaches the essential genes a deletion collection cannot hold, which is a third of the genome a deletion-based campaign would miss.
Lian 2017 (CRISPR-AID, beta-carotene)	19	17	Three perturbation modalities in one cell, activation, interference and deletion, which is combinatorial input by modality rather than by gene.
Hughes 2000 (compendium of expression profiles)	20	18	The original designed-perturbation expression compendium, and the independent measurement that makes cross-laboratory replication of the deletion transcriptome testable against the built Kemmeren 2014.
Lenstra 2011 (chromatin regulator deletion expression)	21	19	The same design over chromatin regulators rather than sequence-specific factors, so the two together cover both halves of transcriptional control by deletion.
Sun 2013 (mRNA synthesis and decay rates across deletion strains)	22	20	A designed single-gene perturbation crossed with a transcriptome, and the only one that resolves the level into a synthesis rate and a decay rate. Rates are what a per-cell snapshot cannot recover on its own.
Jariani 2020 (yeast scRNA-seq through lag phase)	23	21	Per-cell states through a carbon-source shift, so the readout is the distribution across an unsynchronized population rather than its mean.
Guo 2018 (CRISPR-Cas9 tiling of essential genes)	24	22	Guide tiling produces a graded allelic series across one gene, so guide position rather than gene identity carries the effect.
Urbonaite 2021 (yeastDrop-Seq under drug treatment)	25	23	Single-drug arms against their combination with a per-cell readout, the environmental analog of a combinatorial genetic perturbation.
Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	26	24	The high-sensitivity yeast protocol, strand and isoform aware. Sensitivity per cell is the constraint on how small a perturbation effect a campaign can resolve.
Gasch 2017 (single-cell stress heterogeneity)	27	25	Separates intrinsic from extrinsic per-cell variation in an isogenic population under stress, which is the null a perturbation effect is measured against.
Jaffe 2019 (multiplexed CRISPR interference epistasis)	28	26	Two knockdowns in the same cell with a measured interaction, which is the only combinatorial CRISPR interference design located in yeast.
Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	29	27	The platform companion of Jackson 2020: 173,361 cells of one strain, which is what sets the per-cell variance a perturbation has to exceed.
Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	30	28	Combinatorial barcoding in yeast at 96 to 384 multiplexed genotypes or environments per run. The throughput route by which a campaign becomes affordable.
Mulleder 2012 (prototrophic deletion collection)	31	29	The prototrophic deletion collection. Jackson 2020 used a prototrophic background because auxotrophy blocks minimal and nitrogen-limited media, so this is the strain resource a campaign in defined media needs.
Su 2023 (single-cell transcriptomes under four stresses)	32	30	Full-length rather than three-prime tag counting, so isoform-level readout is on the table for a campaign that needs it.

Dataset	Was	Now	Reason
Wang 2022 (single-cell transcriptomes across replicative aging)	33	31	The only yeast single-cell set with age as the condition axis, which is a covariate any pooled campaign carries whether or not it measures it.
Aulakh 2025 (genome-scale ionome)	36	32	A metabolic readout on the whole deletion collection, which is the genotype axis the supported expression compendium already covers.
Leutert 2023 (phosphoproteome x 101 conditions)	39	33	Enzyme regulation acts faster than transcription, so a phosphosite layer over 101 conditions is what explains flux changes an expression table cannot.
Brauer 2008 (growth-rate-controlled chemostat transcriptome)	40	34	Expression at controlled growth rate under each nutrient limitation, which is the condition axis Boer 2010 and Hackett 2016 measure metabolites and flux on.
Hackett 2016 (SIMMER multi-omic flux)	41	35	Flux, metabolite and transcript measured in one study under the same nutrient limitations, so the join is internal rather than across papers.
Zhu 2014 (kinase / phosphatase lipidomics)	42	36	A lipidome over 129 signaling deletions, most of which carry an expression profile in the supported compendium.
Gerke 2017 (urea-cycle mQTL)	43	37	Metabolite quantitative trait loci on a segregant cross that also has an expression map, so a metabolite locus can be read through its transcript.
Ambroset 2014 (metabolite QTL)	44	38	A 74-metabolite panel on a sequenced segregant panel, the widest metabolite vector available on a recombinant genotype axis.
Blank 2005 (13C metabolic flux)	45	39	The only row measuring flux rather than a concentration, on deletion mutants that also have a transcriptome in the supported set.
Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	47	40	Expression under the exact nitrogen-limited conditions that Jackson 2020 profiles single cells in, so the deletion effect separates from the medium effect.
Martin-Perez 2017 (protein half-lives)	49	41	Protein turnover per gene, the second of those two terms, spanning three orders of magnitude and therefore not replaceable by a global constant.
Boer 2010 (metabolome across nutrient limitations)	50	42	The metabolite half of the chemostat nutrient-limitation series whose transcriptome half is Brauer 2008, same laboratory and same conditions.
Tengolics 2024 (domestication metabolome)	51	43	Metabolite levels across sequenced isolates, the panel the supported Caudal 2024 transcriptomes and Muenzner 2024 proteomes also sit on.
Eder 2020 (flux QTL)	52	44	Flux mapped to segregant genotypes, which is the flux counterpart of the expression and protein maps on the same panel type.
Yu 2021 (proteome and metabolome under nitrogen limitation)	54	45	Both layers measured in one study under nitrogen limitation, so it calibrates the protein-to-metabolite step that cross-study joins assume.
de Boer 2020 (100M random promoters)	55	46	Unchanged criteria; moved by 9 as rows around it were promoted into a band.
Lee 2014 (HIP-HOP fitness signatures)	57	47	Unchanged criteria; moved by 10 as rows around it were promoted into a band.
Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	58	48	Unchanged criteria; moved by 10 as rows around it were promoted into a band.
Galardini 2019 (four backgrounds x 38 conditions)	59	49	Unchanged criteria; moved by 10 as rows around it were promoted into a band.
Dutta 2026 (barcoded natural-isolate chemical response)	61	50	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Li 2016 (tRNA fitness landscape)	62	51	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Domingo 2018 (tRNA double-mutant landscape)	63	52	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Zhang 2021 (YETI titratable overexpression)	64	53	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Peeters 2021 (fermentation-trait QTL atlas)	65	54	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Van Leeuwen 2024 (short-chain organic acids)	66	55	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Chica 2026 (AutoDRY autophagy screen)	67	56	Unchanged criteria; moved by 11 as rows around it were promoted into a band.

Dataset	Was	Now	Reason
Sopko 2006 (genome-scale overexpression toxicity)	68	57	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Liu 2021 (tryptophan / isobutanol tolerance)	69	58	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Kuroda 2019 (isobutanol-specific tolerance)	70	59	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Turco 2023 (Yeast Phenome)	71	60	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Rossi 2021 (ChIP-exo protein architecture)	72	61	Unchanged criteria; moved by 11 as rows around it were promoted into a band.
Bloom 2025 (global epistasis in a yeast cross)	74	62	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Gasch 2000 (environmental stress response)	75	63	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Decourty 2021 (RNA-metabolism genetic interactions)	76	64	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Cuperus 2017 (random 5' UTR library)	77	65	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Faure 2022 (doubledeepPCA)	78	66	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Chong 2015 / CYCLoPs (single-cell proteome dynamics)	79	67	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Teyssonniere 2024 (species-wide trait survey)	80	68	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Diss 2018 (Fos-Jun interaction double mutants)	81	69	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Warringer 2011 (trait variation across the SGRP panel)	82	70	Unchanged criteria; moved by 12 as rows around it were promoted into a band.
Piotrowski 2017 (MOSAIC diagnostic panel)	73	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as EnvChemgenPiotrowski2017Dataset; it is now a join partner rather than a candidate.
Hale 2024 (CRISPRi x natural variation)	5	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as CrispriSegregantHale2024Dataset; it is now a join partner rather than a candidate.
Parsons 2006 (bioactive-compound profiling)	60	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as EnvChemgenParsons2006Dataset; it is now a join partner rather than a candidate.
Peter 2018 (1,011 isolate genomes + phenome)	2	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as IsolateGrowthPeter2018Dataset; it is now a join partner rather than a candidate.
Muenzner 2024 (natural-isolate proteome)	1	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as two datasets (ProteomeMuenzner2024Dataset and ProteinTurnoverMuenzner2024Dataset); it is now a join partner rather than a candidate.
Puddu 2019 (WGS of the deletion collection)	7	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as GenomeArchitecturePuddu2019Dataset; it is now a join partner rather than a candidate.
N'Guessan 2025 (segregant scRNA-seq eQTL)	8	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as Nguessan2025Dataset; it is now a join partner rather than a candidate.
Vaishnav 2022 (regulatory DNA fitness landscape)	56	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as PooledPromoterExpressionVaishnav2022Dataset; it is now a join partner rather than a candidate.
Boocock 2025 (single-cell eQTL mapping)	4	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as Boocock2025Dataset, 79,516 records over four segregant panels; it is now a join partner rather than a candidate.
De Chiara 2022 (domestication and the life cycle)	3	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as three datasets (DoublingTimeDeChiara2022Dataset, CellYieldDeChiara2022Dataset, LifeCycleDeChiara2022Dataset); it is now a join partner rather than a candidate.

Dataset	Was	Now	Reason
Hu 2007 (TF deletion expression compendium)	10	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as MicroarrayScoreHu2007Dataset; it is now a join partner rather than a candidate.
Jackson 2020 (TF-deletion single-cell atlas)	6	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as TfDeletionPseudobulkJackson2020Dataset; it is now a join partner rather than a candidate.
Hackett 2020 (IDEA inducible-TF transcriptome time series)	9	–	Served since the previous pass in KG 4.0 (release 2026.10.17-4295f7ca) as InductionTimecourseHackett2020Dataset; it is now a join partner rather than a candidate.

3 The first fifty, and ten extra ✗

Table 4: The final fifty, then ten extra in rank order in case a row above proves unreachable. Every stat the ranking used is here beside the reason. *Band, tier*: *N* is the next-build band (the author's order, Sec. 2.2), *P* the perturb-seq band, *M* molecular layers, *S* scale, applied before tier (Sec. 2); within a band the tier rule orders, then measurements. *Genotypes* and *Env* are the perturbation and condition axes. *Inst.* is genotype×environment records, † where it is the product of the two axes rather than a reported count and ‡ where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, the quantity rows are ranked on. *Sequence basis* is the route to each strain's total genomic content; a row with no route is excluded (Table 7). *Time* names the row's time dimension where it has one; a dash means steady state or endpoint. A • marks a row high on a Perturb-seq axis; superscript **B** marks a row blocked on data access, **L** one with a loader in flight. Citations, links and data locations are in Table 5; joins in Table 6.

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
1	**Albert 2018 (eQTL in 1,012 segregants)** • Natural variation; mRNA abundance (RNA-seq)	G, 1	1,012 sequenced segregants	1	1.0×10^3	5.8×10^6	segregant-WGS	–	Transcriptomes on a thousand sequenced recombinants, so genotype maps to a genome-wide expression vector rather than one trait. The single best existing analog to a Perturb-seq readout on a high-dimensional genotype axis.
2	**Jakobson 2025 (genome-to-proteome map)** Natural variation; protein abundance, >6,400 pQTL	G, 1	800 F6 segregants	1	800	1.6×10^6	segregant-WGS	–	Natural alleles mapped to enzyme abundance. A lower-risk engineering lever than a heterologous construct, and the proteome counterpart to the metabolite-QTL panels.
3	**Teyssonniere 2024 (species-wide proteome against transcriptome)** Natural variation; protein abundance, paired with transcript abundance	G, 2	942 natural isolates	1	942	1.9×10^6	isolate-WGS	–	Quantitative proteomes for 942 sequenced isolates, the same panel whose transcriptomes the built Caudal 2024 supplies, so the strain-level pairing is one to one rather than an overlap. It is also the published answer to what that pairing shows: mRNA and protein correlate weakly at the population gene level, and the variants associated with protein levels overlap those associated with transcript levels by 3 percent. That makes it the calibration for any RNA-to-protein inference rather than only another proteome. DE-DUPLICATION OPEN: it shares authors and a panel with Muenzner 2024 at 796 isolates, and whether the two are independent acquisitions or one re-served is not established here and must be settled before both are built.
4	**Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)** Natural variation; transcript, protein and phosphosite abundance	G, 2	112 genomically defined strains (BY x RM panel)	1	112	1.1×10^6	segregant-WGS	–	The only row that measures transcript, protein and protein phosphorylation on one set of sequenced strains, 1,862 proteins and 2,116 phosphopeptides over 988 proteins. The pairing it yields is per strain rather than per gene: for each of the 112 genotypes the mRNA level and the protein level of the same gene are both observed, which is what turns the RNA-to-protein question from a population correlation into a within-strain residual. It also reports that genetic variants changing a protein's phosphorylation are mostly different from those changing its abundance, so the phosphosite layer is not a proxy for the protein layer.
5	**Smith 2008 (gene-environment interaction eQTL)** • Natural variation; mRNA abundance	G, 3	109 segregants	glucose / ethanol	218†	1.3×10^6	segregant-WGS	–	Transcriptomes on the same segregants in two carbon sources, so an expression QTL can be shown to be condition-dependent. Small, and it is the design Boocock 2025 scales up.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
6	**Skelly 2013 (expression variation across isolates)** Natural variation; transcriptome + proteome	G, 3	22 sequenced isolates	1	22	1.8×10^5	isolate-WGS	–	Transcriptome and proteome on the same isolates, which is the paired anchor the RNA-to-protein inference argument needs. Few strains, but the pairing is measured rather than joined across studies.
7	**McManus 2014 (ribosome profiling, allele-specific)** Modality / backbone; ribosome occupancy paired with mRNA abundance	G, 3	S. cerevisiae, S. paradoxus and their F1 hybrid	1	3	3.3×10^4	reference-only	–	Translation efficiency measured gene by gene alongside the mRNA abundance of the same gene, which is the first of the two terms that make protein level differ from transcript level. Its result is the reason the term is needed: ribosome occupancy is more conserved than transcript abundance, so translation buffers divergence in mRNA rather than tracking it. The S. paradoxus arm is out of species and would be stored separately or dropped; the cerevisiae arm and the hybrid's cerevisiae alleles are in scope.
8	**Foss 2007 (BY x RM segregant proteome)** Natural variation; protein abundance (label-free MS)	G, 3	BY x RM segregants; the count is not stated in the abstract and was not confirmed	1	–‡	–	segregant-WGS	–	The first segregant proteome, and the independent replicate of Grossbach 2022 on the same cross fifteen years earlier and on a different platform. Its finding is the one this whole group turns on: the loci that influence protein abundance differ from those that influence transcript levels. Ranked below Grossbach because Grossbach measures the transcript layer on the same strains and this does not, and because neither the segregant count nor the proteome depth was confirmed this pass.
9	**Dong 2021 (MAGIC + SAM biosensor)** CRISPR library screen; SAM biosensor fluorescence (FACS)	P, 2	100,000 guides (a/i/d)	1	1.0×10^5‡	1.0×10^5	S288C+guide	–	MAGIC re-run with a metabolite biosensor instead of a growth selection, so the label is product concentration rather than fitness. The pattern to copy for every precursor we care about, and the sibling of the already-built Lian 2019 furfural screen.
10	**Momen-Roknabadi 2020 (inducible CRISPRi library)** CRISPR library screen; per-guide fitness under induction	P, 2	genome-wide guide library	nutrient limitations	9.0×10^4‡	9.0×10^4	S288C+guide	–	A second CRISPRi library on a different vector and guide-design rule. Whether a model trained on one library transfers to the other is a cheap, decisive test of guide-level overfitting. Not a Perturb-seq row: one edit per cell and a scalar fitness readout is low-dimensional on both axes, however large the library.
11	**McGlincy 2021 (genome-scale CRISPRi library)** • CRISPR library screen; per-guide fitness	P, 2	61,094 guides, 10/gene	continuous culture	6.1×10^4	6.1×10^4	S288C+guide	–	Ten guides per gene gives a graded knockdown axis instead of a binary null, and it covers essential genes. This is the library a yeast Perturb-seq would most plausibly be built on.
12	**Bao 2018 (CHAnGE single-nucleotide library)** CRISPR library screen; variant fitness (pooled barcode seq)	P, 2	tens of thousands of designed variants	inhibitor selection	6.0×10^4‡	6.0×10^4	S288C+ designed-edit	–	Single-nucleotide resolution rather than whole-gene nulls. The only genotype axis in this table finer than an ORF, and the one that matches how a real strain-design edit is specified.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
13	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)** Tolerance / robustness; dose-graded knockdown fitness under alcohol stress	P, 2	genome-scale tunable RNAi	isobutanol, 1-butanol	2.0×10^4‡	2.0×10^4	S288C+guide	–	Two related alcohols at graded knockdown strength. The dosage axis is the signal a binary screen throws away, and Hsp70 emerged only because of it.
14	**Roy 2018 (multiplexed precision editing)** CRISPR library screen; variant fitness	P, 2	thousands of designed edits	1	1.6×10^4‡	1.6×10^4	S288C+ designed-edit	–	Genomic rather than plasmid barcodes, which removes the barcode-swapping artifact that quietly corrupts pooled fitness data. The methodological complement to CHAnGE.
15	**Newman 2006 (protein noise, GFP library)** • Modality / backbone; protein abundance and cell-to-cell noise	P, 2	2,500 GFP-tagged strains	rich / minimal	5.0×10^3†	1.0×10^4	S288C+tag	–	Measures the distribution rather than the mean, per protein, in two media. Noise is the quantity a per-cell readout recovers and a bulk one destroys, so this sets what a single-cell design should expect to see.
16	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)** Tolerance / robustness; fitness under acid stress	P, 2	1,100 essential genes	acetic acid vs control	2.2×10^3†	2.2×10^3	S288C+guide	–	Acetic acid is the dominant hydrolysate stressor, and this reaches the essential genes a deletion collection structurally cannot.
17	**Lian 2017 (CRISPR-AID, beta-carotene)** CRISPR library screen; beta-carotene titer	P, 2	combinatorial a/i/d triples	1	200‡	200	engineered-chassis	–	The tri-functional chassis MAGIC was built on, read out on beta-carotene. Small, but it is a combinatorial genotype with a measured product titer, which is the exact record type inverse strain design needs.
18	**Hughes 2000 (compendium of expression profiles)** Expression / single cell; mRNA log ratio (two-color array)	P, 3	300 mutations and chemical treatments	1	300	1.8×10^6	S288C-KO	–	The original deletion expression compendium, and the oldest independent measurement of the phenotype the built Kemmeren 2014 measures. Ingesting it makes cross-laboratory replication of the deletion transcriptome testable on gene-by-gene profiles fourteen years apart and on different array platforms, which is a check the built set cannot run against itself. The 300 profiles mix deletions, titratable alleles and compound treatments, so the compound arm carries a reference genotype and the mutant arm an S288C-KO one; the split was not confirmed this pass.
19	**Lenstra 2011 (chromatin regulator deletion expression)** • Expression / single cell; mRNA abundance	P, 3	165 chromatin-regulator deletions	1	165	9.9×10^5	S288C-KO	–	Transcriptomes for the chromatin machinery specifically, from the group behind the built Kemmeren compendium and in the same format. Chromatin regulators are the class whose deletion most reshapes the expression landscape.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
20	**Sun 2013 (mRNA synthesis and decay rates across deletion strains)** Expression / single cell; mRNA level, synthesis rate and decay rate	P, 3	46 deletion strains of mRNA degradation and metabolism genes	1	46	8.3×10^5	S288C-KO	rates derived from metabolic labeling; the rates are stored, not the series	Comparative dynamic transcriptome analysis decomposes a transcript level into the synthesis rate and the decay rate that produce it, across 46 single deletions. That is the transcript-side counterpart of Martin-Perez 2017's protein half-lives, so the two together give both turnover terms. Its finding also warns what a steady-state compendium hides: a change in degradation rate is generally compensated by a change in synthesis rate, so mRNA level is buffered and two strains with the same level can have different kinetics. Overlap with the other deletion-strain expression sets is on the strain axis, not the measurement: the built Kemmeren 2014 (1,484 deletions), Hughes 2000, Hu 2007 and Lenstra 2011 all store a steady-state level per gene, and none stores a rate, so this row is a second layer on shared strains rather than a second copy of one. The 46 strains are single deletions of mRNA degradation and metabolism genes and are expected to be inside the Kemmeren set by gene identity; the exact shared-strain count needs the GEO strain list, which was not fetched, and the abstract does not say the strains came from that collection.
21	**Jariani 2020 (yeast scRNA-seq through lag phase)** • Expression / single cell; single-cell transcriptome	P, 3	isogenic wild type	glucose to maltose shift, time course through lag phase	6‡	3.6×10^4	reference-only	time course through the lag phase of a glucose-to-maltose shift	A yeast-adapted 10x protocol that resolves the transcriptional states of individual cells during a carbon-source shift, so the readout is the distribution across a population that is not synchronized rather than its mean. One genotype, so the value is the protocol and the per-cell variance structure a campaign has to budget against.
22	**Guo 2018 (CRISPR-Cas9 tiling of essential genes)** • CRISPR library screen; per-guide fitness	P, 3	3e4 tiling guides	1	3.0×10^4‡	3.0×10^4	S288C+guide	–	Tiles guides across genes rather than sampling a few per gene, so the readout resolves within-gene functional regions. A finer input axis than a gene-level library.
23	**Urbonaite 2021 (yeastDrop-Seq under drug treatment)** • Expression / single cell; single-cell transcriptome	P, 3	isogenic wild type	mycophenolic acid, guanine, both, untreated	4	2.4×10^4	reference-only	–	A yeast-optimized Drop-seq with a small chemical condition axis, including a two-compound arm. One genotype, so it enters as a platform reference: the pair of single-drug arms against their combination is a per-cell readout of a two-factor environment, which is the environmental analog of the combinatorial genetic perturbation a Perturb-seq needs.
24	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** • Expression / single cell; single-cell transcriptome	P, 3	clonal wild type	baseline + stress	2‡	1.2×10^4	reference-only	–	Establishes within-clone transcript correlation, which is the noise floor any per-strain Perturb-seq mean has to clear. Directly sizes the cells-per-guide question.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
25	**Gasch 2017 (single-cell stress heterogeneity)** • Expression / single cell; single-cell transcriptome	P, 3	wild type	stress vs unstressed	2‡	1.2×10^4	reference-only	–	Separates intrinsic from extrinsic expression heterogeneity under stress. Sets what fraction of Perturb-seq variance is recoverable signal rather than cell-state noise.
26	**Jaffe 2019 (multiplexed CRISPR interference epistasis)** • CRISPR library screen; fitness / genetic interaction	P, 3	1e4 guide pairs	1	1.0×10^4‡	1.0×10^4	S288C+guide	–	Two guides per cell, so the perturbation space is combinatorial rather than one edit at a time. This is the shape a Perturb-seq input axis should have, measured here against fitness only.
27	**Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)** • Expression / single cell; single-cell transcriptome (173,361 cells)	P, 3	wild type	one time course, sampled continuously	1	6.0×10^3	reference-only	one time course, sampled continuously	The second scYeast pre-training set: 173,361 cells from one wild-type strain sampled continuously through a time course without metabolic labeling, so the strain-by-condition count is one and the value is the per-cell distribution of transcriptional states, not a genotype axis. A reference backbone for any single-cell expression head, and the companion of Jackson 2020 on the same platform.
28	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)** • Expression / single cell; single-cell transcriptome	P, 3	method	method	–‡	–	reference-only	–	The platform the costing model in experiments/024 is built on. Not a dataset to ingest; the reason it is here is that a yeast Perturb-seq proposal has to name its chemistry.
29	**Mulleder 2012 (prototrophic deletion collection)** Modality / backbone; strain resource (no phenotype)	P, 3	5,185 prototrophic + 839 titratable essential	1	–‡	–	S288C-KO	–	The background the already-built Mulleder 2016 and Messner 2023 were measured in, and mulleder2016.py already cites it. Listed not as an ingestion target but because that loader records the prototrophy-restoring markers as unmodeled: they are a GeneAddition this schema does not yet carry, and auxotrophy markers distort metabolite pools.
30	**Su 2023 (single-cell transcriptomes under four stresses)** Expression / single cell; single-cell transcriptome (117 cells, full length)	P, 4	wild type	isosmotic, hypoosmotic, glucose starvation, amino acid starvation	4	2.4×10^4	reference-only	–	Full-length single-cell RNA-seq of 117 cells across four stress conditions in one strain; scYeast's stress-state classification benchmark (Fan 2027, Sec. 4.1.6). Tiny, but it is a full-length protocol rather than 3' tag counting, which matters for isoform-level heads.
31	**Wang 2022 (single-cell transcriptomes across replicative aging)** Expression / single cell; single-cell transcriptome (125 cells)	P, 4	wild type	2 h, 16 h, 36 h of aging	3	6.6×10^3	reference-only	three ages: 2 h, 16 h and 36 h	Three age groups of one strain, 37, 43 and 45 cells each, about 2,200 genes detected per cell. Small, but it is the only yeast single-cell set with age as the condition axis, and scYeast used it to infer age-specific regulatory relationships (Fan 2027, Sec. 4.1.4).
32	**Aulakh 2025 (genome-scale ionome)** Metabolite / precursor; intracellular metal-ion content (ICP-MS)	M, 1	4,800 YKO	1	4.8×10^3	4.8×10^4	S288C-KO	–	The only gene-indexed ionome. Metal-cofactor supply gates iron-sulfur and zinc-dependent pathway enzymes, and no built dataset measures it.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
33	**Leutert 2023 (phosphoproteome x 101 conditions)** Expression / single cell; phosphosite abundance	M, 2	wild type	101 conditions	101	5.0×10^5	reference-only	–	Post-translational enzyme control acts faster than transcription and is invisible to every other layer here. Condition axis only.
34	**Brauer 2008 (growth-rate-controlled chemostat transcriptome)** Expression / single cell; mRNA abundance	M, 2	wild type	36 chemostats, 6 limitations	36	2.2×10^5	reference-only	–	Separates growth rate from nutrient identity by design, which no batch-culture dataset can. A large share of any expression response is a growth-rate confound, and this is the reference for removing it.
35	**Hackett 2016 (SIMMER multi-omic flux)** Expression / single cell; paired metabolome + proteome + flux	M, 2	wild type	25 chemostat states	25	7.5×10^4	reference-only	–	The only jointly measured metabolome, proteome and fluxome. No genotype axis, so it is a mechanism prior rather than training data, but it is the only place the three layers are measured on the same cells.
36	**Zhu 2014 (kinase / phosphatase lipidomics)** Metabolite / precursor; lipid class and species abundance	M, 2	129 kinase + phosphatase KOs	1	129	2.6×10^4	S288C-KO	–	A gene-indexed lipidome on a regulatory gene set. Complements the already-built da Silveira 2014 lipidomics, which has no kinase axis, and reads out the malonyl-CoA sink directly.
37	**Gerke 2017 (urea-cycle mQTL)** Natural variation; untargeted metabolite abundance	M, 2	147 diploid segregants	1	147	1.5×10^4	segregant-WGS	–	A second mQTL panel from a different cross (oak x wine), so cross-background generalization of metabolite QTL can be tested rather than assumed.
38	**Ambroset 2014 (metabolite QTL)** Natural variation; 74 metabolite concentrations (LC-MS/MS)	M, 2	100 segregants	1	100	7.4×10^3	segregant-WGS	–	The founding metabolite-QTL panel. Pairs with Cooper and Mulleder as the natural-variation view of a phenotype space the deletion collection covers by engineering.
39	**Blank 2005 (13C metabolic flux)** Metabolite / precursor; intracellular flux distribution (13C-MFA)	M, 2	200 viable null mutants	glucose minimal	200‡	6.0×10^3	S288C-KO	–	The only real flux measurement tied to single-gene deletions. Everything else in the built set is a concentration or a titer, which is a proxy for flux rather than flux.
40	**Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)** Expression / single cell; bulk transcriptome	M, 3	wild type, prototrophic	nitrogen sources x steady-state growth rates, plus an upshift time course	20‡	1.2×10^5	reference-only	an upshift time course beside the steady states	Chemostat transcriptomes under controlled nitrogen limitation from the same laboratory, and on the same media axis, as Jackson 2020's NLIM-GLN, NLIM-PRO, NLIM-NH4 and NLIM-UREA conditions. That shared axis is what makes the wild-type baseline of the single-cell atlas separable from the deletion effect, and it is the nitrogen counterpart of Brauer 2008's carbon-limited growth-rate series. Counts are from the citation rather than the released series and need confirming.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
41	**Martin-Perez 2017 (protein half-lives)** Modality / backbone; protein turnover rate	M, 3	wild type, exponential growth	1	1	3.2×10^3	reference-only	half-lives derived from a labeling time course; the rates are stored	Turnover rates for 3,160 proteins, the second term that makes protein level differ from transcript level. Half-lives span three orders of magnitude and are not normally distributed, so a single global scaling from mRNA to protein cannot be right, and this is the per-gene coefficient that replaces it. The paper also reports that localization, complex membership and connectivity predict turnover better than sequence features do, which names the covariates such a model needs.
42	**Boer 2010 (metabolome across nutrient limitations)** Metabolite / precursor; intracellular metabolite concentrations	M, 3	wild type	nutrient limitations x growth rates	25	2.5×10^3	reference-only	–	The metabolite counterpart to Brauer on the same chemostat design, so metabolite pools are separated from growth rate the same way. Covers central-carbon nodes the deletion metabolomics rows measure at low power.
43	**Tengolics 2024 (domestication metabolome)** Natural variation; 19 amino acids + 78 other metabolites	M, 3	17 S. cerevisiae populations	1	17	1.6×10^3	isolate-WGS	–	Shows which metabolic traits already moved under domestication, which is a prior on which are tractable to engineer further. Small n.
44	**Eder 2020 (flux QTL)** Natural variation; modeled intracellular flux	M, 3	segregant panel (n unconfirmed)	wine fermentation	100‡	100	segregant-WGS	–	Flux as a QTL phenotype, with PDB1 and VID30 validated. The label is modeled rather than measured, which caps how far it can be trusted.
45	**Yu 2021 (proteome and metabolome under nitrogen limitation)** Metabolite / precursor; proteome + metabolome	M, 4	wild type	nitrogen-limited chemostats	10‡	2.5×10^4	reference-only	–	Nitrogen limitation is the standard trigger for redirecting carbon to product, and this measures both layers under it. Condition axis only, and small.
46	**de Boer 2020 (100M random promoters)** • Regulatory DNA; expression driven (YFP sort-seq)	S, 1	>100M random promoters	complex + defined	1.0×10^8	1.0×10^8	S288C+ reporter-locus	–	The largest sequence-to-phenotype dataset in yeast by orders of magnitude, and every genotype is an exactly known 80mer. The perturbation space is continuous rather than a gene list, which is what a model of regulatory grammar needs and no deletion collection can supply.
47	**Lee 2014 (HIP-HOP fitness signatures)** [B] Tolerance / robustness; chemogenomic fitness score	S, 1	11,000 (het + hom)	3,356 compounds	1.3×10^7	1.3×10^7	S288C-KO	–	Largest chemical-genetic matrix in existence, on the exact YKO genotype axis Costanzo and Kuzmin already use. Biggest single instance-count gain available. BLOCKED: WS15 already attempted it and it stands as awaiting author matrices, so the Science SI and the lab portal did not yield per-strain values. Unblocking is an author request, not a loader.
48	**Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)** • Natural variation; fitness (Bar-seq)	S, 1	100,000 barcoded segregants	8 conditions	8.0×10^5†	8.0×10^5	segregant-WGS	–	The largest genotyped recombinant panel in yeast, each segregant barcoded and low-coverage sequenced against known parents. Two orders of magnitude more genotypes than Bloom 2013 on the same kind of cross, which is what makes small-effect and epistatic loci detectable.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
49	**Galardini 2019 (four backgrounds x 38 conditions)** Natural variation; colony-size fitness (S-score)	S, 1	3,786 KOs x 4 backgrounds	38 conditions	5.8×10^5†	5.8×10^5	isolate-WGS	–	The same knockout built independently in four sequenced backgrounds. Quantifies the 18.5 percent of deletion phenotypes that do not transfer, which is the generalization risk the whole substrate is exposed to.
50	**Dutta 2026 (barcoded natural-isolate chemical response)** Natural variation; pooled-barcode fitness	S, 1	520 natural isolates	>600 compounds	3.1×10^5†	3.1×10^5	isolate-WGS	–	Natural sequence diversity crossed with a chemogenomic panel, on isolates drawn from the sequenced 1,011 collection. The same strains already carry Caudal 2024 transcriptomes and Muenzner 2024 proteomes, so it is a three-modality join on one genotype axis.
Ten extra, in rank order: rows 51–60 replace a row above that proves unreachable.									
51	**Li 2016 (tRNA fitness landscape)** • Deep mutational scan; fitness	S, 1	65,000 tRNA variants	1	6.5×10^4‡	6.5×10^4	S288C+ designed-edit	–	A near-complete fitness landscape of one native yeast gene. Every genotype is a known sequence and the sampling is exhaustive rather than one allele per gene, which is the opposite of what a deletion collection provides.
52	**Domingo 2018 (tRNA double-mutant landscape)** • Deep mutational scan; fitness	S, 1	23,000 single + double mutants	1	2.3×10^4‡	2.3×10^4	S288C+ designed-edit	–	Pairwise epistasis at nucleotide resolution inside one gene. The higher-order structure Costanzo and Kuzmin map between genes, measured within a gene, which is the resolution inverse design actually needs.
53	**Zhang 2021 (YETI titratable overexpression)** Metabolite / precursor; fitness across an induction gradient	S, 1	1,022 essential + 4,668 non-essential	beta-estradiol dose series	2.3×10^4‡	2.3×10^4	S288C+tag	–	Native-locus, continuously titratable overexpression. The dosage axis pathway engineering actually uses, and the only gain-of-function library that is not binary.
54	**Peeters 2021 (fermentation-trait QTL atlas)** Natural variation; ethanol / glycerol / isobutanol titer, stress resistance, aroma	S, 1	1,125 sequenced segregants	18 traits	2.0×10^4†	2.0×10^4	segregant-WGS	–	The only large sequenced panel with direct isobutanol, ethanol and glycerol titers. Hits the natural-isolate and the isobutanol asks at once.
55	**Van Leeuwen 2024 (short-chain organic acids)** Tolerance / robustness; fitness per acid	S, 1	4,800 YKO	acetic, butyric, octanoic	1.4×10^4†	1.4×10^4	S288C-KO	–	Three chemically homologous acids on one strain panel, so shared and acid-specific mechanisms separate within a single study rather than across incomparable screens.
56	**Chica 2026 (AutoDRY autophagy screen)** Modality / backbone; autophagy flux (deep-learning image classifier)	S, 1	4,760 KO + 1,159 DAmP	2 (nitrogen starvation, replete)	1.2×10^4†	1.2×10^4	S288C-KO	–	Largest single-condition imaging screen with a live Dryad deposit, and it covers essential-adjacent genes via DAmP rather than stopping at the non-essential set.
57	**Sopko 2006 (genome-scale overexpression toxicity)** Metabolite / precursor; growth-rate reduction on induction	S, 1	5,280 ORFs	GAL induced vs glucose	1.1×10^4†	1.1×10^4	S288C+ORF-plasmid	–	769 genes are growth-inhibitory when overexpressed. Overexpression toxicity is the constraint that caps titer when a pathway enzyme is pushed, and nothing in the built set measures it.

Table 4, continued

#	Dataset (class; phenotype)	Band, tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Time	Why
58	**Liu 2021 (tryptophan / isobutanol tolerance)** Tolerance / robustness; colony-size fitness (ScreenMill imaging)	S, 1	5,006 YKO	+/- isobutanol	1.0×10^4†	1.0×10^4	S288C-KO	–	A second, independent full-collection isobutanol screen with a different readout. Paired with Kuroda it gives a same-phenotype, same-genotype, different-method replicate.
59	**Kuroda 2019 (isobutanol-specific tolerance)** Tolerance / robustness; isobutanol-specific fitness	S, 1	4,800 YKO	isobutanol vs ethanol	9.6×10^3†	9.6×10^3	S288C-KO	–	The isobutanol screen, designed against an ethanol comparator so the alcohol-general response subtracts out. Its GLN3 hit raised production 4.9-fold, a rare tolerance-to-titer translation.
60	**Turco 2023 (Yeast Phenome)** [L] Tolerance / robustness; growth / fitness per environment	S, 2	5,000	7,536 environments	3.8×10^7†	3.8×10^7	S288C-KO	–	The widest environment axis in yeast, 96 percent chemical. Held out of tier 1 because it is a meta-aggregation: its instance count re-serves primary screens, several already built, so the net-new fraction is unknown until it is de-duplicated against the rest of this table. A loader already exists and retains 49 growth screens, so this is in flight rather than net-new work; Lee 2014 is not among those screens, so YeastPhenome does not backfill row 1.

Table 5: Sources for Table 4, in the same order. *Why it is ordered here* is the band reason for a perturb-seq or molecular-layers row, and the tier and measurement rule for a scale row; what the row buys is the *Why* column of Table 4. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

#	Dataset, citation and link	Why it is ordered here	Data
1	**Albert 2018 (eQTL in 1,012 segregants)** Albert FW, Bloom JS, Siegel J, Day L, Kruglyak L. eLife 2018;7:e35471. doi:10.7554/eLife.35471	State across genomes, first: bulk RNA-seq of 1,012 sequenced BY x RM segregants, the largest genotype-indexed transcriptome in yeast, and the bulk partner of the served Boocock 2025, whose pooled panel stores 393 of these same segregants.	GEO /eLife data availability (unconfirmed)
2	**Jakobson 2025 (genome-to-proteome map)** Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198	State across genomes: protein abundance across the same sequenced segregant panel Albert 2018 profiles by transcriptome, 800 F6 segregants, so protein and transcript loci are measurable on one genotype set.	Science data-availability statement (accession unconfirmed)
3	**Teyssonniere 2024 (species-wide proteome against transcriptome)** Teyssonniere EM, Trebulle P, Muenzner J, Loegler V, Ludwig D, Amari F, Mulleder M, Friedrich A, Hou J, Ralser M, Schacherer J. Proc Natl Acad Sci USA 2024;121:e2319211121. doi:10.1073/pnas.2319211121	State across genomes: proteomes for 942 of the 1,011 isolates whose transcriptomes the served Caudal 2024 holds and whose proteomes the served Muenzner 2024 holds for 796, so the de-duplication against Muenzner is the first task and the pairing with Caudal is one to one.	PNAS SI; PMC11087752. PRIDE identifier not fetched this pass
4	**Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)** Grossbach J, Gillet L, Clement-Ziza M, Schmalohr CL, Schubert OT, Schutter M, Mawer JSP, Barnes CA, Bludau I, Weith M, Tessarz P, Graef M, Aebersold R, Beyer A. Mol Syst Biol 2022;18:e10712. doi:10.15252/msb.202110712	State across genomes: transcript, protein and phosphosite on one set of 112 sequenced BY x RM strains, the only row where all three layers are observed within a genotype.	Mol Syst Biol SI + PRIDE, identifier not fetched this pass
5	**Smith 2008 (gene-environment interaction eQTL)** Smith EN, Kruglyak L. PLoS Biol 2008;6:e83. doi:10.1371/journal.pbio.0060083	State across genomes: mRNA abundance of 109 BY x RM segregants in glucose and in ethanol, the one segregant transcriptome with a condition axis, so genotype by environment is a within-panel contrast.	GEO (unconfirmed)
6	**Skelly 2013 (expression variation across isolates)** Skelly DA, Merrihew GE, Riffle M, et al., MacCoss MJ, Akey JM. Genome Res 2013;23:1496-1504. doi:10.1101/gr.155762.113	State across genomes: transcriptome and proteome on the same 22 sequenced isolates, the cheap and the expensive layer paired within one experiment on natural genomes.	Genome Research SI (unconfirmed)
7	**McManus 2014 (ribosome profiling, allele-specific)** McManus CJ, May GE, Spealman P, Shteyman A. Genome Res 2014;24:422-430. doi:10.1101/gr.164996.113	State across genomes: ribosome occupancy beside mRNA abundance for S. cerevisiae, S. paradoxus and their F1 hybrid, so translation is read allele by allele across two genomes; the paradoxus arm is out of species and is stored apart or dropped.	GEO, series not fetched this pass
8	**Foss 2007 (BY x RM segregant proteome)** Foss EJ, Radulovic D, Shaffer SA, Ruderfer DM, Bedalov A, Goodlett DR, Kruglyak L. Nat Genet 2007;39:1369-1375. doi:10.1038/ng.2007.22	State across genomes: the first segregant proteome on the BY x RM cross and the independent replicate of Grossbach 2022 fifteen years and two platforms earlier.	Nat Genet supplementary tables; not fetched
9	**Dong 2021 (MAGIC + SAM biosensor)** Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327. doi:10.1016/j.ymben.2021.03.005	A genome-wide CRISPR activation, interference and deletion library sorted on a metabolite biosensor, so the label is product concentration rather than fitness. The pattern a sorted Perturb-seq would reuse.	Metab Eng SI (accession unconfirmed)

Table 5, continued

#	Dataset, citation and link	Why it is ordered here	Data
10	**Momen-Roknabadi 2020 (inducible CRISPRi library)** Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723. doi:10.1038/s42003-020-01452-9	A genome-scale inducible CRISPR interference library with a released per-guide matrix. Induction control is what lets a knockdown be applied after a cell is captured rather than during outgrowth.	Addgene + Commun Biol SI
11	**McGlincy 2021 (genome-scale CRISPRi library)** McGlincy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205. doi:10.1186/s12864-021-07518-0	The second independently designed genome-scale yeast CRISPR interference library with a released per-guide matrix, so guide design and gene effect can be separated.	ingolia-lab.org/yeast-crispri + Addgene + BMC SI
12	**Bao 2018 (CHAnGE single-nucleotide library)** Bao Z, HamediRad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508. doi:10.1038/nbt.4132	A guide-indexed genome-wide library whose edits sit below the open reading frame, so the perturbation is an allele rather than a null.	Nat Biotechnol SI (raw sequencing location unconfirmed)
13	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)** Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654. PMID 27654654	A dose-graded knockdown axis. Graded rather than binary perturbation is what turns a per-cell readout into a dose-response curve.	AMB SI (DOI is an open verification item)
14	**Roy 2018 (multiplexed precision editing)** Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520. doi:10.1038/nbt.4137	Multiplexed designed edits per cell, which is combinatorial input in the sense the Perturb-seq specification requires.	Nat Biotechnol SI (accession unconfirmed)
15	**Newman 2006 (protein noise, GFP library)** Newman JRS, Ghaemmaghami S, Ihmels J, Breslow DK, Noble M, DeRisi JL, Weissman JS. Nature 2006;441:840-846. doi:10.1038/nature04785	Per-cell protein abundance distributions across a genome-scale tagged collection. Expression noise per gene is what sets how large a perturbation effect has to be before a per-cell readout can resolve it.	Nature SI
16	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)** Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21. doi:10.1128/mSystems.00410-21	Knockdown reaches the essential genes a deletion collection cannot hold, which is a third of the genome a deletion-based campaign would miss.	mSystems open-access SI
17	**Lian 2017 (CRISPR-AID, beta-carotene)** Lian J, HamediRad M, Hu S, Zhao H. Nat Commun 2017;8:1688. doi:10.1038/s41467-017-01695-x	Three perturbation modalities in one cell, activation, interference and deletion, which is combinatorial input by modality rather than by gene.	Nat Commun SI
18	**Hughes 2000 (compendium of expression profiles)** Hughes TR, Marton MJ, Jones AR, Roberts CJ, Stoughton R, et al., Friend SH. Cell 2000;102:109-126. doi:10.1016/S0092-8674(00)00015-5	The original designed-perturbation expression compendium, and the independent measurement that makes cross-laboratory replication of the deletion transcriptome testable against the built Kemmeren 2014.	Rosetta compendium, distributed with the paper; not fetched
19	**Lenstra 2011 (chromatin regulator deletion expression)** Lenstra TL, Benschop JJ, Kim T, et al., Holstege FCP. Mol Cell 2011;42:536-549. doi:10.1016/j.molcel.2011.03.026	The same design over chromatin regulators rather than sequence-specific factors, so the two together cover both halves of transcriptional control by deletion.	GEO (unconfirmed)
20	**Sun 2013 (mRNA synthesis and decay rates across deletion strains)** Sun M, Schwalb B, Pirkl N, Maier KC, Schenk A, Failmezger H, Tresch A, Cramer P. Mol Cell 2013;52:52-62. doi:10.1016/j.molcel.2013.09.010	A designed single-gene perturbation crossed with a transcriptome, and the only one that resolves the level into a synthesis rate and a decay rate. Rates are what a per-cell snapshot cannot recover on its own.	GEO, series not fetched this pass

Table 5, continued

#	Dataset, citation and link	Why it is ordered here	Data
21	**Jariani 2020 (yeast scRNA-seq through lag phase)** Jariani A, Vermeersch L, Cerulus B, Perez-Samper G, Voordeckers K, Van Brussel T, Thienpont B, Lambrechts D, Verstrepen KJ. eLife 2020;9:e55320. doi:10.7554/eLife.55320	Per-cell states through a carbon-source shift, so the readout is the distribution across an unsynchronized population rather than its mean.	GEO GSE144820 and GSE116246; from the collection sweep, not fetched
22	**Guo 2018 (CRISPR-Cas9 tiling of essential genes)** Guo X, Chavez A, Tung A, et al., Church GM, Shalem O. Nat Biotechnol 2018;36:540-546. doi:10.1038/nbt.4147	Guide tiling produces a graded allelic series across one gene, so guide position rather than gene identity carries the effect.	Nat Biotechnol SI (unconfirmed)
23	**Urbonaite 2021 (yeastDrop-Seq under drug treatment)** Urbonaite G, Lee JTH, Liu P, Parras GG, Hemberg M, Acar M. Commun Biol 2021;4:1372. doi:10.1038/s42003-021-02895-4	Single-drug arms against their combination with a per-cell readout, the environmental analog of a combinatorial genetic perturbation.	GEO GSE165686; Zenodo 10.5281/zenodo.4767298 and 10.5281/zenodo.4762526; from the collection sweep, not fetched
24	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692. doi:10.1038/s41564-018-0346-9	The high-sensitivity yeast protocol, strand and isoform aware. Sensitivity per cell is the constraint on how small a perturbation effect a campaign can resolve.	Nat Microbiol SI /GEO
25	**Gasch 2017 (single-cell stress heterogeneity)** Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050. doi:10.1371/journal.pbio.2004050	Separates intrinsic from extrinsic per-cell variation in an isogenic population under stress, which is the null a perturbation effect is measured against.	PLoS Biology SI /GEO
26	**Jaffe 2019 (multiplexed CRISPR interference epistasis)** Jaffe M, Sherlock G, Levy SF. G3 2019;9:2843-2852. doi:10.1534/g3.119.400323	Two knockdowns in the same cell with a measured interaction, which is the only combinatorial CRISPR interference design located in yeast.	G3 open-access SI (unconfirmed)
27	**Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)** Jackson CA, Beheler-Amass M, Tjarnberg A, Suresh I, Hickey AS, Bonneau R, Gresham D. bioRxiv 2023. 10.1101/2023.09.21.558277. doi:10.1101/2023.09.21.558277	The platform companion of Jackson 2020: 173,361 cells of one strain, which is what sets the per-cell variance a perturbation has to exceed.	GEO GSE242556 (from the scYeast Methods, Sec. 4.1.3)
28	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)** Brettner L, et al. Yeast 2024. doi:10.1002/yea.3927. doi:10.1002/yea.3927	Combinatorial barcoding in yeast at 96 to 384 multiplexed genotypes or environments per run. The throughput route by which a campaign becomes affordable.	Yeast (Wiley) SI
29	**Mulleder 2012 (prototrophic deletion collection)** Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178. doi:10.1038/nbt.2442	The prototrophic deletion collection. Jackson 2020 used a prototrophic background because auxotrophy blocks minimal and nitrogen-limited media, so this is the strain resource a campaign in defined media needs.	EUROSCARF; Addgene kit 40276
30	**Su 2023 (single-cell transcriptomes under four stresses)** Su Y, Xu C, Shea J, DeStephanis D, Su Z. BMC Genomics 2023;24:88. doi:10.1186/s12864-023-09184-w	Full-length rather than three-prime tag counting, so isoform-level readout is on the table for a campaign that needs it.	GEO GSE201386 (from the scYeast Methods)

Table 5, continued

#	Dataset, citation and link	Why it is ordered here	Data
31	**Wang 2022 (single-cell transcriptomes across replicative aging)** Wang J, Sang Y, Jin S, Wang X, Azad GK, McCormick MA, Kennedy BK, Li Q, Wang J, Zhang X, et al. Aging Cell 2022;21:e13712. doi:10.1111/acel.13712	The only yeast single-cell set with age as the condition axis, which is a covariate any pooled campaign carries whether or not it measures it.	GEO GSE210032 (from the scYeast Methods)
32	**Aulakh 2025 (genome-scale ionome)** Aulakh SK, et al. Cell Syst 2025;16:101319. doi:10.1016/j.cels.2025.101319	A metabolic readout on the whole deletion collection, which is the genotype axis the supported expression compendium already covers.	Cell Systems SI (accession unconfirmed)
33	**Leutert 2023 (phosphoproteome x 101 conditions)** Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773. doi:10.1038/s41594-023-01115-3	Enzyme regulation acts faster than transcription, so a phosphosite layer over 101 conditions is what explains flux changes an expression table cannot.	PRIDE PXD034997
34	**Brauer 2008 (growth-rate-controlled chemostat transcriptome)** Brauer MJ, Huttenhower C, Airoldi EM, et al., Botstein D. Mol Biol Cell 2008;19:352-367. doi:10.1091/mbc.e07-08-0779	Expression at controlled growth rate under each nutrient limitation, which is the condition axis Boer 2010 and Hackett 2016 measure metabolites and flux on.	GEO (unconfirmed)
35	**Hackett 2016 (SIMMER multi-omic flux)** Hackett SR, Zanotelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786. doi:10.1126/science.aaf2786	Flux, metabolite and transcript measured in one study under the same nutrient limitations, so the join is internal rather than across papers.	Science SI (accession unconfirmed)
36	**Zhu 2014 (kinase / phosphatase lipidomics)** Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872. PMC4196872	A lipidome over 129 signaling deletions, most of which carry an expression profile in the supported compendium.	Mol Biol Cell SI Table S4 (Excel)
37	**Gerke 2017 (urea-cycle mQTL)** Gerke J, et al., Fay JC. Genetics 2017;206:2199. doi:10.1534/genetics.117.201107	Metabolite quantitative trait loci on a segregant cross that also has an expression map, so a metabolite locus can be read through its transcript.	Genetics (GSA) SI
38	**Ambroset 2014 (metabolite QTL)** Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142. doi:10.1371/journal.pgen.1004142	A 74-metabolite panel on a sequenced segregant panel, the widest metabolite vector available on a recombinant genotype axis.	PLOS Genetics SI
39	**Blank 2005 (13C metabolic flux)** Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49. doi:10.1186/gb-2005-6-6-r49	The only row measuring flux rather than a concentration, on deletion mutants that also have a transcriptome in the supported set.	Genome Biology open-access SI
40	**Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)** Airoldi EM, Miller D, Athanasiadou R, Brandt N, Abdul-Rahman F, Neymotin B, Hashimoto T, Bahmani T, Gresham D. Mol Biol Cell 2016;27:1383-1396. doi:10.1091/mbc.E14-05-1013	Expression under the exact nitrogen-limited conditions that Jackson 2020 profiles single cells in, so the deletion effect separates from the medium effect.	GEO, series not confirmed this pass
41	**Martin-Perez 2017 (protein half-lives)** Martin-Perez M, Villen J. Cell Syst 2017;5:283-294.e5. doi:10.1016/j.cels.2017.08.008	Protein turnover per gene, the second of those two terms, spanning three orders of magnitude and therefore not replaceable by a global constant.	Cell Systems SI; PRIDE identifier not fetched this pass

Table 5, continued

#	Dataset, citation and link	Why it is ordered here	Data
42	**Boer 2010 (metabolome across nutrient limitations)** Boer VM, Crutchfield CA, Bradley PH, Botstein D, Rabinowitz JD. Mol Biol Cell 2010;21:198-211. doi:10.1091/mbc.e09-07-0597	The metabolite half of the chemostat nutrient-limitation series whose transcriptome half is Brauer 2008, same laboratory and same conditions.	MBoC SI (unconfirmed)
43	**Tengolics 2024 (domestication metabolome)** Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121. doi:10.1073/pnas.2313354121	Metabolite levels across sequenced isolates, the panel the supported Caudal 2024 transcriptomes and Muenzner 2024 proteomes also sit on.	PNAS Dataset S11
44	**Eder 2020 (flux QTL)** Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164. PMID 32034164	Flux mapped to segregant genotypes, which is the flux counterpart of the expression and protein maps on the same panel type.	Journal SI (venue and accession unconfirmed)
45	**Yu 2021 (proteome and metabolome under nitrogen limitation)** Yu R, Vorontsov E, Sihlbom C, Nielsen J. eLife 2021;10:e65722. doi:10.7554/eLife.65722	Both layers measured in one study under nitrogen limitation, so it calibrates the protein-to-metabolite step that cross-study joins assume.	eLife data availability (unconfirmed)
46	**de Boer 2020 (100M random promoters)** de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A. Nat Biotechnol 2020;38:56-65. doi:10.1038/s41587-019-0315-8	Scale band: tier 1, ordered by measurements (100,000,000).	GEO (de Boer 2020); DREAM 2022 release of 6,739,258 promoters
47	**Lee 2014 (HIP-HOP fitness signatures)** Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217	Scale band: tier 1, ordered by measurements (13,000,000).	Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap
48	**Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)** Nguyen Ba AN, Lawrence KR, Rego-Costa A, Gopalakrishnan S, Temko D, Michor F, Desai MM. eLife 2022;11:e73983. doi:10.7554/eLife.73983	Scale band: tier 1, ordered by measurements (800,000).	eLife data availability; SRA (unconfirmed)
49	**Galardini 2019 (four backgrounds x 38 conditions)** Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831. doi:10.15252/msb.20198831	Scale band: tier 1, ordered by measurements (575,472).	GEO GSE123118 + github.com/mgalardini/2018koyeast
50	**Dutta 2026 (barcoded natural-isolate chemical response)** Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026. doi:10.1038/s41467-026-73532-z	Scale band: tier 1, ordered by measurements (312,000).	Nat Commun SI; isolates from the 1,011 panel (ENA)
51	**Li 2016 (tRNA fitness landscape)** Li C, Qian W, Maclean CJ, Zhang J. Science 2016;352:837-840. doi:10.1126/science.aae0568	Scale band: tier 1, ordered by measurements (65,000).	Science SI (unconfirmed)
52	**Domingo 2018 (tRNA double-mutant landscape)** Domingo J, Diss G, Lehner B. Nature 2018;558:117-121. doi:10.1038/s41586-018-0170-7	Scale band: tier 1, ordered by measurements (23,000).	Nature SI (unconfirmed)

Table 5, continued

#	Dataset, citation and link	Why it is ordered here	Data
53	**Zhang 2021 (YETI titratable overexpression)** Zhang Y, Ho Yee Chow M, Ling Y, et al., Andrews BJ. Mol Syst Biol 2021;17:e10321. doi:10.15252/msb.202010321	Scale band: tier 1, ordered by measurements (22,760).	Mol Syst Biol SI (repository accession unconfirmed)
54	**Peeters 2021 (fermentation-trait QTL atlas)** Peeters B, Reijbroek F, Verbaet J, et al., Jarosz DF, Verstrepen KJ. 2021. PMID 34727964. PMID 34727964	Scale band: tier 1, ordered by measurements (20,250).	Journal SI; raw reads at SRA/ENA (accession unconfirmed)
55	**Van Leeuwen 2024 (short-chain organic acids)** Shared and specific genetic determinants of tolerance to acetic, butyric and octanoic acid. PMC10903034. PMC10903034	Scale band: tier 1, ordered by measurements (14,400).	Journal SI (open-access PMC)
56	**Chica 2026 (AutoDRY autophagy screen)** Chica N, et al. Nat Cell Biol 2026;28:465-479. doi:10.1038/s41556-025-01837-0	Scale band: tier 1, ordered by measurements (11,838).	Dryad 10.5061/dryad.cfxpnvxdh
57	**Sopko 2006 (genome-scale overexpression toxicity)** Sopko R, Huang D, Preston N, et al., Boone C, Andrews B. Mol Cell 2006;21:319-330. doi:10.1016/j.molcel.2005.12.011	Scale band: tier 1, ordered by measurements (10,560).	Mol Cell SI tables
58	**Liu 2021 (tryptophan / isobutanol tolerance)** Liu H-L, Wang CH-T, Chiang EP-I, Huang C-C, Li W-H. Biotechnol Biofuels 2021;14:200. doi:10.1186/s13068-021-02048-z	Scale band: tier 1, ordered by measurements (10,012).	GEO GSE175794 (companion RNA-seq); screen in BMC SI
59	**Kuroda 2019 (isobutanol-specific tolerance)** Kuroda K, Hammer SK, Watanabe Y, Montano Lopez J, Fink GR, Stephanopoulos G, Ueda M, Avalos JL. Cell Syst 2019;9:534-547.e5. doi:10.1016/j.cels.2019.10.006	Scale band: tier 1, ordered by measurements (9,600).	ArrayExpress E-MTAB-8175 (RNA-seq); screen data in Cell Syst SI
60	**Turco 2023 (Yeast Phenome)** Turco G, Chang C, Wang RY, et al., Boone C, Andrews BJ, Roth FP. Sci Adv 2023;9:eadg5702. doi:10.1126/sciadv.adg5702	Scale band: tier 2, ordered by measurements (37,680,000).	yeastphenome.org (Zenodo 10.5281/zenodo.7714347); loader torchcell/datasets/scerevisiae/yeastphenome.py, 49 screens, not yet in the built set

4 What each candidate joins to ✗

A dataset's value in a substrate is not only what it measures. It is also what becomes measurable once it sits beside what is already held. Table 6 names 89 such pairs across the sixty rows of Table 4, 53 of the sixty carrying at least one, each with the axis the two share.

4.1 A join needs a shared axis, not a shared theme ✗

The **join key** column is required rather than decorative. Two metabolomics papers are not joinable because both measure metabolites; they are joinable when they measure overlapping analytes on overlapping strains. The keys in use are the deletion collection, the sequenced 1,011-isolate panel, a segregant panel from one cross, a guide library over one gene set, a media formulation, a compound set, and a single gene. A pair with none of these is a theme and is not listed.

Fifty-five of the 89 pairs, on 40 of the sixty rows, have a partner that is already served, so that join becomes runnable on the day the candidate is ingested rather than after two ingestions. Thirteen of those partners became served since the previous pass, which is why pairs that were candidate-to-candidate then, Teyssonniere 2024 against Muenzner 2024 or Hughes 2000 against Hu 2007, are now one ingestion from running.

Two kinds of key are distinguished in the table and the difference decides what a join can do. A *strain-level* key means both layers were measured on the same genotypes, so the join is a paired observation and the residual between the layers is per strain. A *gene-level* key means one side is a single genome-wide coefficient, a half-life or a translation rate, that applies to every strain alike. A gene-level layer cannot explain why two strains differ; it explains why two genes differ, and the two are not interchangeable.

4.2 The joins that motivate the ordering ✗

Four groups carry most of the weight, and each is a specific question rather than an accumulation.

State on one genome axis. The eight rows of the first band are joins by construction. Albert 2018 and Jakobson 2025 pair transcript and protein on one segregant class, and Albert 2018 pairs with the served Boocock 2025 on the 393 pooled segregants the two share. Teyssonniere 2024 pairs with the served Caudal 2024 strain by strain, 942 against 943 isolates, and with the served Muenzner 2024 on the protein layer, where the pairing is also the de-duplication question. Grossbach 2022 and Foss 2007 are the same cross fifteen years apart. Smith 2008 is the segregant transcriptome with a condition axis, so it pairs with every other BY x RM row on genotype and with Jackson 2020 on media.

Transcript against protein, and the two terms between them. Steady-state transcript and steady-state protein are both well covered, by the served Kemmeren 2014 and Messner 2023 and by the state rows, and they disagree. Two rows make the disagreement something a model can carry a term for rather than absorb as error. McManus 2014 gives translation efficiency beside mRNA abundance and finds ribosome occupancy more conserved than transcript abundance, so translation buffers rather than tracks. Martin-Perez 2017 gives turnover for 3,160 proteins, spanning three orders of magnitude, which is why a single global mRNA-to-protein scaling cannot be right. Sun 2013 supplies the transcript-side counterpart, decomposing mRNA level into synthesis and decay across 46 deletion strains.

Cross-laboratory replication of the deletion transcriptome. Kemmeren 2014 is served and is the largest genotype-indexed transcriptome with designed perturbations, and the served Hu 2007 now gives it one independent check on 263 transcription-factor deletions. Hughes 2000, Lenstra 2011 and Sun 2013 add three more laboratories across fourteen years of array and sequencing platforms, so agreement can be measured gene by gene on shared strains rather than assumed. That check matters here specifically: the served Nadal-Ribelles 2025 does not replicate its own genotype assignment across batches, so a second opinion on what a deletion does to a transcriptome is not a formality.

Metabolism against expression on one genotype axis. Blank 2005 measures flux through central carbon metabolism on deletion mutants that already carry an expression profile in the served Kemmeren 2014 compendium, which is the only place in this table where the transcript-to-flux step can be fitted rather than assumed. Aulakh 2025 adds a second metabolic panel, the ionome, on the whole collection. Gerke 2017, Ambroset 2014 and Eder 2020 put metabolite and flux quantitative trait loci on segregant crosses that Albert 2018 and Jakobson 2025 give an expression and a protein map, so a metabolite locus can be read against both.

A condition axis measured on more than one layer. Brauer 2008 and Boer 2010 are the transcript and metabolite halves of one chemostat series, same laboratory and same limitations, and Hackett 2016 adds flux over the same conditions. Airoldi 2016 supplies the nitrogen-limited counterpart, and its media are the media the served Jackson

2020 profiles single cells in. That last pairing is what makes a deletion effect in the single-cell atlas a contrast against a wild-type baseline in the same medium rather than an absolute.

Table 6: Named joins between a candidate and either a supported dataset or another candidate, for the sixty rows of Table 4. # is the candidate's rank there. A partner in **bold** is already built, so that row's join becomes runnable the moment the candidate is ingested; an unbolded partner is itself a candidate, so the join costs two ingestions. *Join key* is the shared axis that makes the pair addressable; a pair with no such axis is a theme and is not listed. None of these transfers has been measured.

#	Candidate	Partner	Join key	What the join yields
1	Albert 2018 (eQTL in 1,012 segregants)	Jakobson 2025 (genome-to-proteome map)	the same segregant panel	The transcript half of a paired transcript and protein map.
		Gerke 2017 (urea-cycle mQTL)	segregant genotype class	A metabolite locus read through the transcripts of the pathway that produces it.
2	Jakobson 2025 (genome-to-proteome map)	Albert 2018 (eQTL in 1,012 segregants)	the same sequenced segregant panel	Protein and transcript quantitative trait loci on one genotype set, which is the direct measurement of how much of protein variation transcript variation explains.
		Messner 2023 (proteome)	protein abundance, natural variation against deletion	Whether the proteome response to a deleted gene resembles the response to a natural allele of it.
3	Teyssonniere 2024 (species-wide proteome against transcriptome)	**Caudal 2024 (pan-transcriptome)**	the same sequenced isolates, 942 against 943	Transcript and protein for the same strain, one to one rather than by overlap. The published result on this pair is that the two correlate weakly and share 3 percent of their associated variants, so it is the calibration an inference model is scored against.
		Muenzner 2024 (natural-isolate proteomes)	the 1,011-isolate panel, protein layer	De-duplication before either is built: the two share authors and a panel, and whether they are independent acquisitions is unsettled.
4	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	Albert 2018 (eQTL in 1,012 segregants)	BY x RM segregants, transcript layer	112 strains with transcript and protein measured together against 1,012 with transcript alone, so the within-strain RNA-to-protein residual fitted on the small panel can be applied to the large one.
		Jakobson 2025 (genome-to-proteome map)	segregant genotype class, protein layer	Two segregant proteomes on different panels and platforms, which is what separates a protein quantitative trait locus from a batch.
		Leutert 2023 (phosphoproteome x 101 conditions)	phosphosite identity	Phosphorylation driven by genotype against phosphorylation driven by condition, on one site vocabulary.
6	Skelly 2013 (expression variation across isolates)	**Muenzner 2024 (natural-isolate proteomes)**	natural isolates, paired against joined modalities	A within-experiment paired transcript and protein measurement to calibrate the cross-study join.
7	McManus 2014 (ribosome profiling, allele-specific)	**Messner 2023 (proteome)**	gene identity, translation rate against protein abundance	A per-gene translation term to put against measured protein level, which is the coefficient a model otherwise has to learn blind.
		Martin-Perez 2017 (protein half-lives)	gene identity, synthesis against degradation	Both halves of protein turnover, so steady-state abundance can be decomposed rather than only predicted.
8	Foss 2007 (BY x RM segregant proteome)	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	the same cross, protein layer, fifteen years and two platforms apart	Cross-laboratory replication of segregant protein abundance, which neither measurement can establish alone.
9	Dong 2021 (MAGIC + SAM biosensor)	**Lian 2019 (MAGIC CRISPR-AID)**	the same tri-functional library, fitness against biosensor readout	One library, two labels: growth selection and product concentration, which is how a tolerance screen is converted into a production screen.
		Cachera 2023 (CRI-SPA betaxanthin)	metabolite biosensor or colorimetric product readout on a genome-scale library	Two product-labeled genome-scale screens on different chemistry.
10	Momen-Roknabadi 2020 (inducible CRISPRi library)	McGlincy 2021 (genome-scale CRISPRi library)	independently designed guide libraries over the same genes	The paired half of the library-transfer test.
		SGD (essentiality)	essential genes, knockdown against deletion	A graded phenotype where a deletion collection records only absence.

Table 6, continued

#	Candidate	Partner	Join key	What the join yields
11	McGlincy 2021 (genome-scale CRISPRi library)	Momen-Roknabadi 2020 (inducible CRISPRi library)	independently designed guide libraries over the same genes	Library-to-library transfer, which is the cheapest decisive test of whether a model learned gene function or guide-design idiosyncrasy.
		Smith 2016 (CRISPRi chem-genetic)	CRISPR interference over the same gene set	A knockdown fitness prior for the chemical-genetic conditions already built.
12	Bao 2018 (CHAnGE single-nucleotide library)	**Lian 2019 (MAGIC CRISPR-AID)**	guide-indexed genome-wide library from the same laboratory	Sub-gene alleles against whole-gene activation, interference and deletion on one platform lineage.
		Li 2016 (tRNA fitness landscape)	designed single-nucleotide variants with a fitness label	Genome-wide shallow variant coverage against single-gene exhaustive coverage.
13	Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	**Lopez 2024 (isobutanol screen, private)**	isobutanol, knockdown against deletion	A graded knockdown axis over the phenotype the built biosensor screen measures as a titer proxy.
		Kuroda 2019 (isobutanol-specific tolerance)	isobutanol tolerance on the same collection	Dose-graded knockdown against whole-gene deletion for one phenotype.
14	Roy 2018 (multiplexed precision editing)	Bao 2018 (CHAnGE single-nucleotide library)	designed edits, single against multiplexed	Whether multiplexed edit effects are the sum of their single-edit effects.
15	Newman 2006 (protein noise, GFP library)	**Messner 2023 (proteome)**	protein abundance across the genome, per cell against population	Which proteins have a population mean that no single cell is near, which decides where a mean-level model is the wrong object.
		Gasch 2017 (single-cell stress heterogeneity)	per-cell variation in an isogenic population	Noise at the protein and transcript layers on the same question.
16	Mukherjee 2021 (CRISPRi essential genes x acetic acid)	**Mormino 2022 (CRISPRi acetic-acid)**	acetic acid, CRISPR interference	The genome-scale extension of a twelve-gene screen already built.
		Mota 2024 (weak-acid screen)	weak-acid stress on deletion strains	Essential-gene coverage for a phenotype the deletion collection can only sample from the non-essential side.
17	Lian 2017 (CRISPR-AID, beta-carotene)	**Ozaydin 2013 (beta-carotene screen)**	beta-carotene titer	Three perturbation modalities against a whole-collection deletion screen on the same product.
18	Hughes 2000 (compendium of expression profiles)	**Kemmeren 2014 (deletion compendium)**	deletion strains, gene by gene, fourteen years and two array platforms apart	Cross-laboratory replication of the deletion transcriptome, which the built set cannot test against itself and which Nadal-Ribelles 2025 failed cross-batch.
		Hu 2007 (TF-deletion error-model scores)	deletion strains with a bulk expression readout	A third independent compendium, so agreement can be measured across three laboratories rather than asserted from two.
19	Lenstra 2011 (chromatin regulator deletion expression)	**Kemmeren 2014 (deletion compendium)**	single deletions with a bulk expression readout, different gene class	Chromatin regulators beside sequence-specific factors on one expression readout.
		Sameith 2015 (dm)	regulator deletions, single against double	Whether chromatin-regulator effects combine the way transcription-factor effects do.
20	Sun 2013 (mRNA synthesis and decay rates across deletion strains)	**Kemmeren 2014 (deletion compendium)**	single deletion strains, gene by gene	A steady-state transcript level beside the synthesis and decay rates that produce it, so two strains with the same level but different kinetics stop looking identical.
		Martin-Perez 2017 (protein half-lives)	turnover, transcript side against protein side, gene by gene	Both half-life terms in one place, which is what an RNA-to-protein model needs in order to have a reason for the two to disagree.
21	Jariani 2020 (yeast scRNA-seq through lag phase)	**Jackson 2020 (TF-deletion pseudobulk)**	carbon-source shift with a per-cell readout	A transition sampled densely in time where the atlas samples eleven steady states.

Table 6, continued

#	Candidate	Partner	Join key	What the join yields
22	Guo 2018 (CRISPR-Cas9 tiling of essential genes)	**SGD (essentiality)**	essential genes	A graded allelic series where the built record is binary.
23	Urbonaite 2021 (yeastDrop-Seq under drug treatment)	**Hoepfner 2014 (HIP/HOP atlas)**	chemical treatment of yeast	A per-cell readout for compound response where the built atlas has a pooled fitness score.
24	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	**Nadal-Ribelles 2025 (Perturb-seq)**	same laboratory, protocol against application	The sensitivity ceiling of the platform the built genome-scale single-cell dataset was produced on.
		Gasch 2017 (single-cell stress heterogeneity)	isogenic per-cell variation in BY4741	Two independent measurements of how much an unperturbed yeast population varies, which is the floor a perturbation must clear.
25	Gasch 2017 (single-cell stress heterogeneity)	Gasch 2000 (environmental stress response)	the same stress conditions, bulk against single cell	How much of the canonical stress response is a population average of cells that are individually in different states.
26	Jaffe 2019 (multiplexed CRISPR interference epistasis)	**Costanzo 2016 (dmi)**	gene pairs, knockdown against deletion	Whether digenic interaction measured between two nulls is recovered between two knockdowns, which decides if the built interaction data can supervise a knockdown campaign.
		Kuzmin 2018 (tmi)	higher-order gene combinations	The knockdown analog of trigenic interaction.
27	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	**Jackson 2020 (TF-deletion pseudobulk)**	same platform and laboratory, wild type against deletions	The unperturbed per-cell variance the perturbed atlas is read against.
28	Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	**Jackson 2020 (TF-deletion pseudobulk)**	multiplexed genotypes per single-cell run	The throughput ceiling for a barcoded-genotype design, measured rather than assumed.
29	Mulleder 2012 (prototrophic deletion collection)	**Mulleder 2016 (amino-acid metabolome)**	the same prototrophic deletion strains	The strain resource the built amino-acid metabolome was measured on, which is what makes defined-medium phenotyping interpretable.
		Jackson 2020 (TF-deletion pseudobulk)	prototrophy as a precondition for minimal media	A ready-made prototrophic library for a campaign that needs nitrogen or carbon limitation.
30	Su 2023 (single-cell transcriptomes under four stresses)	Gasch 2000 (environmental stress response)	osmotic and starvation stress	Full-length per-cell transcripts against the bulk stress-response reference.
31	Wang 2022 (single-cell transcriptomes across replicative aging)	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	wild-type per-cell states over time	Replicative age as a covariate on a platform that does not measure it.
32	Aulakh 2025 (genome-scale ionome)	**Mulleder 2016 (amino-acid metabolome)**	the whole deletion collection, metabolite readout	Two orthogonal metabolic panels on one genotype axis.
		Kemmeren 2014 (deletion compendium)	the same deletion collection	Element levels against the transcriptional response of the same knockout.
33	Leutert 2023 (phosphoproteome x 101 conditions)	Gasch 2000 (environmental stress response)	overlapping stress conditions	Post-translational regulation against transcriptional regulation for one condition set, on the layer that acts first.
		Messner 2023 (proteome)	protein identity	Modification state against abundance, which abundance alone cannot separate.
34	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	Boer 2010 (metabolome across nutrient limitations)	the same chemostat limitations	The transcript half of a paired transcript and metabolite series.

Table 6, continued

#	Candidate	Partner	Join key	What the join yields
		Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	growth rate under limitation, carbon against nitrogen	Whether the growth-rate expression program is nutrient-general.
35	Hackett 2016 (SIMMER multi-omic flux)	Boer 2010 (metabolome across nutrient limitations)	the same nutrient-limited chemostats	Metabolite concentration and flux under one condition set, which is what a kinetic model needs and neither supplies alone.
		Brauer 2008 (growth-rate-controlled chemostat transcriptome)	nutrient limitation at controlled growth rate	Expression, metabolite and flux over one condition axis.
36	Zhu 2014 (kinase / phosphatase lipidomics)	**da Silveira 2014 (lipidomics)**	lipid species on deletion strains	A second lipidome on a signaling-focused genotype set.
		Kemmeren 2014 (deletion compendium)	kinase and phosphatase deletions	Lipid composition against the transcriptional response of the same signaling mutant.
		Xue 2025 (free fatty acids, private)	fatty-acid metabolism	Lipid class distribution against free fatty-acid titer, the two sides of the malonyl-CoA sink.
37	Gerke 2017 (urea-cycle mQTL)	Albert 2018 (eQTL in 1,012 segregants)	segregant genotype class	A metabolite locus and the expression loci in the same interval.
		Mulleder 2016 (amino-acid metabolome)	nitrogen and amino-acid metabolites	Natural-allele against deletion effects on one metabolic module.
38	Ambroset 2014 (metabolite QTL)	Peeters 2021 (fermentation-trait QTL atlas)	segregant panels with fermentation traits	74 metabolites against 18 mapped traits, so a trait locus can be read as a metabolite change.
		Zelezniak 2018 (metabolome)	metabolite panel	Recombinant against deletion genotypes on a metabolite readout.
39	Blank 2005 (13C metabolic flux)	**Kemmeren 2014 (deletion compendium)**	deletion strains present in both	Flux against expression for the same knockout, which is the only place the transcript-to-flux step can be fitted rather than assumed.
		Mulleder 2016 (amino-acid metabolome)	deletion strains, flux against pool size	Whether a changed pool reflects changed flux or changed demand.
40	Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	**Jackson 2020 (TF-deletion pseudobulk)**	the same nitrogen-limited media	A wild-type bulk baseline for the medium the single-cell atlas perturbs in.
		Yu 2021 (proteome and metabolome under nitrogen limitation)	nitrogen limitation	Transcript, protein and metabolite under one limitation.
41	Martin-Perez 2017 (protein half-lives)	**Messner 2023 (proteome)**	gene identity, half-life against abundance	Which proteins are abundant because they are made fast and which because they are destroyed slowly, a distinction abundance alone cannot make.
		Kemmeren 2014 (deletion compendium)	gene identity, protein half-life against transcript response	Whether a transcript change reaches the protein layer at all, which for a long-lived protein it largely does not.
42	Boer 2010 (metabolome across nutrient limitations)	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	the same chemostat limitations and growth rates	The metabolite and transcript halves of one experimental series.
		Zelezniak 2018 (metabolome)	metabolite panel	Condition-driven against genotype-driven metabolite variation.

Table 6, continued

#	Candidate	Partner	Join key	What the join yields
43	Tengolics 2024 (domestication metabolome)	**Caudal 2024 (pan-transcriptome)**	sequenced natural isolates	Metabolite against transcript variation across the same population.
44	Eder 2020 (flux QTL)	Blank 2005 (13C metabolic flux)	flux measurement, natural variation against deletion	Whether flux control points found by deletion are the loci natural variation actually moves.
		Albert 2018 (eQTL in 1,012 segregants)	segregant genotype class	Flux loci against expression loci on comparable panels.
45	Yu 2021 (proteome and metabolome under nitrogen limitation)	Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	nitrogen limitation	The transcript layer for the protein and metabolite layers measured here.
46	de Boer 2020 (100M random promoters)	**Vaishnav 2022 (promoter sort-seq)**	the same random promoter library, expression against fitness	Two labels on one sequence set, which turns a sequence-to-expression model into a sequence-to-fitness model without new data.
		Renganaath 2020 (natural promoter-variant MPRA)	promoter sequence with a measured expression readout	Whether a model trained on random sequence predicts the effect of a real allele, the cheapest decisive transfer test in the table.
47	Lee 2014 (HIP-HOP fitness signatures)	**Hoepfner 2014 (HIP/HOP atlas)**	the same heterozygous and homozygous collections, compound response	Two chemical-genetic matrices on one genotype axis, which is enough compound overlap to measure cross-screen reproducibility directly.
		Hillenmeyer 2008 (FitDb HIP, het)	the same collections and readout	A third independent screen of the same design.
48	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	**Bloom 2019 (16-cross segregant panel)**	segregant genotype class, panel size	One cross at 100,000 progeny against sixteen founders at 14,000, which separates panel size from allelic diversity.
49	Galardini 2019 (four backgrounds x 38 conditions)	**Costanzo 2021 (condition-SGA)**	deletion strains across conditions	The same conditional fitness question asked in four backgrounds rather than one.
		Hale 2024 (CRISPRi x segregants)	perturbation held fixed, background varied	Deletion against knockdown for the background-dependence question.
50	Dutta 2026 (barcoded natural-isolate chemical response)	**Caudal 2024 (pan-transcriptome)**	the sequenced 1,011-isolate panel	Chemical response and transcriptome on one isolate set.
		Hoepfner 2014 (HIP/HOP atlas)	compound response, isolates against deletions	Whether compound sensitivity found by gene deletion predicts sensitivity across natural genetic backgrounds.
51	Li 2016 (tRNA fitness landscape)	Domingo 2018 (tRNA double-mutant landscape)	the same tRNA gene, single against double mutants	Whether a single-mutant landscape predicts the double-mutant one, which is the within-gene form of the epistasis question.
58	Liu 2021 (tryptophan / isobutanol tolerance)	Kuroda 2019 (isobutanol-specific tolerance)	isobutanol tolerance on the same deletion collection	Same genotypes, same phenotype, different readout, which is a method-effect control no single screen provides.
59	Kuroda 2019 (isobutanol-specific tolerance)	**Lopez 2024 (isobutanol screen, private)**	isobutanol on the deletion collection	Tolerance against production for one product, on one genotype axis.
60	Turco 2023 (Yeast Phenome)	**Hoepfner 2014 (HIP/HOP atlas)**	aggregated growth screens against a primary screen	De-duplication, which is the precondition for knowing what the aggregate adds.

5 What was dropped ✗

Table 7: Considered and dropped. *no-sequence* is the hard gate: without a route to the strain's genomic content there is no genotype to map a phenotype from.

Dataset or group	Rule	Reason
Snoek 2015 robot-assisted genome shuffling (ethanol tolerance)	no-sequence	Shuffled progeny are recombinants of unrecorded parental segments with no per-strain sequencing, so the genotype cannot be reconstructed.
Adaptive laboratory evolution / EMS mutagenesis tolerance panels (e.g. farnesol-FPP selection)	no-sequence	Mutations are unmapped, so a strain's genome is unknown even in principle without whole-genome resequencing the study did not do.
Avalos lab isobutanol-biosensor deletion screen	not-a-dataset	Described only in a PhD thesis and a DOE report. Track for publication; not citable as a dataset.
Kitamoto/Kaneko ester and higher-alcohol screen	not-a-dataset	Citation could not be pinned to a specific paper. A placeholder, not a located dataset.
Fuhrer/Zamboni flow-injection metabolomics	off-species	The validated genome-scale application is in E. coli. No yeast deletion-collection version was located.
Anglada-Girotto 2022 (CRISPRi + metabolomics)	off-species	352 CRISPRi genes against 1,342 drug-induced metabolic changes, but in E. coli. Verified from the mirrored PDF.
Wildenhain 2016 (Sci Data chemical-genetic matrix)	already-built	A data descriptor, not a second experiment. Its own Data Citation 1 is NCBI PubChem BioAssay AID 1159580, which is exactly what torchcell/datasets/scerevisiae/wildenhain2015.py already ingests: 242 strains, 5,518 compounds, 492,126 interaction tests.
O'Duibhir 2014 expression compendium	already-built	The paper's expression data IS the Kemmeren 2014 compendium plus PCA transforms, already built, as oduibhir2014.py states; only its growth-rate readout was new and that is built too. Caught by check_candidate_overlap.py rather than by inspection, which is the first time the script found a duplicate before a person did.
Ozturk 2022 proteome-effects screen	off-species	3,308 deletions in S. pombe. Out of species scope.
Chen 2024 K. phaffii CRISPR fitness screen; Y1000+ multi-species panel	off-species	Not S. cerevisiae. Revisit if the generalization axis widens to other hosts.
iIsor850 genome-scale model (Issatchenkia orientalis)	not-a-dataset	A reconstruction, not an experiment, and off species. No genome-wide I. orientalis phenotype screen was located; the CABBI knockout library is unpublished.
CABBI 13C-MFA kinetic-model repository	not-a-dataset	16 knockout strains total, and the released fluxes are K-FIT model fits recapitulating 75-77 percent of measurements, not raw measurement. Blank 2005 supersedes it on both counts.
YEASTRACT+ (Teixeira 2023; Monteiro 2020)	not-a-dataset	A literature-curated knowledge base of 304,547 documented TF-target associations, each tagged with the evidence type (DNA binding or expression) and with the environmental condition in which it was observed, so a condition-scoped regulatory graph is recoverable from it. It is a prior over the genome, not a genotype-by-environment measurement, and it belongs in the graph layer beside GO and the metabolic model rather than in this table. Both papers are mirrored; the flat files are shared on request by the maintainers and are not yet held, and scYeast (Fan 2027) built its attention prior from the documented-targets subset (about 170,000 pairs).
Sc2.0 / Sc3.0 synthetic-genome design	not-a-dataset	Genome-engineering programs, not gene-indexed perturbation-phenotype data.
Lee 2014 / Hoepfner 2014 / Vanacloig-Pedros 2022 / Messner 2023 / Mulleder 2016 / Lian 2019 / Mormino 2022 / Nadal-Ribelles 2025 and 12 others	already-built	Named as top candidates by the 2026-07 triage pass and BUILT since. Only Lee 2014 remains outstanding and is kept as row 4.
Bloom 2019 (16-parent cross)	already-built	Ranked 12 in the previous pass and built since as the 50th supported dataset: 13,950 sequenced segregants over 38 traits, L0-L4 verified. Its place in this pass is as a join partner, not a candidate.

Dataset or group	Rule	Reason
TF-target prior networks used by Jackson 2020 (YEASTRACT, ATAC-motif of Castro 2019, Bussemaker affinity of Ward 2008, Tchourine 2018 gold standard)	not-a-dataset	Four regulator-by-gene matrices: 11,486 unsigned YEASTRACT edges over 3,912 genes and 152 TFs, 71,865 signed ATAC-motif edges over 5,551 genes and 138 TFs, a dense 6,516 by 123 affinity matrix, and the 1,403-edge signed gold standard. Each is a prior over the genome with no genotype and no environment, so each belongs in the graph layer beside GO and YEASTRACT+, not in a genotype-by-environment table. All four ship as TSVs in the eLife Source code 1 archive.
Tchourine 2018 bulk expression compendium (2,577 observations)	not-a-dataset	The benchmark Jackson 2020 compares its single-cell network against. It is a re-aggregation of public bulk series, and the strain behind an observation lives in each source series' metadata rather than in the released matrix, so admitting it is a de-duplication and re-curation task against Kemmeren 2014, Gasch 2000 and Brauer 2008 rather than one loader. Revisit if the per-sample genotype table is located.
Albert 2014 (X-pQTL, single-cell protein abundance in large populations)	not-a-dataset	Albert FW, Treusch S, Shockley AH, Bloom JS, Kruglyak L. Nature 2014;506:494-497. Considered as the BY x RM segregant proteome and it is not one: protein level is read from a green fluorescent protein fusion for one gene at a time in a large unsequenced pool, and the mapping comes from sorting that pool rather than from genotyped individuals. There is no strain-by-protein matrix to store, so it cannot enter as a proteome row. The segregant proteome rows here are Grossbach 2022 and Foss 2007.
Scholes 2019 bulk RNA-seq control (GEO GSE135430)	not-a-dataset	Jackson 2020's external bulk reference point. One wild-type BY4741 genotype in one condition, and the study's own variable is the RNA isolation protocol, so there is neither a genotype nor an environment axis to record.
Brandner 2025 (mapSPLiT, CRISPRa and CRISPRi single-cell transcriptomes in E. coli and P. putida)	not-a-dataset	The only microbial dataset found that reaches both Perturb-seq axes on purpose: 52 transcription factors targeted by 118 guides, pooled, with multi-guide combinations included to resolve genetic interactions, read out as a single-cell transcriptome. Pseudomonas putida is a single-cell bioproduction host and so is in scope for the generalization axis. It is a preprint and no deposited accession was found, so there is no scriptable data route and it is excluded rather than ranked. It is the closest existing design template for the campaign and should be revisited on publication.
Mammalian Perturb-seq (Replogle 2022, Zhu 2026, Dixit 2016, Datlinger 2017, Yao 2024)	off-species	The format references, and off species. Replogle perturbs every expressed gene by CRISPR interference across more than 2.5 million cells; Zhu 2026 crosses a genome-scale perturbation with several stimulation time points across about 22 million cells, which is the perturbation-by-context tensor a yeast campaign is an analog of; Yao 2024 pools random multi-perturbation composites over 598 genes and decompresses them, which is the one published route to second-order genetic interactions at a tractable cell count. None supplies yeast data; all four inform the design.
Bacterial single-cell transcriptomics (Wang 2023 M3-seq, Ma 2023 BacDrop, McNulty 2023 ProBac-seq, Kuchina 2021 microSPLiT)	off-species	Off species, and the axis is environmental rather than genetic: hundreds of thousands of cells over stress, antibiotic and growth-stage conditions in isogenic backgrounds. No perturbation-by-readout matrix to ingest.
Fulcher 2024 (nanoSPLITS), Taniguchi 2010, Baronas 2026, Leonaviciene 2023 and 2020, Datlinger 2021, Macosko 2015, Rosenberg 2018	off-species	Platform and paired-modality papers, all off species. nanoSPLITS measures a transcriptome and a mass-spectrometry proteome from the same single cell, which is the precedent for inferring an expensive layer from a cheap one on paired anchors; Taniguchi 2010 is the reference for protein and mRNA copy-number noise in a microbial host. Neither releases a perturbation matrix.
Reviews, protocols and single-cell statistics in the collection (Nadal-Ribelles 2024, Sun 2023, Larson 2013, Zun 2023, Gaisser 2024, Squair 2021, Zhang 2020, Svensson 2020, Grun 2014, Robinson 2008, McCarthy 2012, Hart 2013, Wang 2026)	not-a-dataset	Thirteen items with no perturbation data of their own. They are the sizing and false-discovery discipline for the campaign, not rows: pseudobulk replicate variation, sequencing depth per cell, dispersion estimation, and whether droplet counts are zero-inflated.

Table 7 holds the 25 rows that were considered and never ranked: twelve that are not a dataset (a prior, a method, a knowledge base), seven that are off species, four that were already served before the previous pass, and two with no

sequence basis. The thirteen rows that were ranked last pass and are served now are not in this table; they are in Table 3, because they were ranked, were built in rank order, and their removal is a move rather than an exclusion.

5.1 Two rows are not untouched candidates ✗

Checking a name against the served list is not sufficient. A dataset can carry a loader, a failed retrieval attempt, or an already-ingested deposit under a different paper's name.

Lee 2014, row 47, is blocked rather than available. An earlier ingestion attempt is recorded as awaiting author matrices, which means the Science supplementary tables and the Nislow and Giaever lab portal did not yield per-strain values. Its rank is still earned: 3,356 compounds against the full heterozygous and homozygous collections really is the largest chemical-genetic matrix in existence, and it really is on the YKO genotype axis the served set already uses. What changes is the nature of the work. Row 47 is an author request, not a loader, and the 1.3×10^7 instances it contributes to the totals in Sec. 7 should not be counted as reachable until that request succeeds.

Turco 2023, row 60, already has a loader. `torchcell/datasets/scerevisiae/yeastphenome.py` consumes the curated YeastPhenome release and retains 49 growth screens, so this row is in flight rather than net new. It is not in the served set, which is why it appears here at all. Its de-duplication problem is partly solved already, since the loader excludes the primaries served directly, along with heterozygous-diploid and expression screens. Lee 2014 is not among its 49 screens, so ingesting YeastPhenome does not backfill row 47.

5.2 Overlap is checked by script, and the check is not yet strong ✗

Rows that are already held have been caught by inspection before and by script once, so the check is `experiments/database/scripts/check_candidate_overlap.py`, run against the candidate JSON. It applies three keys of increasing strength: the candidate DOI cited in a served loader, an accession token shared with one, and the candidate PMID appearing in an aggregator's source-PMID column.

Against the 55 served loaders it reports one overlap, Turco 2023, which Sec. 5.1 already records as a loader in flight, and no overlap with SynLethDB across 3,658 source PMIDs spanning both the lethality and rescue layers. The honest limit is coverage. The script resolves a PMID for 124 of 154 rows and finds a parseable accession in 11. Accession is the strongest key, so it is running on about one row in fourteen. Recording each row's deposit identifier is the single highest-value edit available to this list.

5.3 Confidence, and the candidate gate ✗

Rows carry a confidence value in the generated JSON. *sourced* means the figures trace to something read in a pass or to the sourced triage note, and covers 65 of the 154 rows, 34 of the sixty printed. *recall* means the row was written from domain knowledge, and covers 89 of the 154, 26 of the sixty. For a recall row the dataset is real and the description is sound, but the citation details, the exact counts and the accession all need confirming before a loader is written.

The new first band sharpens that warning rather than softening it. Six of the eight state rows are recall rows, Albert 2018 among them, and none of the eight has a deposit identifier fetched: Albert 2018's GEO series, Jakobson 2025's accession, the PRIDE identifiers of Teyssonniere 2024 and Grossbach 2022, Smith 2008's GEO series and McManus 2014's are all marked unconfirmed in Table 5. Treat the top of the table as a claim to verify first, not a queue to start work from; the candidate gate is where that verification is recorded.

Table 11 in Sec. 8 gives each row's verdict from the gate's verdict store. Eight of the 154 rows have been gated: one admissible, one admissible with typed gaps, three blocked on provision and three blocked on a wall. None of the eight state rows has a verdict yet, so gating them is the first step of the next build week.

6 Before a loader is written ✗

Every figure in Table 4 is an axis count, not a verified record count, and several are weaker than that. Three qualifications apply, in decreasing order of how much damage they would do if ignored.

Instance counts marked † are products, not counts. A screen reported as 4,800 strains against 3 conditions is entered as 14,400 instances. Real screens drop strains per condition, so the released matrix is smaller, sometimes by a third. Thirteen of the sixty rows carry the mark.

Measurements assume a dense vector. A proteome row is counted as instances times the number of proteins quantified, but a label-free proteome quantifies a different subset of proteins in each sample. The served Messner 2023 ranges from 1,441 to 1,849 proteins per record, and a candidate proteome will be sparser than its headline depth in the same way.

Estimates marked ‡ are order-of-magnitude. Twenty-one of the sixty rows have an instance count that is an estimate rather than a reported figure or a product of reported axes, and among the eight state rows Foss 2007 has none at all.

7 Summary statistics ✗

Table 8: Summary of the final sixty, split at the fifty line. *Top 50* and *Extra* count rows; *Genotypes*, *Instances* and *Meas.* sum the row axes over all sixty and are lower bounds, since a row with no count contributes nothing. The sixty rows name 89 joins, 55 of them to a dataset already built.

	Top 50	Extra	Genotypes	Instances	Meas.
By band					
state across genomes	8	0	3,000	3.1×10^3	1.2×10^7
perturb-seq	23	0	327,436	4.0×10^5	4.1×10^6
molecular layers	14	0	5,500	5.7×10^3	1.1×10^6
scale	5	10	100,252,284	1.5×10^8	1.5×10^8
By tier					
tier 1	8	9	100,253,896	1.1×10^8	1.2×10^8
tier 2	18	1	287,527	3.8×10^7	4.2×10^7
tier 3	21	0	46,794	4.1×10^4	5.4×10^6
tier 4	3	0	3	17	5.6×10^4
By class					
CRISPR library screen	8	0	307,294	3.7×10^5	3.7×10^5
Deep mutational scan	0	2	88,000	8.8×10^4	8.8×10^4
Expression / single cell	15	0	522	715	4.7×10^6
Metabolite / precursor	5	2	16,101	3.8×10^4	1.4×10^5
Modality / backbone	4	1	14,447	1.7×10^4	5.8×10^4
Natural variation	14	1	120,150	1.7×10^6	1.4×10^7
Regulatory DNA	1	0	100,000,000	1.0×10^8	1.0×10^8
Tolerance / robustness	3	4	41,706	5.1×10^7	5.1×10^7
By sequence basis					
S288C+ORF-plasmid	0	1	5,280	1.1×10^4	1.1×10^4
S288C+designed-edit	2	2	164,000	1.6×10^5	1.6×10^5
S288C+guide	7	0	242,194	3.1×10^5	3.1×10^5
S288C+reporter-locus	1	0	100,000,000	1.0×10^8	1.0×10^8
S288C+tag	1	1	8,190	2.8×10^4	3.3×10^4
S288C-KO	8	5	48,189	5.1×10^7	5.4×10^7
engineered-chassis	1	0	200	200	200
isolate-WGS	5	0	16,645	8.9×10^5	2.9×10^6
reference-only	16	0	17	243	1.1×10^6
segregant-WGS	9	1	103,505	8.2×10^5	1.1×10^7
Perturb-seq axis					
high on both axes	1	0	1,012	1.0×10^3	5.8×10^6
input axis only	5	2	100,289,094	1.0×10^8	1.0×10^8
output axis only	9	0	2,779	5.4×10^3	2.4×10^6
neither axis	35	8	295,335	5.2×10^7	6.0×10^7
Other attributes					
carries a time axis	6	0	51	77	10.0×10^5
has a join to a built dataset	38	2	100,455,061	1.5×10^8	1.6×10^8
has a join to another candidate only	11	2	87,236	9.2×10^4	7.3×10^6
no named join	1	6	45,923	1.0×10^5	1.4×10^6
figures sourced this pass	26	8	355,000	5.2×10^7	5.4×10^7
figures from recall, to confirm	24	2	100,233,220	1.0×10^8	1.2×10^8
blocked on data access	1	0	11,000	1.3×10^7	1.3×10^7
loader in flight	0	1	5,000	3.8×10^7	3.8×10^7
instances a reported count	26	0	100,080,698	1.1×10^8	1.3×10^8
instances a product of the axes	6	7	151,303	3.9×10^7	4.1×10^7
instances an estimate	18	3	356,219	4.4×10^5	6.5×10^5
Total	50	10	100,588,220	1.5×10^8	1.7×10^8

The fifty are the three priority bands in full, 8 state rows, 23 perturb-seq rows and 14 molecular-layers rows, plus the first five scale rows. The ten extra are all scale, and the fifty's five scale rows include the two largest datasets in the

whole ranking, de Boer 2020 and Lee 2014, which is why the instance sums in Table 8 are carried mostly by five rows. The banded rows are vector-valued: a few hundred strains each carrying a transcriptome, a proteome or a per-cell distribution, so they hold few instances and many measurements, and ranking on instances alone would have ordered the two blocks the other way round. The state band is the extreme case, 3,000 genotypes and 1.2×10^7 measurements.

Two class facts follow from the bands. Natural variation and expression are the joint largest classes in the sixty at fifteen rows each, and 16 of the sixty are **reference-only**, meaning the environment rather than a strain edit carries the perturbation, which is what a condition-crossed expression layer looks like. Thirteen of the sixty sit on the S288C deletion collection, the axis most of the served set already uses, so those thirteen are the cheapest joins in the table.

7.1 Time ✗

Six of the sixty rows carry a time dimension (Table 4, *Time* column): Jariani 2020's lag-phase course, Jackson 2023's continuous course, Wang 2022's three ages, Airoldi 2016's upshift course, and the two rate rows, Sun 2013 and Martin-Perez 2017, whose rates are derived from a labeling series that is not itself released. Hackett 2020's induction series, the row the previous pass's time discussion was anchored at, is served as 1,752 induction time-course records, one environment per time point. The steady-state chemostat rows (Brauer 2008, Boer 2010, Hackett 2016, Yu 2021) vary dilution rate, which is a growth rate, not a time axis, and are not counted.

What the schema holds is `Environment.duration_hours` and `Environment.duration_generations`, treatment durations that make two exposure lengths two distinct environments, which is how Hackett 2020 was ingested. It is a flattening: a model sees time as a categorical condition, and nothing on the record says that eight environments are one series in order. A first-class time field on the record and a series identity that groups its points are the two schema additions a $c(t)$ model needs, and the six rows above are the ones that would exercise them. They are recorded here as an open schema item rather than folded into this list's ranking, since the ranking rule does not yet score a time axis.

8 The full pool ✗

Table 9 is the whole pool the sixty were cut from, every row the generating script holds, in rank order with its class, both axes, instance and measurement counts, phenotype, sequence basis and a clickable link. All 154 rows are inside the 200-row cut, so all are recommended; the 46-row shortfall is open. Table 10 gives every row's citation, link, the reason for its position and where the per-record values live, in the same order; for the first sixty rows that reason is also the *Why* column of Table 4. Table 11 gives the candidate-gate verdict of every ranked row from the verdict store. The ranking is produced by `experiments/database/scripts/build_candidate_datasets_table.py` and is also written to `experiments/database/results/candidates/candidate_datasets.json`.

Table 9: Ranked candidates to take the database from 70 supported datasets to 270. *Genotypes* and *Env* are the perturbation and condition axes. *Instances* is the number of genotype×environment records, marked † where it is the product of the two axes rather than a figure the paper reports, and ‡ where it is an order-of-magnitude estimate. *Meas.* is instances times phenotype dimensionality, which is what a vector-valued panel actually contributes and what rows are ranked on. *Sequence basis* is how the total genomic content of one strain would be reconstructed; a row with no such route is excluded rather than ranked (Table 7). Tier is defined in Sec. 2. *Link* resolves to the source and is clickable; the full citation, the reason for the position and the data location are in Table 10. A • marks a row bearing on the Perturb-seq proposal (the first band of Sec. 2). A superscript **B** marks a row already attempted and blocked on data access, and **L** one that already has a loader in flight; neither is an untouched candidate. Rows 1–154 are the recommended set; the pool holds 154 rows, 46 short of the 200 this pass adds, so the shortfall is open discovery, not a reserve.

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
1	**Albert 2018 (eQTL in 1,012 segregants)** •	Natural variation	1,012 sequenced segregants	1	1.0×10^3	5.8×10^6	mRNA abundance (RNA-seq)	segregant-WGS	doi:10.7554/eLife.35471
2	**Jakobson 2025 (genome-to-proteome map)**	Natural variation	800 F6 segregants	1	800	1.6×10^6	protein abundance, >6,400 pQTL	segregant-WGS	doi:10.1126/science.adu3198
3	**Teyssonniere 2024 (species-wide proteome against transcriptome)**	Natural variation	942 natural isolates	1	942	1.9×10^6	protein abundance, paired with transcript abundance	isolate-WGS	doi:10.1073/pnas.2319211121
4	**Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)**	Natural variation	112 genomically defined strains (BY x RM panel)	1	112	1.1×10^6	transcript, protein and phosphosite abundance	segregant-WGS	doi:10.15252/msb.202110712
5	**Smith 2008 (gene-environment interaction eQTL)** •	Natural variation	109 segregants	glucose / ethanol	218†	1.3×10^6	mRNA abundance	segregant-WGS	doi:10.1371/journal.pbio.0060083
6	**Skelly 2013 (expression variation across isolates)**	Natural variation	22 sequenced isolates	1	22	1.8×10^5	transcriptome + proteome	isolate-WGS	doi:10.1101/gr.155762.113
7	**McManus 2014 (ribosome profiling, allele-specific)**	Modality / backbone	S. cerevisiae, S. paradoxus and their F1 hybrid	1	3	3.3×10^4	ribosome occupancy paired with mRNA abundance	reference-only	doi:10.1101/gr.164996.113
8	**Foss 2007 (BY x RM segregant proteome)**	Natural variation	BY x RM segregants; the count is not stated in the abstract and was not confirmed	1	–‡	–	protein abundance (label-free MS)	segregant-WGS	doi:10.1038/ng.2007.22
9	**Dong 2021 (MAGIC + SAM biosensor)**	CRISPR library screen	100,000 guides (a/i/d)	1	1.0×10^5‡	1.0×10^5	SAM biosensor fluorescence (FACS)	S288C+guide	doi:10.1016/j.ymben.2021.03.005
10	**Momen-Roknabadi 2020 (inducible CRISPRi library)**	CRISPR library screen	genome-wide guide library	nutrient limitations	9.0×10^4‡	9.0×10^4	per-guide fitness under induction	S288C+guide	doi:10.1038/s42003-020-01452-9
11	**McGlincy 2021 (genome-scale CRISPRi library)** •	CRISPR library screen	61,094 guides, 10/gene	continuous culture	6.1×10^4	6.1×10^4	per-guide fitness	S288C+guide	doi:10.1186/s12864-021-07518-0
12	**Bao 2018 (CHAnGE single-nucleotide library)**	CRISPR library screen	tens of thousands of designed variants	inhibitor selection	6.0×10^4‡	6.0×10^4	variant fitness (pooled barcode seq)	S288C+designed-edit	doi:10.1038/nbt.4132

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
13	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)**	Tolerance / robustness	genome-scale tunable RNAi	isobutanol, 1-butanol	2.0×10^4‡	2.0×10^4	dose-graded knockdown fitness under alcohol stress	S288C+guide	PMID 27654654
14	**Roy 2018 (multiplexed precision editing)**	CRISPR library screen	thousands of designed edits	1	1.6×10^4‡	1.6×10^4	variant fitness	S288C+designed-edit	doi:10.1038/nbt.4137
15	**Newman 2006 (protein noise, GFP library)** •	Modality / backbone	2,500 GFP-tagged strains	rich / minimal	5.0×10^3†	1.0×10^4	protein abundance and cell-to-cell noise	S288C+tag	doi:10.1038/nature04785
16	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)**	Tolerance / robustness	1,100 essential genes	acetic acid vs control	2.2×10^3†	2.2×10^3	fitness under acid stress	S288C+guide	doi:10.1128/mSystems.00410-21
17	**Lian 2017 (CRISPR-AID, beta-carotene)**	CRISPR library screen	combinatorial a/i/d triples	1	200‡	200	beta-carotene titer	engineered-chassis	doi:10.1038/s41467-017-01695-x
18	**Hughes 2000 (compendium of expression profiles)**	Expression / single cell	300 mutations and chemical treatments	1	300	1.8×10^6	mRNA log ratio (two-color array)	S288C-KO	doi:10.1016/S0092-8674(00)00015-5
19	**Lenstra 2011 (chromatin regulator deletion expression)** •	Expression / single cell	165 chromatin-regulator deletions	1	165	9.9×10^5	mRNA abundance	S288C-KO	doi:10.1016/j.molcel.2011.03.026
20	**Sun 2013 (mRNA synthesis and decay rates across deletion strains)**	Expression / single cell	46 deletion strains of mRNA degradation and metabolism genes	1	46	8.3×10^5	mRNA level, synthesis rate and decay rate	S288C-KO	doi:10.1016/j.molcel.2013.09.010
21	**Jariani 2020 (yeast scRNA-seq through lag phase)** •	Expression / single cell	isogenic wild type	glucose to maltose shift, time course through lag phase	6‡	3.6×10^4	single-cell transcriptome	reference-only	doi:10.7554/eLife.55320
22	**Guo 2018 (CRISPR-Cas9 tiling of essential genes)** •	CRISPR library screen	3e4 tiling guides	1	3.0×10^4‡	3.0×10^4	per-guide fitness	S288C+guide	doi:10.1038/nbt.4147
23	**Urbonaite 2021 (yeastDrop-Seq under drug treatment)** •	Expression / single cell	isogenic wild type	mycophenolic acid, guanine, both, untreated	4	2.4×10^4	single-cell transcriptome	reference-only	doi:10.1038/s42003-021-02895-4
24	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** •	Expression / single cell	clonal wild type	baseline + stress	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1038/s41564-018-0346-9
25	**Gasch 2017 (single-cell stress heterogeneity)** •	Expression / single cell	wild type	stress vs unstressed	2‡	1.2×10^4	single-cell transcriptome	reference-only	doi:10.1371/journal.pbio.2004050
26	**Jaffe 2019 (multiplexed CRISPR interference epistasis)** •	CRISPR library screen	1e4 guide pairs	1	1.0×10^4‡	1.0×10^4	fitness / genetic interaction	S288C+guide	doi:10.1534/g3.119.400323

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
27	**Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics) •**	Expression / single cell	wild type	one time course, sampled continuously	1	6.0×10^3	single-cell transcriptome (173,361 cells)	reference-only	doi:10.1101/2023.09.21.558277
28	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq) •**	Expression / single cell	method	method	–‡	–	single-cell transcriptome	reference-only	doi:10.1002/yea.3927
29	**Mulleder 2012 (prototrophic deletion collection)**	Modality / backbone	5,185 prototrophic + 839 titratable essential	1	–‡	–	strain resource (no phenotype)	S288C-KO	doi:10.1038/nbt.2442
30	**Su 2023 (single-cell transcriptomes under four stresses)**	Expression / single cell	wild type	isosmotic, hypoosmotic, glucose starvation, amino acid starvation	4	2.4×10^4	single-cell transcriptome (117 cells, full length)	reference-only	doi:10.1186/s12864-023-09184-w
31	**Wang 2022 (single-cell transcriptomes across replicative aging)**	Expression / single cell	wild type	2 h, 16 h, 36 h of aging	3	6.6×10^3	single-cell transcriptome (125 cells)	reference-only	doi:10.1111/acel.13712
32	**Aulakh 2025 (genome-scale ionome)**	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^4	intracellular metal-ion content (ICP-MS)	S288C-KO	doi:10.1016/j.cels.2025.101319
33	**Leutert 2023 (phosphoproteome x 101 conditions)**	Expression / single cell	wild type	101 conditions	101	5.0×10^5	phosphosite abundance	reference-only	doi:10.1038/s41594-023-01115-3
34	**Brauer 2008 (growth-rate-controlled chemostat transcriptome)**	Expression / single cell	wild type	36 chemostats, 6 limitations	36	2.2×10^5	mRNA abundance	reference-only	doi:10.1091/mbc.e07-08-0779
35	**Hackett 2016 (SIMMER multi-omic flux)**	Expression / single cell	wild type	25 chemostat states	25	7.5×10^4	paired metabolome + proteome + flux	reference-only	doi:10.1126/science.aaf2786
36	**Zhu 2014 (kinase / phosphatase lipidomics)**	Metabolite / precursor	129 kinase + phosphatase KOs	1	129	2.6×10^4	lipid class and species abundance	S288C-KO	PMC4196872
37	**Gerke 2017 (urea-cycle mQTL)**	Natural variation	147 diploid segregants	1	147	1.5×10^4	untargeted metabolite abundance	segregant-WGS	doi:10.1534/genetics.117.201107
38	**Ambroset 2014 (metabolite QTL)**	Natural variation	100 segregants	1	100	7.4×10^3	74 metabolite concentrations (LC-MS/MS)	segregant-WGS	doi:10.1371/journal.pgen.1004142
39	**Blank 2005 (13C metabolic flux)**	Metabolite / precursor	200 viable null mutants	glucose minimal	200‡	6.0×10^3	intracellular flux distribution (13C-MFA)	S288C-KO	doi:10.1186/gb-2005-6-6-r49

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
40	**Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)**	Expression / single cell	wild type, prototrophic	nitrogen sources x steady-state growth rates, plus an upshift time course	20‡	1.2×10^5	bulk transcriptome	reference-only	doi:10.1091/mbc.E14-05-1013
41	**Martin-Perez 2017 (protein half-lives)**	Modality / backbone	wild type, exponential growth	1	1	3.2×10^3	protein turnover rate	reference-only	doi:10.1016/j.cels.2017.08.008
42	**Boer 2010 (metabolome across nutrient limitations)**	Metabolite / precursor	wild type	nutrient limitations x growth rates	25	2.5×10^3	intracellular metabolite concentrations	reference-only	doi:10.1091/mbc.e09-07-0597
43	**Tengolics 2024 (domestication metabolome)**	Natural variation	17 S. cerevisiae populations	1	17	1.6×10^3	19 amino acids + 78 other metabolites	isolate-WGS	doi:10.1073/pnas.2313354121
44	**Eder 2020 (flux QTL)**	Natural variation	segregant panel (n unconfirmed)	wine fermentation	100‡	100	modeled intracellular flux	segregant-WGS	PMID 32034164
45	**Yu 2021 (proteome and metabolome under nitrogen limitation)**	Metabolite / precursor	wild type	nitrogen-limited chemostats	10‡	2.5×10^4	proteome + metabolome	reference-only	doi:10.7554/eLife.65722
46	**de Boer 2020 (100M random promoters)** •	Regulatory DNA	>100M random promoters	complex + defined	1.0×10^8	1.0×10^8	expression driven (YFP sort-seq)	S288C+reporter-locus	doi:10.1038/s41587-019-0315-8
47	**Lee 2014 (HIP-HOP fitness signatures)** [B]	Tolerance / robustness	11,000 (het + hom)	3,356 compounds	1.3×10^7	1.3×10^7	chemogenomic fitness score	S288C-KO	doi:10.1126/science.1250217
48	**Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)** •	Natural variation	100,000 barcoded segregants	8 conditions	8.0×10^5†	8.0×10^5	fitness (Bar-seq)	segregant-WGS	doi:10.7554/eLife.73983
49	**Galardini 2019 (four backgrounds x 38 conditions)**	Natural variation	3,786 KOs x 4 backgrounds	38 conditions	5.8×10^5†	5.8×10^5	colony-size fitness (S-score)	isolate-WGS	doi:10.15252/msb.20198831
50	**Dutta 2026 (barcoded natural-isolate chemical response)**	Natural variation	520 natural isolates	>600 compounds	3.1×10^5†	3.1×10^5	pooled-barcode fitness	isolate-WGS	doi:10.1038/s41467-026-73532-z
End of wave 1. Rows 1–50 are the recommended next builds.									
51	**Li 2016 (tRNA fitness landscape)** •	Deep mutational scan	65,000 tRNA variants	1	6.5×10^4‡	6.5×10^4	fitness	S288C+designed-edit	doi:10.1126/science.aae0568
52	**Domingo 2018 (tRNA double-mutant landscape)** •	Deep mutational scan	23,000 single + double mutants	1	2.3×10^4‡	2.3×10^4	fitness	S288C+designed-edit	doi:10.1038/s41586-018-0170-7
53	**Zhang 2021 (YETI titratable overexpression)**	Metabolite / precursor	1,022 essential + 4,668 non-essential	beta-estradiol dose series	2.3×10^4‡	2.3×10^4	fitness across an induction gradient	S288C+tag	doi:10.15252/msb.202010321

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
54	**Peeters 2021 (fermentation-trait QTL atlas)**	Natural variation	1,125 sequenced segregants	18 traits	2.0×10^4†	2.0×10^4	ethanol / glycerol / isobutanol titer, stress resistance, aroma	segregant-WGS	PMID 34727964
55	**Van Leeuwen 2024 (short-chain organic acids)**	Tolerance / robustness	4,800 YKO	acetic, butyric, octanoic	1.4×10^4†	1.4×10^4	fitness per acid	S288C-KO	PMC10903034
56	**Chica 2026 (AutoDRY autophagy screen)**	Modality / backbone	4,760 KO + 1,159 DAmP	2 (nitrogen starvation, replete)	1.2×10^4†	1.2×10^4	autophagy flux (deep-learning image classifier)	S288C-KO	doi:10.1038/s41556-025-01837-0
57	**Sopko 2006 (genome-scale overexpression toxicity)**	Metabolite / precursor	5,280 ORFs	GAL induced vs glucose	1.1×10^4†	1.1×10^4	growth-rate reduction on induction	S288C+ORF-plasmid	doi:10.1016/j.molcel.2005.12.011
58	**Liu 2021 (tryptophan / isobutanol tolerance)**	Tolerance / robustness	5,006 YKO	+/- isobutanol	1.0×10^4†	1.0×10^4	colony-size fitness (ScreenMill imaging)	S288C-KO	doi:10.1186/s13068-021-02048-z
59	**Kuroda 2019 (isobutanol-specific tolerance)**	Tolerance / robustness	4,800 YKO	isobutanol vs ethanol	9.6×10^3†	9.6×10^3	isobutanol-specific fitness	S288C-KO	doi:10.1016/j.cels.2019.10.006
60	**Turco 2023 (Yeast Phenome)** [L]	Tolerance / robustness	5,000	7,536 environments	3.8×10^7†	3.8×10^7	growth / fitness per environment	S288C-KO	doi:10.1126/sciadv.adg5702
61	**Rossi 2021 (ChIP-exo protein architecture)**	Modality / backbone	400 tagged proteins	1	400	2.4×10^6	genome-wide binding location, near-bp resolution	S288C+tag	doi:10.1038/s41586-021-03314-8
62	**Bloom 2025 (global epistasis in a yeast cross)** •	Natural variation	large sequenced segregant panel	conditions	1.0×10^6‡	1.0×10^6	fitness	segregant-WGS	PMID 40679398
63	**Gasch 2000 (environmental stress response)**	Expression / single cell	wild type	9 stress series, multi-timepoint	150‡	9.0×10^5	mRNA abundance time course	reference-only	doi:10.1091/mbc.11.12.4241
64	**Decourty 2021 (RNA-metabolism genetic interactions)**	Modality / backbone	700,000 double mutants	1	7.0×10^5	7.0×10^5	genetic-interaction score	S288C-KO	doi:10.1093/nar/gkab680
65	**Cuperus 2017 (random 5' UTR library)**	Regulatory DNA	500,000 random 5' UTRs	1	5.0×10^5‡	5.0×10^5	protein output (translation)	S288C+reporter-locus	doi:10.1101/gr.224964.117
66	**Faure 2022 (doubledeepPCA)** •	Deep mutational scan	5e5 double mutants, 2 domains	1	5.0×10^5‡	5.0×10^5	binding and abundance	engineered-chassis	doi:10.1038/s41586-022-04586-4
67	**Chong 2015 / CYCLoPs (single-cell proteome dynamics)** •	Modality / backbone	4,100 GFP-tagged ORFs	cell-cycle / stress	2.5×10^4‡	4.2×10^5	abundance + localization class	S288C+tag	doi:10.1016/j.cell.2015.04.051
68	**Teyssonniere 2024 (species-wide trait survey)**	Natural variation	1,011 isolates	200 traits	2.0×10^5‡	2.0×10^5	growth traits, expressivity and complexity	isolate-WGS	doi:10.1371/journal.pgen.1011119

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
69	**Diss 2018 (Fos-Jun interaction double mutants) •**	Deep mutational scan	120,000 double mutants	1	1.2×10^5‡	1.2×10^5	protein-protein interaction strength	engineered-chassis	doi:10.7554/eLife.32472
70	**Warringer 2011 (trait variation across the SGRP panel)**	Natural variation	200 sequenced isolates	200 conditions	4.0×10^4‡	1.2×10^5	growth-curve parameters	isolate-WGS	doi:10.1371/journal.pgen.1002111
End of wave 2. Rows 51–70 are the bench, promoted as soon as a wave-1 row proves unreachable.									
71	**Keren 2013 (promoter activity across conditions)**	Regulatory DNA	900 native promoter reporters	10 conditions	9.0×10^3†	9.0×10^4	promoter activity, time-resolved	S288C+reporter-locus	doi:10.1038/msb.2013.59
72	**Puchta 2016 (snoRNA U3 landscape) •**	Deep mutational scan	60,000 U3 variants	1	6.0×10^4‡	6.0×10^4	fitness	S288C+designed-edit	doi:10.1126/science.aaf0965
73	**Yoshikawa 2011 (deletion + overexpression phenome)**	Tolerance / robustness	4,800 KO + matched overexpression	carbon sources / stress	5.8×10^4‡	5.8×10^4	growth-rate fitness	S288C-KO	doi:10.1534/g3.111.000695
74	**Ho 2009 (MoBY-ORF barcoded overexpression)**	Metabolite / precursor	5,100 barcoded ORFs	compound panel	5.1×10^4‡	5.1×10^4	fitness per compound	S288C+ORF-plasmid	doi:10.1038/nbt.1534
75	**Bloom 2013 (1,008 BYxRM segregants)**	Natural variation	1,008 segregants, 11,623 markers	46 traits	4.6×10^4†	4.6×10^4	quantitative growth traits	segregant-WGS	doi:10.1038/nature11867
76	**Xiao 2014 (genome-wide RNAi furfural tolerance)**	Tolerance / robustness	genome-scale RNAi library	furfural vs control	2.0×10^4‡	2.0×10^4	knockdown enrichment under furfural	S288C+guide	doi:10.1186/1754-6834-7-78
77	**Douglas 2012 (barcoded overexpression fitness)**	Metabolite / precursor	genome-scale barcoded ORFs	baseline + stress	2.0×10^4‡	2.0×10^4	pooled competitive fitness on induction	S288C+ORF-plasmid	doi:10.1534/g3.112.003400
78	**She 2018 (natural variant effects across a large isolate panel)**	Natural variation	500 isolates	40 conditions	2.0×10^4‡	2.0×10^4	growth traits	isolate-WGS	doi:10.1016/j.cell.2017.12.015
79	**Costello 2020 (bioreactor Bar-seq)**	Tolerance / robustness	pooled YKO library	feed schemes / pH	1.9×10^4‡	1.9×10^4	fitness under fed-batch cultivation	S288C-KO	doi:10.1186/s12934-020-01423-z
80	**Gurvitz (fatty-acid utilization screen)**	Metabolite / precursor	4,800 viable YKO	oleate, myristate, acetate	1.4×10^4†	1.4×10^4	fatty-acid utilization competence	S288C-KO	embopress.org/doi/pdf/10.1038/msb4100051
81	**Mira 2010 (acetic acid tolerance, full collection)**	Tolerance / robustness	EUROSCARF collection	70-110 mM acetic acid, pH 4.5	1.4×10^4†	1.4×10^4	acetic-acid tolerance	S288C-KO	doi:10.1186/1475-2859-9-79
82	**Gameiro 2025 (AID2 degron libraries)**	Modality / backbone	>5,600 tagged ORFs	baseline + hydroxyurea	1.1×10^4†	1.1×10^4	degradation efficiency; fitness under stress	S288C+tag	doi:10.1083/jcb.202409007
83	**Si 2017 (RAGE RNAi, isobutanol titer)**	CRISPR library screen	genome-scale RNAi, iterative rounds	isobutanol selection	1.0×10^4‡	1.0×10^4	construct enrichment; isobutanol titer for top strains	S288C+guide	doi:10.1038/ncomms15187

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
84	**HamediRad 2018 (RAGE xylose utilization)**	CRISPR library screen	genome-scale RNAi	xylose-limited	1.0×10^4‡	1.0×10^4	enrichment; xylose consumption rate	S288C+guide	doi:10.1002/bit.26570
85	**Pereira 2014 (wheat-straw hydrolysate)**	Tolerance / robustness	EUROSCARF deletion collection	hydrolysate vs control	9.6×10^3†	9.6×10^3	tolerance to industrial hydrolysate	S288C-KO	doi:10.1007/s10295-014-1519-z
86	**Endo 2008 (vanillin tolerance)**	Tolerance / robustness	diploid deletion collection	vanillin vs control	9.6×10^3†	9.6×10^3	vanillin sensitivity	S288C-KO	doi:10.1128/AEM.01541-08
87	**Sharifpoor 2012 (kinome synthetic dosage lethality)**	Modality / backbone	92 kinase queries, genome-wide array	1	8.7×10^3	8.7×10^3	genetic interaction / SDL score	S288C-KO	doi:10.1101/gr.129213.111
88	**Mavor 2016 (ubiquitin DMS across conditions)**	Deep mutational scan	1,400 ubiquitin variants	chemical stresses	7.0×10^3†	7.0×10^3	fitness	S288C+designed-edit	doi:10.7554/eLife.15802
89	**Flynn 2020 (Hsp90 DMS across environments)**	Deep mutational scan	1,000 Hsp90 variants	stress conditions	6.0×10^3†	6.0×10^3	fitness	S288C+designed-edit	doi:10.7554/eLife.53810
90	**Ho 2018 (unified absolute protein abundance)**	Modality / backbone	wild type	1	5.9×10^3	5.9×10^3	absolute abundance, molecules/cell	reference-only	doi:10.1016/j.cels.2017.12.004
91	**Renganaath 2020 (natural promoter-variant MPRA)** •	Regulatory DNA	5,832 natural variants, 2,503 promoters	1	5.8×10^3	5.8×10^3	allele-specific promoter activity	S288C+reporter-locus	doi:10.7554/eLife.62669
92	**Chang 2012 (organic-acid production screen)**	Metabolite / precursor	4,800 YKO	1	4.8×10^3	4.8×10^3	acetate / pyruvate / succinate halo (pH indicator)	S288C-KO	PMID 22277779
93	**Trikka 2015 (carotenogenic heterozygous screen)**	Metabolite / precursor	4,700 heterozygous deletions	1	4.7×10^3	4.7×10^3	carotenoid color; sclareol titer for top strains	S288C-KO/het	doi:10.1186/s12934-015-0246-0
94	**Sardi 2018 (natural variation in hydrolysate tolerance)**	Tolerance / robustness	sequenced isolate / segregant panel	hydrolysate toxins	2.0×10^3‡	2.0×10^3	tolerance	isolate-WGS	doi:10.1371/journal.pgen.1007217
95	**Breslow 2008 (DAmP hypomorphic collection)**	Modality / backbone	842 haploid + 970 diploid essential	1	1.8×10^3	1.8×10^3	competitive fitness	S288C+tag	doi:10.1038/nmeth.1234
96	**Kofoed 2015 (barcoded ts alleles)**	Modality / backbone	600+ barcoded ts alleles	permissive / restrictive	1.2×10^3†	1.2×10^3	ts severity (growth ratio)	S288C+designed-edit	doi:10.1534/g3.115.019174
97	**Zhang 2022 (GCE-SCRaMbLE recombination outcomes)** •	Combinatorial genome	200 sequenced SCRaMbLE genomes	1	200‡	200	rearrangement structure + fitness	isolate-WGS	doi:10.1038/s41467-022-33606-0
98	**Cubillos 2017 (nitrogen-consumption QTL)**	Natural variation	165 sequenced F12 segregants	nitrogen-limited fermentation	165	165	nitrogen consumption	segregant-WGS	doi:10.1534/g3.117.042127
99	**Hillenmeyer 2010 (systematic fitness reanalysis)**	Tolerance / robustness	het + hom collections	400 conditions	2.4×10^6‡	2.4×10^6	fitness-defect score	S288C-KO	doi:10.1186/gb-2010-11-3-r30

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
100	**Braberg 2020 (point-mutant E-MAP)**	Modality / backbone	350 point mutants x genome-wide array	1	5.0×10^5	5.0×10^5	residue-level genetic interaction	S288C+designed-edit	doi:10.1126/science.aaz4910
101	**Breker 2013 (GFP library under stress)** •	Modality / backbone	4,159 GFP-tagged strains	stress conditions	1.2×10^4†	2.1×10^5	abundance + localization, per cell	S288C+tag	doi:10.1083/jcb.201301120
102	**Tkach 2012 (protein relocalization under DNA damage)** •	Modality / backbone	GFP collection	DNA-damaging agents	1.2×10^4†	2.0×10^5	localization and abundance change	S288C+tag	doi:10.1038/ncb2549
103	**Ehrenreich 2010 (X-QTL bulk segregant mapping)**	Natural variation	1e6 pooled segregants	17 selective conditions	17	1.9×10^5	allele-frequency shift (pooled)	segregant-WGS	doi:10.1038/nature08923
104	**Denervaud 2013 (microfluidic GFP dynamics)** •	Modality / backbone	1,000 GFP strains	dynamic environments	4.0×10^3†	6.8×10^4	time-resolved single-cell abundance	S288C+tag	doi:10.1073/pnas.1308265110
105	**Brogaard 2012 (chemical nucleosome map)**	Modality / backbone	wild type	1	1	6.7×10^4	nucleosome center positions, bp resolution	reference-only	doi:10.1038/nature11142
106	**Xia 2022 (proteome allocation across conditions)**	Modality / backbone	wild type	chemostat conditions	30‡	6.0×10^4	absolute proteome, resource allocation	reference-only	doi:10.1038/s41467-022-30513-2
107	**Melamed 2013 (PAB1 RRM domain scan)**	Deep mutational scan	40,000 PAB1 variants	1	4.0×10^4‡	4.0×10^4	fitness	S288C+designed-edit	doi:10.1261/rna.040709.113
108	**Yvert 2013 (single-cell trait variation across isolates)** •	Natural variation	37 isolates	1	37	1.9×10^4	single-cell morphology (CalMorph)	isolate-WGS	doi:10.1186/1752-0509-7-54
109	**Lahtvee 2017 (absolute proteome across conditions)**	Modality / backbone	wild type	9 growth conditions	9	1.8×10^4	absolute protein abundance + turnover	reference-only	doi:10.1016/j.cels.2017.03.003
110	**Weill 2018 (SWAT libraries)**	Modality / backbone	5,500 acceptor-tagged strains	1	5.5×10^3	1.6×10^4	abundance, localization, topology	S288C+tag	doi:10.1038/s41592-018-0044-9
111	**Kinsler 2020 (barcoded pleiotropy across environments)**	Natural variation	300 sequenced adaptive clones	45 environments	1.4×10^4†	1.4×10^4	fitness	isolate-WGS	doi:10.7554/eLife.61271
112	**Pelechano 2013 (transcript isoform landscape)**	Modality / backbone	wild type	glucose / galactose	2	1.2×10^4	transcript isoform boundaries and abundance	reference-only	doi:10.1038/nature12121
113	**Deutschbauer 2005 (haploinsufficiency, genome-wide)**	Tolerance / robustness	heterozygous diploid collection	rich / minimal	1.2×10^4†	1.2×10^4	haploinsufficiency fitness	S288C-KO/het	doi:10.1534/genetics.104.036871
114	**Valenti 2025 (SWAT degron + GFP collection)**	Modality / backbone	proteome-wide SWAT library	depleted / not	1.0×10^4‡	1.0×10^4	depletion efficiency + abundance/localization	S288C+tag	doi:10.1083/jcb.202409050

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
115	**Sousa 2013 (acetate resistance overexpression screen)**	Tolerance / robustness	genome-wide overexpression library	sodium acetate vs control	1.0×10^4†	1.0×10^4	acetate resistance	S288C+ORF-plasmid	PMID 23262128
116	**Sadhu 2016 (CRISPR-directed causal variant mapping)**	Natural variation	1,000 targeted recombinants	conditions	1.0×10^4‡	1.0×10^4	fitness	segregant-WGS	doi:10.1126/science.aaf5124
117	**Weinberg 2016 (ribosome profiling, improved)**	Modality / backbone	wild type	1	1	6.0×10^3	ribosome occupancy, translation efficiency	reference-only	doi:10.1016/j.celrep.2016.01.043
118	**Ghaemmaghami 2003 (absolute protein abundance)**	Modality / backbone	4,251 TAP-tagged ORFs	1	4.3×10^3	4.3×10^3	absolute abundance (molecules/cell)	S288C+tag	doi:10.1038/nature02046
119	**Huh 2003 (GFP localization atlas)**	Modality / backbone	4,156 GFP-tagged ORFs	1	4.2×10^3	4.2×10^3	subcellular localization (22 classes)	S288C+tag	doi:10.1038/nature02026
120	**Michaelis 2023 (protein interactome)**	Modality / backbone	proteome-scale AP-MS baits	1	4.0×10^3‡	4.0×10^3	protein-protein interaction + interface	S288C+tag	doi:10.1038/s41586-023-06739-5
121	**Christiano 2014 (protein turnover)**	Modality / backbone	wild type	1	3.9×10^3‡	3.9×10^3	protein half-life	reference-only	doi:10.1016/j.celrep.2014.10.065
122	**Kitzman 2015 (GAL4 DMS)**	Deep mutational scan	1,200 GAL4 variants	selective / permissive	2.4×10^3†	2.4×10^3	transcription-factor activity	S288C+designed-edit	doi:10.1038/nmeth.3223
123	**Sharon 2012 (designed promoter library)**	Regulatory DNA	2,000 designed promoters	1	2.0×10^3‡	2.0×10^3	expression (fluorescence)	S288C+reporter-locus	doi:10.1038/nbt.2205
124	**Sadhu 2018 (CRISPR-directed mitotic recombination)**	CRISPR library screen	1,500 recombinant genotypes	1	1.5×10^3‡	1.5×10^3	fitness / trait	segregant-WGS	doi:10.7554/eLife.41090
125	**Duveau 2021 (TDH3 promoter allele series)**	Regulatory DNA	250 promoter alleles	carbon sources	1.0×10^3‡	1.0×10^3	expression level, noise, fitness	S288C+designed-edit	doi:10.7554/eLife.67806
126	**Strope 2015 (100 clinical and natural genomes)**	Natural variation	100 sequenced isolates	phenotype panel	1.0×10^3‡	1.0×10^3	growth and clinical-relevant traits	isolate-WGS	doi:10.1101/gr.185538.114
127	**Schulte 2023 (mitochondrial complexome)**	Modality / backbone	900 mitochondrial proteins	1	900	900	complex assembly profile	reference-only	doi:10.1038/s41586-022-05641-w
128	**Metzger 2015 (promoter mutation effects on expression and noise)**	Regulatory DNA	235 promoter mutants	1	235‡	470	expression mean and noise	S288C+designed-edit	doi:10.1038/nature14424
129	**Venkataram 2016 (barcoded adaptive lineages, sequenced)**	Natural variation	400 sequenced adaptive clones	glucose-limited	400‡	400	fitness	isolate-WGS	doi:10.1016/j.cell.2016.08.002
130	**Jakociunas 2021 (degron-tuned terpene flux)**	Metabolite / precursor	30 degron strains	induction levels	90‡	270	farnesol / geraniol / nerolidol titer	engineered-chassis	doi:10.1002/bit.27735

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
131	**Shen 2016 (SCRaMbLE genome minimization)**	Combinatorial genome	100 SCRaMbLEd genomes	selective / permissive	200‡	200	viability and fitness	isolate-WGS	doi:10.1101/gr.193433.115
132	**Blount 2018 (SCRaMbLE for phenotype improvement)**	Combinatorial genome	100 rearranged genomes	selection conditions	200‡	200	growth and product phenotype	isolate-WGS	doi:10.1038/s41467-018-03143-w
133	**Liti 2009 (population genomics of S. cerevisiae and S. paradoxus)**	Natural variation	70 sequenced isolates	1	70‡	70	genome sequence and population structure	isolate-WGS	doi:10.1038/nature07743
134	**Yue 2017 (wild vs domesticated assemblies)**	Natural variation	12 long-read assemblies	n/a	12	12	genome structure, TE and subtelomere dynamics	isolate-WGS	doi:10.1038/ng.3847
135	**Zackrisson 2016 (Scan-o-matic growth curves)**	Modality / backbone	YKO-compatible platform	panel-dependent	–‡	–	full growth curve (rate, lag, yield)	S288C-KO	doi:10.1534/g3.116.032342
136	**Spellman 1998 (cell-cycle transcriptome)**	Expression / single cell	wild type, synchronized	4 synchronization methods	77	4.6×10^5	mRNA abundance time course	reference-only	doi:10.1091/mbc.9.12.3273
137	**Levy 2008 (morphology and phenotypic capacitance)** •	Modality / backbone	1,700 deletion strains	1	1.7×10^3‡	3.4×10^5	single-cell morphology variability	S288C-KO	doi:10.1371/journal.pbio.0060264
138	**Causton 2001 (environmental change transcriptome)**	Expression / single cell	wild type	environmental changes	40‡	2.4×10^5	mRNA abundance	reference-only	doi:10.1091/mbc.12.2.323
139	**Gasch 2001 (DNA damage response transcriptome)**	Expression / single cell	wild type + checkpoint mutants	damaging agents	24‡	1.4×10^5	mRNA abundance	S288C-KO	doi:10.1091/mbc.12.10.2987
140	**Handfield 2013 (image-based localization change)**	Modality / backbone	GFP collection	baseline / perturbed	8.0×10^3‡	1.4×10^5	localization change score	S288C+tag	doi:10.1371/journal.pcbi.1003085
141	**Kaplan 2009 (nucleosome sequence preferences)**	Modality / backbone	wild type (in vitro + in vivo)	in vitro / in vivo	2	1.3×10^5	nucleosome occupancy	reference-only	doi:10.1038/nature07667
142	**Giaever 2004 (chemogenomic drug-target profiling)**	Tolerance / robustness	het + hom collections	10 compounds	5.9×10^4‡	5.9×10^4	fitness	S288C-KO	doi:10.1073/pnas.0307490100
143	**Paulo 2016 (TMT proteome across carbon sources)**	Modality / backbone	wild type	carbon sources	9	2.2×10^4	protein abundance	reference-only	doi:10.1091/mbc.E16-03-0184
144	**Gibney 2013 (osmotic and glucose stress fitness)**	Tolerance / robustness	YKO collection	osmotic / glucose	1.9×10^4†	1.9×10^4	fitness	S288C-KO	doi:10.1073/pnas.1318100110
145	**Meurer 2018 (genome-wide GFP library, rebuilt)**	Modality / backbone	4,000 tagged strains	1	4.0×10^3‡	1.2×10^4	abundance and localization	S288C+tag	doi:10.1038/s41592-018-0044-9
146	**Bennett 2001 (ionizing-radiation resistance)**	Tolerance / robustness	4,800 YKO	irradiated vs control	9.6×10^3†	9.6×10^3	radiation sensitivity	S288C-KO	doi:10.1038/ng778

Table 9, continued

#	Dataset	Class	Genotypes	Env	Instances	Meas.	Phenotype	Sequence basis	Link
147	**Sugiyama (lactic-acid tolerance)**	Tolerance / robustness	4,800 YKO	4 percent lactic acid vs control	9.6×10^3†	9.6×10^3	acid tolerance	S288C-KO	nisr.or.jp/wp-content/uploads/NISR06sugiyama.pdf
148	**Sousa 2013 (acetic-acid programmed cell death screen)**	Tolerance / robustness	YKO haploid collection	acetic acid vs control	9.6×10^3†	9.6×10^3	death / survival (regulated cell death)	S288C-KO	PMID 24286259
149	**Nagalakshmi 2008 (RNA-seq transcriptome annotation)**	Modality / backbone	wild type	1	1	6.0×10^3	transcript structure and abundance	reference-only	doi:10.1126/science.1158441
150	**Yofe 2016 (SWAT N-terminal library)**	Modality / backbone	1,800 N-terminally tagged strains	1	1.8×10^3‡	5.4×10^3	localization and abundance	S288C+tag	doi:10.1038/nmeth.3795
151	**Hénault 2023 (hybrid and introgression panel)**	Natural variation	hybrids and introgression lines	conditions	5.0×10^3‡	5.0×10^3	fitness	isolate-WGS	doi:10.1038/s41467-024-46155-5
152	**Cubillos 2013 (F12 recombinant panel construction)**	Natural variation	192 sequenced F12 segregants	conditions	3.8×10^3‡	3.8×10^3	growth traits	segregant-WGS	doi:10.1534/genetics.113.155515
153	**Kessi-Perez 2019 (nitrogen-consumption QTL panel)**	Natural variation	sequenced segregants / isolates	nitrogen conditions	200‡	200	nitrogen consumption, fermentation rate	segregant-WGS	doi:10.1186/s13068-019-1440-9
154	**Lian 2016 (combinatorial xylose pathway optimization)**	Metabolite / precursor	combinatorial pathway variants	xylose	100‡	100	ethanol titer and rate on xylose	engineered-chassis	doi:10.1016/j.ymben.2015.11.001

Table 10: Sources for Table 9, in the same order. *Why it is ordered here* is the band reason for a perturb-seq or molecular-layers row, and the tier and measurement rule for a scale row; for the first 60 rows what the row buys is the *Why* column of Table 4. *Data* is where the per-record values live; entries marked unconfirmed were not fetched live and must be checked before a loader is written. Every link is clickable.

#	Dataset, citation and link	Why it is ordered here	Data
1	**Albert 2018 (eQTL in 1,012 segregants)** Albert FW, Bloom JS, Siegel J, Day L, Kruglyak L. eLife 2018;7:e35471. doi:10.7554/eLife.35471	State across genomes, first: bulk RNA-seq of 1,012 sequenced BY x RM segregants, the largest genotype-indexed transcriptome in yeast, and the bulk partner of the served Boocock 2025, whose pooled panel stores 393 of these same segregants.	GEO /eLife data availability (unconfirmed)
2	**Jakobson 2025 (genome-to-proteome map)** Jakobson CM, et al. Science 2025;390:eadu3198. doi:10.1126/science.adu3198	State across genomes: protein abundance across the same sequenced segregant panel Albert 2018 profiles by transcriptome, 800 F6 segregants, so protein and transcript loci are measurable on one genotype set.	Science data-availability statement (accession unconfirmed)
3	**Teyssonniere 2024 (species-wide proteome against transcriptome)** Teyssonniere EM, Trebulle P, Muenzner J, Loegler V, Ludwig D, Amari F, Mulleder M, Friedrich A, Hou J, Ralser M, Schacherer J. Proc Natl Acad Sci USA 2024;121:e2319211121. doi:10.1073/pnas.2319211121	State across genomes: proteomes for 942 of the 1,011 isolates whose transcriptomes the served Caudal 2024 holds and whose proteomes the served Muenzner 2024 holds for 796, so the de-duplication against Muenzner is the first task and the pairing with Caudal is one to one.	PNAS SI; PMC11087752. PRIDE identifier not fetched this pass
4	**Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)** Grossbach J, Gillet L, Clement-Ziza M, Schmalohr CL, Schubert OT, Schutter M, Mawer JSP, Barnes CA, Bludau I, Weith M, Tessarz P, Graef M, Aebersold R, Beyer A. Mol Syst Biol 2022;18:e10712. doi:10.15252/msb.202110712	State across genomes: transcript, protein and phosphosite on one set of 112 sequenced BY x RM strains, the only row where all three layers are observed within a genotype.	Mol Syst Biol SI + PRIDE, identifier not fetched this pass
5	**Smith 2008 (gene-environment interaction eQTL)** Smith EN, Kruglyak L. PLoS Biol 2008;6:e83. doi:10.1371/journal.pbio.0060083	State across genomes: mRNA abundance of 109 BY x RM segregants in glucose and in ethanol, the one segregant transcriptome with a condition axis, so genotype by environment is a within-panel contrast.	GEO (unconfirmed)
6	**Skelly 2013 (expression variation across isolates)** Skelly DA, Merrihew GE, Riffle M, et al., MacCoss MJ, Akey JM. Genome Res 2013;23:1496-1504. doi:10.1101/gr.155762.113	State across genomes: transcriptome and proteome on the same 22 sequenced isolates, the cheap and the expensive layer paired within one experiment on natural genomes.	Genome Research SI (unconfirmed)
7	**McManus 2014 (ribosome profiling, allele-specific)** McManus CJ, May GE, Spealman P, Shteyman A. Genome Res 2014;24:422-430. doi:10.1101/gr.164996.113	State across genomes: ribosome occupancy beside mRNA abundance for S. cerevisiae, S. paradoxus and their F1 hybrid, so translation is read allele by allele across two genomes; the paradoxus arm is out of species and is stored apart or dropped.	GEO, series not fetched this pass
8	**Foss 2007 (BY x RM segregant proteome)** Foss EJ, Radulovic D, Shaffer SA, Ruderfer DM, Bedalov A, Goodlett DR, Kruglyak L. Nat Genet 2007;39:1369-1375. doi:10.1038/ng.2007.22	State across genomes: the first segregant proteome on the BY x RM cross and the independent replicate of Grossbach 2022 fifteen years and two platforms earlier.	Nat Genet supplementary tables; not fetched
9	**Dong 2021 (MAGIC + SAM biosensor)** Dong C, Schultz JC, Liu W, Lian J, Huang L, Xu Z, Zhao H. Metab Eng 2021;66:319-327. doi:10.1016/j.ymben.2021.03.005	A genome-wide CRISPR activation, interference and deletion library sorted on a metabolite biosensor, so the label is product concentration rather than fitness. The pattern a sorted Perturb-seq would reuse.	Metab Eng SI (accession unconfirmed)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
10	**Momen-Roknabadi 2020 (inducible CRISPRi library)** Momen-Roknabadi A, Oikonomou P, Zegans M, Tavazoie S. Commun Biol 2020;3:723. doi:10.1038/s42003-020-01452-9	A genome-scale inducible CRISPR interference library with a released per-guide matrix. Induction control is what lets a knockdown be applied after a cell is captured rather than during outgrowth.	Addgene + Commun Biol SI
11	**McGlincy 2021 (genome-scale CRISPRi library)** McGlincy NJ, Meacham ZA, Reynaud KK, Muller R, Baum R, Ingolia NT. BMC Genomics 2021;22:205. doi:10.1186/s12864-021-07518-0	The second independently designed genome-scale yeast CRISPR interference library with a released per-guide matrix, so guide design and gene effect can be separated.	ingolia-lab.org/yeast-crispri + Addgene + BMC SI
12	**Bao 2018 (CHAnGE single-nucleotide library)** Bao Z, HamediRad M, Xue P, Xiao H, Tasan I, Chao R, Liang J, Zhao H. Nat Biotechnol 2018;36:505-508. doi:10.1038/nbt.4132	A guide-indexed genome-wide library whose edits sit below the open reading frame, so the perturbation is an allele rather than a null.	Nat Biotechnol SI (raw sequencing location unconfirmed)
13	**Crook 2016 (tunable RNAi, isobutanol + 1-butanol)** Crook N, Sun J, Morse N, Schmitz A, Alper HS. Appl Microbiol Biotechnol 2016;100:10005-10018. PMID 27654654. PMID 27654654	A dose-graded knockdown axis. Graded rather than binary perturbation is what turns a per-cell readout into a dose-response curve.	AMB SI (DOI is an open verification item)
14	**Roy 2018 (multiplexed precision editing)** Roy KR, Smith JD, Vonesch SC, et al., St Onge RP. Nat Biotechnol 2018;36:512-520. doi:10.1038/nbt.4137	Multiplexed designed edits per cell, which is combinatorial input in the sense the Perturb-seq specification requires.	Nat Biotechnol SI (accession unconfirmed)
15	**Newman 2006 (protein noise, GFP library)** Newman JRS, Ghaemmaghami S, Ihmels J, Breslow DK, Noble M, DeRisi JL, Weissman JS. Nature 2006;441:840-846. doi:10.1038/nature04785	Per-cell protein abundance distributions across a genome-scale tagged collection. Expression noise per gene is what sets how large a perturbation effect has to be before a per-cell readout can resolve it.	Nature SI
16	**Mukherjee 2021 (CRISPRi essential genes x acetic acid)** Mukherjee V, Lind U, St Onge RP, Blomberg A, Nystrom T. mSystems 2021;6:e00410-21. doi:10.1128/mSystems.00410-21	Knockdown reaches the essential genes a deletion collection cannot hold, which is a third of the genome a deletion-based campaign would miss.	mSystems open-access SI
17	**Lian 2017 (CRISPR-AID, beta-carotene)** Lian J, HamediRad M, Hu S, Zhao H. Nat Commun 2017;8:1688. doi:10.1038/s41467-017-01695-x	Three perturbation modalities in one cell, activation, interference and deletion, which is combinatorial input by modality rather than by gene.	Nat Commun SI
18	**Hughes 2000 (compendium of expression profiles)** Hughes TR, Marton MJ, Jones AR, Roberts CJ, Stoughton R, et al., Friend SH. Cell 2000;102:109-126. doi:10.1016/S0092-8674(00)00015-5	The original designed-perturbation expression compendium, and the independent measurement that makes cross-laboratory replication of the deletion transcriptome testable against the built Kemmeren 2014.	Rosetta compendium, distributed with the paper; not fetched
19	**Lenstra 2011 (chromatin regulator deletion expression)** Lenstra TL, Benschop JJ, Kim T, et al., Holstege FCP. Mol Cell 2011;42:536-549. doi:10.1016/j.molcel.2011.03.026	The same design over chromatin regulators rather than sequence-specific factors, so the two together cover both halves of transcriptional control by deletion.	GEO (unconfirmed)
20	**Sun 2013 (mRNA synthesis and decay rates across deletion strains)** Sun M, Schwalb B, Pirkl N, Maier KC, Schenk A, Failmezger H, Tresch A, Cramer P. Mol Cell 2013;52:52-62. doi:10.1016/j.molcel.2013.09.010	A designed single-gene perturbation crossed with a transcriptome, and the only one that resolves the level into a synthesis rate and a decay rate. Rates are what a per-cell snapshot cannot recover on its own.	GEO, series not fetched this pass

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
21	**Jariani 2020 (yeast scRNA-seq through lag phase)** Jariani A, Vermeersch L, Cerulus B, Perez-Samper G, Voordeckers K, Van Brussel T, Thienpont B, Lambrechts D, Verstrepen KJ. eLife 2020;9:e55320. doi:10.7554/eLife.55320	Per-cell states through a carbon-source shift, so the readout is the distribution across an unsynchronized population rather than its mean.	GEO GSE144820 and GSE116246; from the collection sweep, not fetched
22	**Guo 2018 (CRISPR-Cas9 tiling of essential genes)** Guo X, Chavez A, Tung A, et al., Church GM, Shalem O. Nat Biotechnol 2018;36:540-546. doi:10.1038/nbt.4147	Guide tiling produces a graded allelic series across one gene, so guide position rather than gene identity carries the effect.	Nat Biotechnol SI (unconfirmed)
23	**Urbonaite 2021 (yeastDrop-Seq under drug treatment)** Urbonaite G, Lee JTH, Liu P, Parras GG, Hemberg M, Acar M. Commun Biol 2021;4:1372. doi:10.1038/s42003-021-02895-4	Single-drug arms against their combination with a per-cell readout, the environmental analog of a combinatorial genetic perturbation.	GEO GSE165686; Zenodo 10.5281/zenodo.4767298 and 10.5281/zenodo.4762526; from the collection sweep, not fetched
24	**Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)** Nadal-Ribelles M, Islam S, Wei W, et al., Posas F, Steinmetz LM. Nat Microbiol 2019;4:683-692. doi:10.1038/s41564-018-0346-9	The high-sensitivity yeast protocol, strand and isoform aware. Sensitivity per cell is the constraint on how small a perturbation effect a campaign can resolve.	Nat Microbiol SI /GEO
25	**Gasch 2017 (single-cell stress heterogeneity)** Gasch AP, Yu FB, Hose J, et al., Quake SR. PLoS Biol 2017;15:e2004050. doi:10.1371/journal.pbio.2004050	Separates intrinsic from extrinsic per-cell variation in an isogenic population under stress, which is the null a perturbation effect is measured against.	PLoS Biology SI /GEO
26	**Jaffe 2019 (multiplexed CRISPR interference epistasis)** Jaffe M, Sherlock G, Levy SF. G3 2019;9:2843-2852. doi:10.1534/g3.119.400323	Two knockdowns in the same cell with a measured interaction, which is the only combinatorial CRISPR interference design located in yeast.	G3 open-access SI (unconfirmed)
27	**Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)** Jackson CA, Beheler-Amass M, Tjarnberg A, Suresh I, Hickey AS, Bonneau R, Gresham D. bioRxiv 2023. 10.1101/2023.09.21.558277. doi:10.1101/2023.09.21.558277	The platform companion of Jackson 2020: 173,361 cells of one strain, which is what sets the per-cell variance a perturbation has to exceed.	GEO GSE242556 (from the scYeast Methods, Sec. 4.1.3)
28	**Brettner 2024 (ultra-high-throughput yeast scRNA-seq)** Brettner L, et al. Yeast 2024. doi:10.1002/yea.3927. doi:10.1002/yea.3927	Combinatorial barcoding in yeast at 96 to 384 multiplexed genotypes or environments per run. The throughput route by which a campaign becomes affordable.	Yeast (Wiley) SI
29	**Mulleder 2012 (prototrophic deletion collection)** Mulleder M, Capuano F, Pir P, et al., Ralser M. Nat Biotechnol 2012;30:1176-1178. doi:10.1038/nbt.2442	The prototrophic deletion collection. Jackson 2020 used a prototrophic background because auxotrophy blocks minimal and nitrogen-limited media, so this is the strain resource a campaign in defined media needs.	EUROSCARF; Addgene kit 40276
30	**Su 2023 (single-cell transcriptomes under four stresses)** Su Y, Xu C, Shea J, DeStephanis D, Su Z. BMC Genomics 2023;24:88. doi:10.1186/s12864-023-09184-w	Full-length rather than three-prime tag counting, so isoform-level readout is on the table for a campaign that needs it.	GEO GSE201386 (from the scYeast Methods)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
31	**Wang 2022 (single-cell transcriptomes across replicative aging)** Wang J, Sang Y, Jin S, Wang X, Azad GK, McCormick MA, Kennedy BK, Li Q, Wang J, Zhang X, et al. Aging Cell 2022;21:e13712. doi:10.1111/acel.13712	The only yeast single-cell set with age as the condition axis, which is a covariate any pooled campaign carries whether or not it measures it.	GEO GSE210032 (from the scYeast Methods)
32	**Aulakh 2025 (genome-scale ionome)** Aulakh SK, et al. Cell Syst 2025;16:101319. doi:10.1016/j.cels.2025.101319	A metabolic readout on the whole deletion collection, which is the genotype axis the supported expression compendium already covers.	Cell Systems SI (accession unconfirmed)
33	**Leutert 2023 (phosphoproteome x 101 conditions)** Leutert M, et al., Villen J. Nat Struct Mol Biol 2023;30:1761-1773. doi:10.1038/s41594-023-01115-3	Enzyme regulation acts faster than transcription, so a phosphosite layer over 101 conditions is what explains flux changes an expression table cannot.	PRIDE PXD034997
34	**Brauer 2008 (growth-rate-controlled chemostat transcriptome)** Brauer MJ, Huttenhower C, Airoldi EM, et al., Botstein D. Mol Biol Cell 2008;19:352-367. doi:10.1091/mbc.e07-08-0779	Expression at controlled growth rate under each nutrient limitation, which is the condition axis Boer 2010 and Hackett 2016 measure metabolites and flux on.	GEO (unconfirmed)
35	**Hackett 2016 (SIMMER multi-omic flux)** Hackett SR, Zanotelli VRT, Xu W, et al., Rabinowitz JD. Science 2016;354:aaf2786. doi:10.1126/science.aaf2786	Flux, metabolite and transcript measured in one study under the same nutrient limitations, so the join is internal rather than across papers.	Science SI (accession unconfirmed)
36	**Zhu 2014 (kinase / phosphatase lipidomics)** Zhu Z, Loewen CJR. Mol Biol Cell 2014. PMC4196872. PMC4196872	A lipidome over 129 signaling deletions, most of which carry an expression profile in the supported compendium.	Mol Biol Cell SI Table S4 (Excel)
37	**Gerke 2017 (urea-cycle mQTL)** Gerke J, et al., Fay JC. Genetics 2017;206:2199. doi:10.1534/genetics.117.201107	Metabolite quantitative trait loci on a segregant cross that also has an expression map, so a metabolite locus can be read through its transcript.	Genetics (GSA) SI
38	**Ambroset 2014 (metabolite QTL)** Ambroset C, et al., Fay JC. PLoS Genet 2014;10:e1004142. doi:10.1371/journal.pgen.1004142	A 74-metabolite panel on a sequenced segregant panel, the widest metabolite vector available on a recombinant genotype axis.	PLOS Genetics SI
39	**Blank 2005 (13C metabolic flux)** Blank LM, Kuepfer L, Sauer U. Genome Biol 2005;6:R49. doi:10.1186/gb-2005-6-6-r49	The only row measuring flux rather than a concentration, on deletion mutants that also have a transcriptome in the supported set.	Genome Biology open-access SI
40	**Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)** Airoldi EM, Miller D, Athanasiadou R, Brandt N, Abdul-Rahman F, Neymotin B, Hashimoto T, Bahmani T, Gresham D. Mol Biol Cell 2016;27:1383-1396. doi:10.1091/mbc.E14-05-1013	Expression under the exact nitrogen-limited conditions that Jackson 2020 profiles single cells in, so the deletion effect separates from the medium effect.	GEO, series not confirmed this pass
41	**Martin-Perez 2017 (protein half-lives)** Martin-Perez M, Villen J. Cell Syst 2017;5:283-294.e5. doi:10.1016/j.cels.2017.08.008	Protein turnover per gene, the second of those two terms, spanning three orders of magnitude and therefore not replaceable by a global constant.	Cell Systems SI; PRIDE identifier not fetched this pass

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
42	**Boer 2010 (metabolome across nutrient limitations)** Boer VM, Crutchfield CA, Bradley PH, Botstein D, Rabinowitz JD. Mol Biol Cell 2010;21:198-211. doi:10.1091/mbc.e09-07-0597	The metabolite half of the chemostat nutrient-limitation series whose transcriptome half is Brauer 2008, same laboratory and same conditions.	MBoC SI (unconfirmed)
43	**Tengolics 2024 (domestication metabolome)** Tengolics R, Szappanos B, Mulleder M, et al., Ralser M, Papp B. PNAS 2024;121:e2313354121. doi:10.1073/pnas.2313354121	Metabolite levels across sequenced isolates, the panel the supported Caudal 2024 transcriptomes and Muenzner 2024 proteomes also sit on.	PNAS Dataset S11
44	**Eder 2020 (flux QTL)** Eder M, Nidelet T, Sanchez I, Camarasa C, Legras JL, Dequin S. PMID 32034164. PMID 32034164	Flux mapped to segregant genotypes, which is the flux counterpart of the expression and protein maps on the same panel type.	Journal SI (venue and accession unconfirmed)
45	**Yu 2021 (proteome and metabolome under nitrogen limitation)** Yu R, Vorontsov E, Sihlbom C, Nielsen J. eLife 2021;10:e65722. doi:10.7554/eLife.65722	Both layers measured in one study under nitrogen limitation, so it calibrates the protein-to-metabolite step that cross-study joins assume.	eLife data availability (unconfirmed)
46	**de Boer 2020 (100M random promoters)** de Boer CG, Vaishnav ED, Sadeh R, Abeyta EL, Friedman N, Regev A. Nat Biotechnol 2020;38:56-65. doi:10.1038/s41587-019-0315-8	Scale band: tier 1, ordered by measurements (100,000,000).	GEO (de Boer 2020); DREAM 2022 release of 6,739,258 promoters
47	**Lee 2014 (HIP-HOP fitness signatures)** Lee AY, St Onge RP, Proctor MJ, et al., Giaever G, Nislow C. Science 2014;344:208-211. doi:10.1126/science.1250217	Scale band: tier 1, ordered by measurements (13,000,000).	Science SI + Nislow/Giaever portal, neither of which yielded per-strain matrices; awaiting author matrices per the WS15 roadmap
48	**Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)** Nguyen Ba AN, Lawrence KR, Rego-Costa A, Gopalakrishnan S, Temko D, Michor F, Desai MM. eLife 2022;11:e73983. doi:10.7554/eLife.73983	Scale band: tier 1, ordered by measurements (800,000).	eLife data availability; SRA (unconfirmed)
49	**Galardini 2019 (four backgrounds x 38 conditions)** Galardini M, Busby BP, Vieitez C, Dunham AS, Typas A, Beltrao P. Mol Syst Biol 2019;15:e8831. doi:10.15252/msb.20198831	Scale band: tier 1, ordered by measurements (575,472).	GEO GSE123118 + github.com/mgalardini/2018koyeast
50	**Dutta 2026 (barcoded natural-isolate chemical response)** Dutta A, Garin M, Loegler V, Brach G, Friedrich A, Yoshimura M, Hirano H, Osada H, Boone C, Yashiroda Y, Hou J, Schacherer J. Nat Commun 2026. doi:10.1038/s41467-026-73532-z	Scale band: tier 1, ordered by measurements (312,000).	Nat Commun SI; isolates from the 1,011 panel (ENA)
51	**Li 2016 (tRNA fitness landscape)** Li C, Qian W, Maclean CJ, Zhang J. Science 2016;352:837-840. doi:10.1126/science.aae0568	Scale band: tier 1, ordered by measurements (65,000).	Science SI (unconfirmed)
52	**Domingo 2018 (tRNA double-mutant landscape)** Domingo J, Diss G, Lehner B. Nature 2018;558:117-121. doi:10.1038/s41586-018-0170-7	Scale band: tier 1, ordered by measurements (23,000).	Nature SI (unconfirmed)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
53	**Zhang 2021 (YETI titratable overexpression)** Zhang Y, Ho Yee Chow M, Ling Y, et al., Andrews BJ. Mol Syst Biol 2021;17:e10321. doi:10.15252/msb.202010321	Scale band: tier 1, ordered by measurements (22,760).	Mol Syst Biol SI (repository accession unconfirmed)
54	**Peeters 2021 (fermentation-trait QTL atlas)** Peeters B, Reijbroek F, Verbaet J, et al., Jarosz DF, Verstrepen KJ. 2021. PMID 34727964. PMID 34727964	Scale band: tier 1, ordered by measurements (20,250).	Journal SI; raw reads at SRA/ENA (accession unconfirmed)
55	**Van Leeuwen 2024 (short-chain organic acids)** Shared and specific genetic determinants of tolerance to acetic, butyric and octanoic acid. PMC10903034. PMC10903034	Scale band: tier 1, ordered by measurements (14,400).	Journal SI (open-access PMC)
56	**Chica 2026 (AutoDRY autophagy screen)** Chica N, et al. Nat Cell Biol 2026;28:465-479. doi:10.1038/s41556-025-01837-0	Scale band: tier 1, ordered by measurements (11,838).	Dryad 10.5061/dryad.cfxpnvxdh
57	**Sopko 2006 (genome-scale overexpression toxicity)** Sopko R, Huang D, Preston N, et al., Boone C, Andrews B. Mol Cell 2006;21:319-330. doi:10.1016/j.molcel.2005.12.011	Scale band: tier 1, ordered by measurements (10,560).	Mol Cell SI tables
58	**Liu 2021 (tryptophan / isobutanol tolerance)** Liu H-L, Wang CH-T, Chiang EP-I, Huang C-C, Li W-H. Biotechnol Biofuels 2021;14:200. doi:10.1186/s13068-021-02048-z	Scale band: tier 1, ordered by measurements (10,012).	GEO GSE175794 (companion RNA-seq); screen in BMC SI
59	**Kuroda 2019 (isobutanol-specific tolerance)** Kuroda K, Hammer SK, Watanabe Y, Montano Lopez J, Fink GR, Stephanopoulos G, Ueda M, Avalos JL. Cell Syst 2019;9:534-547.e5. doi:10.1016/j.cels.2019.10.006	Scale band: tier 1, ordered by measurements (9,600).	ArrayExpress E-MTAB-8175 (RNA-seq); screen data in Cell Syst SI
60	**Turco 2023 (Yeast Phenome)** Turco G, Chang C, Wang RY, et al., Boone C, Andrews BJ, Roth FP. Sci Adv 2023;9:eadg5702. doi:10.1126/sciadv.adg5702	Scale band: tier 2, ordered by measurements (37,680,000).	yeastphenome.org (Zenodo 10.5281/ zenodo.7714347); loader torchcell/datasets/ scerevisiae/yeastphenome.py, 49 screens, not yet in the built set
61	**Rossi 2021 (ChIP-exo protein architecture)** Rossi MJ, Kuntala PK, Lai WKM, et al., Pugh BF. Nature 2021;592:309-314. doi:10.1038/s41586-021-03314-8	Scale band: tier 2, ordered by measurements (2,400,000).	yeastepigenome.org; github.com/CEGRcode/ 2021-Rossi_Nature
62	**Bloom 2025 (global epistasis in a yeast cross)** Bloom JS, et al., Kruglyak L. 2025. PMID 40679398. PMID 40679398	Scale band: tier 2, ordered by measurements (1,000,000).	journal SI (unconfirmed; recent)
63	**Gasch 2000 (environmental stress response)** Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Mol Biol Cell 2000;11:4241-4257. doi:10.1091/mbc.11.12.4241	Scale band: tier 2, ordered by measurements (900,000).	SPELL /SGD Expression Connection; Stanford legacy mirror

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
64	**Decourty 2021 (RNA-metabolism genetic interactions)** Decourty L, et al., Saveanu C. Nucleic Acids Res 2021;49:8535-8555. doi:10.1093/nar/gkab680	Scale band: tier 2, ordered by measurements (700,000).	NAR open-access SI (accession unconfirmed)
65	**Cuperus 2017 (random 5' UTR library)** Cuperus JT, Groves B, Kuchina A, Rosenberg AB, Jojic N, Fields S, Seelig G. Genome Res 2017;27:2015-2024. doi:10.1101/gr.224964.117	Scale band: tier 2, ordered by measurements (500,000).	GEO (unconfirmed)
66	**Faure 2022 (doubledeepPCA)** Faure AJ, Domingo J, Schmiedel JM, Hidalgo-Carcedo C, Diss G, Lehner B. Nature 2022;604:175-183. doi:10.1038/s41586-022-04586-4	Scale band: tier 2, ordered by measurements (500,000).	Nature data availability (unconfirmed)
67	**Chong 2015 / CYCLoPs (single-cell proteome dynamics)** Chong YT, Koh JLY, Friesen H, et al., Andrews BJ, Boone C. Cell 2015;161:1413-1424. doi:10.1016/j.cell.2015.04.051	Scale band: tier 2, ordered by measurements (418,200).	thecellvision.org/cyclops (bulk CSV)
68	**Teyssonniere 2024 (species-wide trait survey)** Teyssonniere EM, Shichino Y, Mito M, Friedrich A, Iwasaki S, Schacherer J. PLoS Genet 2024;20:e1011119. doi:10.1371/journal.pgen.1011119	Scale band: tier 2, ordered by measurements (202,200).	PLoS Genetics SI (unconfirmed)
69	**Diss 2018 (Fos-Jun interaction double mutants)** Diss G, Lehner B. eLife 2018;7:e32472. doi:10.7554/eLife.32472	Scale band: tier 2, ordered by measurements (120,000).	eLife data availability (unconfirmed)
70	**Warringer 2011 (trait variation across the SGRP panel)** Warringer J, Zorgo E, Cubillos FA, et al., Blomberg A, Liti G. PLoS Genet 2011;7:e1002111. doi:10.1371/journal.pgen.1002111	Scale band: tier 2, ordered by measurements (120,000).	PLoS Genetics SI (unconfirmed)
71	**Keren 2013 (promoter activity across conditions)** Keren L, Zackay O, Lotan-Pompan M, Barenholz U, Dekel E, Sasson V, Aidelberg G, Bren A, Zeevi D, Weinberger A, Alon U, Milo R, Segal E. Mol Syst Biol 2013;9:701. doi:10.1038/msb.2013.59	Scale band: tier 2, ordered by measurements (90,000).	Mol Syst Biol SI (unconfirmed)
72	**Puchta 2016 (snoRNA U3 landscape)** Puchta O, Cseke B, Czaja H, Tollervey D, Sanguinetti G, Kudla G. Science 2016;352:840-844. doi:10.1126/science.aaf0965	Scale band: tier 2, ordered by measurements (60,000).	Science SI (unconfirmed)
73	**Yoshikawa 2011 (deletion + overexpression phenome)** Yoshikawa K, Tanaka T, Furusawa C, Nagahisa K, Hirasawa T, Shimizu H. G3 2011;1:247-267. doi:10.1534/g3.111.000695	Scale band: tier 2, ordered by measurements (57,600).	G3 open-access SI
74	**Ho 2009 (MoBY-ORF barcoded overexpression)** Ho CH, Magtanong L, Barker SL, et al., Boone C. Nat Biotechnol 2009;27:369-377. doi:10.1038/nbt.1534	Scale band: tier 2, ordered by measurements (51,000).	moby.ccbr.utoronto.ca + Andrews lab portal

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
75	**Bloom 2013 (1,008 BYxRM segregants)** Bloom JS, Ehrenreich IM, Loo WT, Lite TL, Kruglyak L. Nature 2013;494:234-237. doi:10.1038/nature11867	Scale band: tier 2, ordered by measurements (46,368).	Nature SI
76	**Xiao 2014 (genome-wide RNAi furfural tolerance)** Xiao H, Zhao H. Biotechnol Biofuels 2014;7:78. doi:10.1186/1754-6834-7-78	Scale band: tier 2, ordered by measurements (20,000).	BMC open-access SI
77	**Douglas 2012 (barcoded overexpression fitness)** Douglas AC, Smith AM, Sharifpoor S, et al., Andrews BJ, Boone C, Nislow C. G3 2012;2:1279-1289. doi:10.1534/g3.112.003400	Scale band: tier 2, ordered by measurements (20,000).	G3 open-access SI
78	**She 2018 (natural variant effects across a large isolate panel)** She R, Jarosz DF. Cell 2018;172:478-490. doi:10.1016/j.cell.2017.12.015	Scale band: tier 2, ordered by measurements (20,000).	Cell SI (unconfirmed)
79	**Costello 2020 (bioreactor Bar-seq)** Costello Z, Wehrs M, Mukhopadhyay A, et al. Microb Cell Fact 2020;19:167. doi:10.1186/s12934-020-01423-z	Scale band: tier 2, ordered by measurements (19,200).	Microb Cell Fact open-access SI
80	**Gurvitz (fatty-acid utilization screen)** Gurvitz A, et al. Mol Syst Biol (EMBO Press). embopress.org/doi/pdf/10.1038/msb4100051	Scale band: tier 2, ordered by measurements (14,400).	Mol Syst Biol Table I (PDF-embedded)
81	**Mira 2010 (acetic acid tolerance, full collection)** Mira NP, Palma M, Guerreiro JF, Sa-Correia I. Microb Cell Fact 2010;9:79. doi:10.1186/1475-2859-9-79	Scale band: tier 2, ordered by measurements (14,400).	Microb Cell Fact SI (PMC2972246)
82	**Gameiro 2025 (AID2 degron libraries)** Gameiro E, Juarez-Nunez KA, Fung JJ, Shankar S, Luke B, Khmelinskii A. J Cell Biol 2025;224:e202409007. doi:10.1083/jcb.202409007	Scale band: tier 2, ordered by measurements (11,200).	J Cell Biol SI; strains via IMB Mainz / EUROSCARF
83	**Si 2017 (RAGE RNAi, isobutanol titer)** Si T, Chao R, Min Y, Wu Y, Ren W, Zhao H. Nat Commun 2017;8:15187. doi:10.1038/ncomms15187	Scale band: tier 2, ordered by measurements (10,000).	Nat Commun SI (accession unconfirmed)
84	**HamediRad 2018 (RAGE xylose utilization)** HamediRad M, Lian J, Li H, Zhao H. Biotechnol Bioeng 2018;115:1552-1560. doi:10.1002/bit.26570	Scale band: tier 2, ordered by measurements (10,000).	Biotechnol Bioeng SI (accession unconfirmed)
85	**Pereira 2014 (wheat-straw hydrolysate)** Pereira FB, Guimaraes PMR, Gomes DG, et al., Domingues L. J Ind Microbiol Biotechnol 2014;41:1753-1761. doi:10.1007/s10295-014-1519-z	Scale band: tier 2, ordered by measurements (9,600).	Journal SI

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
86	**Endo 2008 (vanillin tolerance)** Endo A, Nakamura T, Ando A, Tokuyasu K, Shima J. Appl Environ Microbiol 2008;74:7175-7185. doi:10.1128/AEM.01541-08	Scale band: tier 2, ordered by measurements (9,600).	AEM SI (PMC2375868)
87	**Sharifpoor 2012 (kinome synthetic dosage lethality)** Sharifpoor S, van Dyk D, Costanzo M, et al., Boone C, Andrews BJ. Genome Res 2012;22:791-801. doi:10.1101/gr.129213.111	Scale band: tier 2, ordered by measurements (8,700).	Genome Research SI
88	**Mavor 2016 (ubiquitin DMS across conditions)** Mavor D, Barlow K, Thompson S, et al., Bolon DNA, Fraser JS. eLife 2016;5:e15802. doi:10.7554/eLife.15802	Scale band: tier 2, ordered by measurements (7,000).	eLife data availability (unconfirmed)
89	**Flynn 2020 (Hsp90 DMS across environments)** Flynn JM, Rossouw A, Cote-Hammarlof P, et al., Bolon DNA. eLife 2020;9:e53810. doi:10.7554/eLife.53810	Scale band: tier 2, ordered by measurements (6,000).	eLife data availability (unconfirmed)
90	**Ho 2018 (unified absolute protein abundance)** Ho B, Baryshnikova A, Brown GW. Cell Syst 2018;6:192-205. doi:10.1016/j.cels.2017.12.004	Scale band: tier 2, ordered by measurements (5,858).	Cell Systems SI
91	**Renganaath 2020 (natural promoter-variant MPRA)** Renganaath K, Cheung R, Day L, Kosuri S, Kruglyak L, Albert FW. eLife 2020;9:e62669. doi:10.7554/eLife.62669	Scale band: tier 2, ordered by measurements (5,832).	eLife data availability; GEO (unconfirmed)
92	**Chang 2012 (organic-acid production screen)** Chang HJ, Suga H, et al. 2012. PMID 22277779. PMID 22277779	Scale band: tier 2, ordered by measurements (4,800).	PDF tables only; manual transcription required
93	**Trikka 2015 (carotenogenic heterozygous screen)** Trikka FA, Nikolaidis A, Athanasakoglou A, et al., Kampranis SC, Makris AM. Microb Cell Fact 2015;14:60. doi:10.1186/s12934-015-0246-0	Scale band: tier 2, ordered by measurements (4,700).	BMC Additional files 1-2
94	**Sardi 2018 (natural variation in hydrolysate tolerance)** Sardi M, Paithane V, Place M, Robinson DE, Hose J, Wohlbach DJ, Gasch AP. PLoS Genet 2018;14:e1007217. doi:10.1371/journal.pgen.1007217	Scale band: tier 2, ordered by measurements (2,000).	PLoS Genetics SI (unconfirmed)
95	**Breslow 2008 (DAmP hypomorphic collection)** Breslow DK, Cameron DM, Collins SR, et al., Weissman JS. Nat Methods 2008;5:711-718. doi:10.1038/nmeth.1234	Scale band: tier 2, ordered by measurements (1,812).	Nat Methods SI; Horizon YSC5050/5090/5093/5094
96	**Kofoed 2015 (barcoded ts alleles)** Kofoed M, Milbury KL, Chiang JH, et al., Hieter P, Stirling PC. G3 2015;5:1879-1887. doi:10.1534/g3.115.019174	Scale band: tier 2, ordered by measurements (1,200).	G3 open-access SI; strains via Dharmacon/Horizon

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
97	**Zhang 2022 (GCE-SCRaMbLE recombination outcomes)** Zhang W, Lazar-Stefanita L, Yamashita H, et al., Boeke JD. Nat Commun 2022;13:5836. doi:10.1038/s41467-022-33606-0	Scale band: tier 2, ordered by measurements (200).	mirrored as zhangSystematicDissectionKey2022; SRA (unconfirmed)
98	**Cubillos 2017 (nitrogen-consumption QTL)** Cubillos FA, Brice C, Molinet J, et al., Martinez C. G3 2017;7:1693-1705. doi:10.1534/g3.117.042127	Scale band: tier 2, ordered by measurements (165).	BioProject PRJNA379146 (RNA-seq); genotypes in Cubillos 2013 Table S1
99	**Hillenmeyer 2010 (systematic fitness reanalysis)** Hillenmeyer ME, Ericson E, Davis RW, Nislow C, Koller D, Giaever G. Genome Biol 2010;11:R30. doi:10.1186/gb-2010-11-3-r30	Scale band: tier 3, ordered by measurements (2,360,000).	Genome Biology SI (unconfirmed)
100	**Braberg 2020 (point-mutant E-MAP)** Braberg H, et al., Krogan NJ. Science 2020;370:eaaz4910. doi:10.1126/science.aaz4910	Scale band: tier 3, ordered by measurements (500,000).	Science SI (accession unconfirmed)
101	**Breker 2013 (GFP library under stress)** Breker M, Gymrek M, Schuldiner M. J Cell Biol 2013;200:839-850. doi:10.1083/jcb.201301120	Scale band: tier 3, ordered by measurements (212,109).	LoQAtE (weizmann.ac.il/molgen/loqate)
102	**Tkach 2012 (protein relocalization under DNA damage)** Tkach JM, Yimit A, Lee AY, et al., Andrews BJ, Brown GW. Nat Cell Biol 2012;14:966-976. doi:10.1038/ncb2549	Scale band: tier 3, ordered by measurements (204,000).	TheCellVision /Nat Cell Biol SI (unconfirmed)
103	**Ehrenreich 2010 (X-QTL bulk segregant mapping)** Ehrenreich IM, Torabi N, Jia Y, Kent J, Martis S, Shapiro JA, Gresham D, Caudy AA, Kruglyak L. Nature 2010;464:1039-1042. doi:10.1038/nature08923	Scale band: tier 3, ordered by measurements (187,000).	Nature SI (unconfirmed)
104	**Denervaud 2013 (microfluidic GFP dynamics)** Denervaud N, Becker J, Delgado-Gonzalo R, et al., Maerkl SJ. PNAS 2013;110:15842-15847. doi:10.1073/pnas.1308265110	Scale band: tier 3, ordered by measurements (68,000).	PNAS SI (unconfirmed)
105	**Brogaard 2012 (chemical nucleosome map)** Brogaard K, Xi L, Wang JP, Widom J. Nature 2012;486:496-501. doi:10.1038/nature11142	Scale band: tier 3, ordered by measurements (67,000).	GEO (unconfirmed)
106	**Xia 2022 (proteome allocation across conditions)** Xia J, Sanchez BJ, Chen Y, Campbell K, Kasvandik S, Nielsen J. Nat Commun 2022;13:2819. doi:10.1038/s41467-022-30513-2	Scale band: tier 3, ordered by measurements (60,000).	Nat Commun SI (unconfirmed)
107	**Melamed 2013 (PAB1 RRM domain scan)** Melamed D, Young DL, Gamble CE, Miller CR, Fields S. RNA 2013;19:1537-1551. doi:10.1261/rna.040709.113	Scale band: tier 3, ordered by measurements (40,000).	journal SI (unconfirmed)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
108	**Yvert 2013 (single-cell trait variation across isolates)** Yvert G, Ohnuki S, Nogami S, Imanaga Y, Fehrmann S, Schacherer J, Ohya Y. BMC Syst Biol 2013;7:54. doi:10.1186/1752-0509-7-54	Scale band: tier 3, ordered by measurements (18,537).	BMC SI (unconfirmed)
109	**Lahtvee 2017 (absolute proteome across conditions)** Lahtvee PJ, Sanchez BJ, Smialowska A, et al., Nielsen J. Cell Syst 2017;4:495-504. doi:10.1016/j.cels.2017.03.003	Scale band: tier 3, ordered by measurements (18,000).	Cell Systems SI (unconfirmed)
110	**Weill 2018 (SWAT libraries)** Weill U, Yofe I, Sass E, et al., Schuldiner M. Nat Methods 2018;15:617-622. doi:10.1038/s41592-018-0044-9	Scale band: tier 3, ordered by measurements (16,500).	Nat Methods SI; strains via EUROSCARF
111	**Kinsler 2020 (barcoded pleiotropy across environments)** Kinsler G, Geiler-Samerotte K, Petrov DA. eLife 2020;9:e61271. doi:10.7554/eLife.61271	Scale band: tier 3, ordered by measurements (13,500).	eLife data availability (unconfirmed)
112	**Pelechano 2013 (transcript isoform landscape)** Pelechano V, Wei W, Steinmetz LM. Nature 2013;497:127-131. doi:10.1038/nature12121	Scale band: tier 3, ordered by measurements (12,000).	ArrayExpress /GEO (unconfirmed)
113	**Deutschbauer 2005 (haploinsufficiency, genome-wide)** Deutschbauer AM, Jaramillo DF, Proctor M, Kumm J, Hillenmeyer ME, Davis RW, Nislow C, Giaever G. Genetics 2005;169:1915-1925. doi:10.1534/genetics.104.036871	Scale band: tier 3, ordered by measurements (11,800).	Genetics SI (unconfirmed)
114	**Valenti 2025 (SWAT degron + GFP collection)** Valenti M, et al. J Cell Biol 2025;224:e202409050. doi:10.1083/jcb.202409050	Scale band: tier 3, ordered by measurements (10,000).	J Cell Biol SI
115	**Sousa 2013 (acetate resistance overexpression screen)** Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 23262128. PMID 23262128	Scale band: tier 3, ordered by measurements (10,000).	journal SI (unconfirmed)
116	**Sadhu 2016 (CRISPR-directed causal variant mapping)** Sadhu MJ, Bloom JS, Day L, Kruglyak L. Science 2016;352:1113-1116. doi:10.1126/science.aaf5124	Scale band: tier 3, ordered by measurements (10,000).	Science SI (unconfirmed)
117	**Weinberg 2016 (ribosome profiling, improved)** Weinberg DE, Shah P, Eichhorn SW, Hussmann JA, Plotkin JB, Bartel DP. Cell Rep 2016;14:1787-1799. doi:10.1016/j.celrep.2016.01.043	Scale band: tier 3, ordered by measurements (6,000).	GEO (unconfirmed)
118	**Ghaemmaghami 2003 (absolute protein abundance)** Ghaemmaghami S, Huh WK, Bower K, et al., Weissman JS. Nature 2003;425:737-741. doi:10.1038/nature02046	Scale band: tier 3, ordered by measurements (4,251).	Nature SI (Excel)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
119	**Huh 2003 (GFP localization atlas)** Huh WK, Falvo JV, Gerke LC, et al., O'Shea EK. Nature 2003;425:686-691. doi:10.1038/nature02026	Scale band: tier 3, ordered by measurements (4,156).	SGD /YeastGFP mirror
120	**Michaelis 2023 (protein interactome)** Michaelis AC, et al., Mann M. Nature 2023;624:192-200. doi:10.1038/s41586-023-06739-5	Scale band: tier 3, ordered by measurements (4,000).	yeast-interactome.biochem.mpg.de + ProteomeXchange
121	**Christiano 2014 (protein turnover)** Christiano R, Nagaraj N, Frohlich F, Walther TC. Cell Rep 2014;9:1959-1965. doi:10.1016/j.celrep.2014.10.065	Scale band: tier 3, ordered by measurements (3,900).	Cell Reports SI (unconfirmed)
122	**Kitzman 2015 (GAL4 DMS)** Kitzman JO, Starita LM, Lo RS, Fields S, Shendure J. Nat Methods 2015;12:203-206. doi:10.1038/nmeth.3223	Scale band: tier 3, ordered by measurements (2,400).	Nat Methods SI (unconfirmed)
123	**Sharon 2012 (designed promoter library)** Sharon E, Kalma Y, Sharp A, et al., Segal E. Nat Biotechnol 2012;30:521-530. doi:10.1038/nbt.2205	Scale band: tier 3, ordered by measurements (2,000).	Nat Biotechnol SI (unconfirmed)
124	**Sadhu 2018 (CRISPR-directed mitotic recombination)** Sadhu MJ, Bloom JS, Day L, Siegel JJ, Kosuri S, Kruglyak L. eLife 2018;7:e41090. doi:10.7554/eLife.41090	Scale band: tier 3, ordered by measurements (1,500).	eLife data availability (unconfirmed)
125	**Duveau 2021 (TDH3 promoter allele series)** Duveau F, Vande Zande P, Metzger BP, et al., Wittkopp PJ. eLife 2021;10:e67806. doi:10.7554/eLife.67806	Scale band: tier 3, ordered by measurements (1,000).	eLife data availability (unconfirmed)
126	**Strope 2015 (100 clinical and natural genomes)** Strope PK, Skelly DA, Kozmin SG, Mahadevan G, Stone EA, Magwene PM, Dietrich FS, McCusker JH. Genome Res 2015;25:762-774. doi:10.1101/gr.185538.114	Scale band: tier 3, ordered by measurements (1,000).	Genome Research SI; SRA (unconfirmed)
127	**Schulte 2023 (mitochondrial complexome)** Schulte U, et al. Nature 2023;614:153-159. doi:10.1038/s41586-022-05641-w	Scale band: tier 3, ordered by measurements (900).	complexomics.org + ProteomeXchange
128	**Metzger 2015 (promoter mutation effects on expression and noise)** Metzger BPH, Yuan DC, Gruber JD, Duveau F, Wittkopp PJ. Nature 2015;521:344-346. doi:10.1038/nature14424	Scale band: tier 3, ordered by measurements (470).	Nature SI (unconfirmed)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
129	**Venkataram 2016 (barcoded adaptive lineages, sequenced)** Venkataram S, Dunn B, Li Y, et al., Petrov DA, Kryazhimskiy S, Sherlock G. Cell 2016;166:1585-1596. doi:10.1016/j.cell.2016.08.002	Scale band: tier 3, ordered by measurements (400).	Cell SI; SRA (unconfirmed)
130	**Jakociunas 2021 (degron-tuned terpene flux)** Jakociunas T, Klitgaard AK, Kristensen M, Jensen MK, Keasling JD. 2021. doi:10.1002/bit.27735	Scale band: tier 3, ordered by measurements (270).	journal SI (unconfirmed)
131	**Shen 2016 (SCRaMbLE genome minimization)** Shen Y, Stracquadanio G, Wang Y, et al., Boeke JD, Bader JS. Genome Res 2016;26:36-49. doi:10.1101/gr.193433.115	Scale band: tier 3, ordered by measurements (200).	Genome Research SI (unconfirmed)
132	**Blount 2018 (SCRaMbLE for phenotype improvement)** Blount BA, Gowers GF, Ho JCH, Ledesma-Amaro R, Jovicevic D, McKiernan RM, Xie ZX, Li BZ, Yuan YJ, Ellis T. Nat Commun 2018;9:1932. doi:10.1038/s41467-018-03143-w	Scale band: tier 3, ordered by measurements (200).	Nat Commun SI (unconfirmed)
133	**Liti 2009 (population genomics of S. cerevisiae and S. paradoxus)** Liti G, Carter DM, Moses AM, et al., Louis EJ. Nature 2009;458:337-341. doi:10.1038/nature07743	Scale band: tier 3, ordered by measurements (70).	SGRP /ENA (unconfirmed)
134	**Yue 2017 (wild vs domesticated assemblies)** Yue JX, Li J, Aigrain L, et al., Liti G. Nat Genet 2017;49:913-924. doi:10.1038/ng.3847	Scale band: tier 3, ordered by measurements (12).	ENA/GenBank (accession unconfirmed)
135	**Zackrisson 2016 (Scan-o-matic growth curves)** Zackrisson M, Hallin J, Ottosson LG, et al., Warringer J, Blomberg A. G3 2016;6:3003-3014. doi:10.1534/g3.116.032342	Scale band: tier 3.	Open-source platform; per-study datasets
136	**Spellman 1998 (cell-cycle transcriptome)** Spellman PT, Sherlock G, Zhang MQ, Iyer VR, Anders K, Eisen MB, Brown PO, Botstein D, Futcher B. Mol Biol Cell 1998;9:3273-3297. doi:10.1091/mbc.9.12.3273	Scale band: tier 4, ordered by measurements (462,000).	SGD /SPELL
137	**Levy 2008 (morphology and phenotypic capacitance)** Levy SF, Siegal ML. PLoS Biol 2008;6:e264. doi:10.1371/journal.pbio.0060264	Scale band: tier 4, ordered by measurements (340,000).	PLoS Biology SI (unconfirmed)
138	**Causton 2001 (environmental change transcriptome)** Causton HC, Ren B, Koh SS, et al., Young RA. Mol Biol Cell 2001;12:323-337. doi:10.1091/mbc.12.2.323	Scale band: tier 4, ordered by measurements (240,000).	SGD /SPELL

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
139	**Gasch 2001 (DNA damage response transcriptome)** Gasch AP, Huang M, Metzner S, Botstein D, Elledge SJ, Brown PO. Mol Biol Cell 2001;12:2987-3003. doi:10.1091/mbc.12.10.2987	Scale band: tier 4, ordered by measurements (144,000).	SGD /SPELL
140	**Handfield 2013 (image-based localization change)** Handfield LF, Chong YT, Simmons J, Andrews BJ, Moses AM. PLoS Comput Biol 2013;9:e1003085. doi:10.1371/journal.pcbi.1003085	Scale band: tier 4, ordered by measurements (136,000).	PLoS Comput Biol SI (unconfirmed)
141	**Kaplan 2009 (nucleosome sequence preferences)** Kaplan N, Moore IK, Fondufe-Mittendorf Y, et al., Segal E. Nature 2009;458:362-366. doi:10.1038/nature07667	Scale band: tier 4, ordered by measurements (134,000).	GEO (unconfirmed)
142	**Giaever 2004 (chemogenomic drug-target profiling)** Giaever G, Flaherty P, Kumm J, et al., Davis RW. PNAS 2004;101:793-798. doi:10.1073/pnas.0307490100	Scale band: tier 4, ordered by measurements (59,000).	PNAS SI (unconfirmed)
143	**Paulo 2016 (TMT proteome across carbon sources)** Paulo JA, O'Connell JD, Gaun A, Gygi SP. Mol Biol Cell 2016;27:2823-2838. doi:10.1091/mbc.E16-03-0184	Scale band: tier 4, ordered by measurements (22,500).	ProteomeXchange (unconfirmed)
144	**Gibney 2013 (osmotic and glucose stress fitness)** Gibney PA, Lu C, Caudy AA, Hess DC, Botstein D. PNAS 2013;110:E4393-E4402. doi:10.1073/pnas.1318100110	Scale band: tier 4, ordered by measurements (19,200).	PNAS SI (unconfirmed)
145	**Meurer 2018 (genome-wide GFP library, rebuilt)** Meurer M, Duan Y, Sass E, Kats I, Herbst K, Buchmuller BC, Dederer V, Huber F, Kirrmaier D, Stefl M, Van Laer K, Dick TP, Lemberg MK, Khmelinskii A, Levy ED, Knop M. Nat Methods 2018;15:617-620. doi:10.1038/s41592-018-0044-9	Scale band: tier 4, ordered by measurements (12,000).	Nat Methods SI (DOI collision with Weill 2018 UNRESOLVED)
146	**Bennett 2001 (ionizing-radiation resistance)** Bennett CB, et al. Nat Genet 2001;29:426-434. doi:10.1038/ng778	Scale band: tier 4, ordered by measurements (9,600).	Nat Genet SI (likely PDF)
147	**Sugiyama (lactic-acid tolerance)** Sugiyama M, Kaneko Y, et al. National Research Institute of Brewing technical report. nisr.or.jp/wp-content/uploads/NISR06sugiyama.pdf	Scale band: tier 4, ordered by measurements (9,600).	NRIB technical report PDF
148	**Sousa 2013 (acetic-acid programmed cell death screen)** Sousa M, Duarte AM, Fernandes TR, et al. 2013. PMID 24286259. PMID 24286259	Scale band: tier 4, ordered by measurements (9,600).	journal SI (unconfirmed)

Table 10, continued

#	Dataset, citation and link	Why it is ordered here	Data
149	**Nagalakshmi 2008 (RNA-seq transcriptome annotation)** Nagalakshmi U, Wang Z, Waern K, Shou C, Raha D, Gerstein M, Snyder M. Science 2008;320:1344-1349. doi:10.1126/science.1158441	Scale band: tier 4, ordered by measurements (6,000).	SRA /GEO (unconfirmed)
150	**Yofe 2016 (SWAT N-terminal library)** Yofe I, Weill U, Meurer M, et al., Knop M, Schuldiner M. Nat Methods 2016;13:371-378. doi:10.1038/nmeth.3795	Scale band: tier 4, ordered by measurements (5,400).	Nat Methods SI; EUROSCARF (unconfirmed)
151	**Hénault 2023 (hybrid and introgression panel)** Henault M, Marsit S, Charron G, Landry CR. Nat Commun / Landry lab hybrid panels. doi:10.1038/s41467-024-46155-5	Scale band: tier 4, ordered by measurements (5,000).	journal SI (unconfirmed)
152	**Cubillos 2013 (F12 recombinant panel construction)** Cubillos FA, Parts L, Salinas F, et al., Liti G. Genetics 2013;195:1141-1155. doi:10.1534/genetics.113.155515	Scale band: tier 4, ordered by measurements (3,840).	Genetics SI Table S1
153	**Kessi-Perez 2019 (nitrogen-consumption QTL panel)** Kessi-Perez EI, Molinet J, Martinez C, et al. 2019/2020 nitrogen-utilization QTL work. doi:10.1186/s13068-019-1440-9	Scale band: tier 4, ordered by measurements (200).	journal SI (unconfirmed)
154	**Lian 2016 (combinatorial xylose pathway optimization)** Lian J, Chao R, Zhao H. Metab Eng 2016;33:139-148. doi:10.1016/j.ymben.2015.11.001	Scale band: tier 4, ordered by measurements (100).	Metab Eng SI (unconfirmed)

Table 11: Candidate-gate verdict per ranked row, rendered from the verdict store (`database/candidates/`). *no verdict* means the row has not been gated.

Rank	Dataset	Verdict	Decided
1	Albert 2018 (eQTL in 1,012 segregants)	no verdict	
2	Jakobson 2025 (genome-to-proteome map)	no verdict	
3	Teyssonniere 2024 (species-wide proteome against transcriptome)	no verdict	
4	Grossbach 2022 (BY x RM transcriptome, proteome and phosphoproteome)	no verdict	
5	Smith 2008 (gene-environment interaction eQTL)	no verdict	
6	Skelly 2013 (expression variation across isolates)	no verdict	
7	McManus 2014 (ribosome profiling, allele-specific)	no verdict	
8	Foss 2007 (BY x RM segregant proteome)	no verdict	
9	Dong 2021 (MAGIC + SAM biosensor)	no verdict	
10	Momen-Roknabadi 2020 (inducible CRISPRi library)	no verdict	
11	McGlincy 2021 (genome-scale CRISPRi library)	no verdict	
12	Bao 2018 (CHAnGE single-nucleotide library)	no verdict	
13	Crook 2016 (tunable RNAi, isobutanol + 1-butanol)	no verdict	
14	Roy 2018 (multiplexed precision editing)	no verdict	
15	Newman 2006 (protein noise, GFP library)	no verdict	
16	Mukherjee 2021 (CRISPRi essential genes x acetic acid)	no verdict	
17	Lian 2017 (CRISPR-AID, beta-carotene)	blocked:provision	2026-10-11
18	Hughes 2000 (compendium of expression profiles)	no verdict	
19	Lenstra 2011 (chromatin regulator deletion expression)	no verdict	
20	Sun 2013 (mRNA synthesis and decay rates across deletion strains)	no verdict	
21	Jariani 2020 (yeast scRNA-seq through lag phase)	no verdict	
22	Guo 2018 (CRISPR-Cas9 tiling of essential genes)	no verdict	
23	Urbonaite 2021 (yeastDrop-Seq under drug treatment)	no verdict	
24	Nadal-Ribelles 2019 (sensitive yeast scRNA-seq)	no verdict	
25	Gasch 2017 (single-cell stress heterogeneity)	no verdict	
26	Jaffe 2019 (multiplexed CRISPR interference epistasis)	no verdict	
27	Jackson 2023 (wild-type scRNA-seq time course for RNA kinetics)	no verdict	
28	Brettner 2024 (ultra-high-throughput yeast scRNA-seq)	no verdict	
29	Mulleder 2012 (prototrophic deletion collection)	no verdict	
30	Su 2023 (single-cell transcriptomes under four stresses)	no verdict	
31	Wang 2022 (single-cell transcriptomes across replicative aging)	no verdict	
32	Aulakh 2025 (genome-scale ionome)	no verdict	
33	Leutert 2023 (phosphoproteome x 101 conditions)	admissible_with_gaps	2026-10-11
34	Brauer 2008 (growth-rate-controlled chemostat transcriptome)	no verdict	
35	Hackett 2016 (SIMMER multi-omic flux)	no verdict	
36	Zhu 2014 (kinase / phosphatase lipidomics)	no verdict	
37	Gerke 2017 (urea-cycle mQTL)	no verdict	
38	Ambroset 2014 (metabolite QTL)	no verdict	
39	Blank 2005 (13C metabolic flux)	no verdict	
40	Airoldi 2016 (nitrogen-limited steady-state and dynamic transcriptome)	no verdict	
41	Martin-Perez 2017 (protein half-lives)	no verdict	
42	Boer 2010 (metabolome across nutrient limitations)	no verdict	
43	Tengolics 2024 (domestication metabolome)	no verdict	
44	Eder 2020 (flux QTL)	no verdict	
45	Yu 2021 (proteome and metabolome under nitrogen limitation)	no verdict	
46	de Boer 2020 (100M random promoters)	no verdict	
47	Lee 2014 (HIP-HOP fitness signatures)	blocked:wall	2026-10-11
48	Nguyen Ba 2022 (barcoded bulk QTL, 100k segregants)	no verdict	
49	Galardini 2019 (four backgrounds x 38 conditions)	no verdict	
50	Dutta 2026 (barcoded natural-isolate chemical response)	no verdict	
51	Li 2016 (tRNA fitness landscape)	no verdict	
52	Domingo 2018 (tRNA double-mutant landscape)	no verdict	
53	Zhang 2021 (YETI titratable overexpression)	no verdict	
54	Peeters 2021 (fermentation-trait QTL atlas)	no verdict	
55	Van Leeuwen 2024 (short-chain organic acids)	no verdict	
56	Chica 2026 (AutoDRY autophagy screen)	no verdict	
57	Sopko 2006 (genome-scale overexpression toxicity)	no verdict	
58	Liu 2021 (tryptophan / isobutanol tolerance)	no verdict	
59	Kuroda 2019 (isobutanol-specific tolerance)	no verdict	
60	Turco 2023 (Yeast Phenome)	admissible	2026-10-11
61	Rossi 2021 (ChIP-exo protein architecture)	no verdict	
62	Bloom 2025 (global epistasis in a yeast cross)	no verdict	
63	Gasch 2000 (environmental stress response)	blocked:provision	2026-10-11
64	Decourty 2021 (RNA-metabolism genetic interactions)	no verdict	
65	Cuperus 2017 (random 5' UTR library)	no verdict	
66	Faure 2022 (doubledeepPCA)	no verdict	

Rank	Dataset	Verdict	Decided
67	Chong 2015 / CYCLoPs (single-cell proteome dynamics)	no verdict	
68	Teyssonniere 2024 (species-wide trait survey)	no verdict	
69	Diss 2018 (Fos-Jun interaction double mutants)	no verdict	
70	Warringer 2011 (trait variation across the SGRP panel)	no verdict	
71	Keren 2013 (promoter activity across conditions)	no verdict	
72	Puchta 2016 (snoRNA U3 landscape)	no verdict	
73	Yoshikawa 2011 (deletion + overexpression phenome)	no verdict	
74	Ho 2009 (MoBY-ORF barcoded overexpression)	no verdict	
75	Bloom 2013 (1,008 BYxRM segregants)	no verdict	
76	Xiao 2014 (genome-wide RNAi furfural tolerance)	no verdict	
77	Douglas 2012 (barcoded overexpression fitness)	no verdict	
78	She 2018 (natural variant effects across a large isolate panel)	blocked:wall	2026-10-11
79	Costello 2020 (bioreactor Bar-seq)	no verdict	
80	Gurvitz (fatty-acid utilization screen)	no verdict	
81	Mira 2010 (acetic acid tolerance, full collection)	no verdict	
82	Gameiro 2025 (AID2 degron libraries)	no verdict	
83	Si 2017 (RAGE RNAi, isobutanol titer)	no verdict	
84	HamediRad 2018 (RAGE xylose utilization)	no verdict	
85	Pereira 2014 (wheat-straw hydrolysate)	no verdict	
86	Endo 2008 (vanillin tolerance)	no verdict	
87	Sharifpoor 2012 (kinome synthetic dosage lethality)	no verdict	
88	Mavor 2016 (ubiquitin DMS across conditions)	no verdict	
89	Flynn 2020 (Hsp90 DMS across environments)	no verdict	
90	Ho 2018 (unified absolute protein abundance)	no verdict	
91	Renganaath 2020 (natural promoter-variant MPRA)	no verdict	
92	Chang 2012 (organic-acid production screen)	no verdict	
93	Trikka 2015 (carotenogenic heterozygous screen)	blocked:provision	2026-10-11
94	Sardi 2018 (natural variation in hydrolysate tolerance)	no verdict	
95	Breslow 2008 (DAmP hypomorphic collection)	no verdict	
96	Kofoed 2015 (barcoded ts alleles)	no verdict	
97	Zhang 2022 (GCE-SCRaMbLE recombination outcomes)	no verdict	
98	Cubillos 2017 (nitrogen-consumption QTL)	no verdict	
99	Hillenmeyer 2010 (systematic fitness reanalysis)	no verdict	
100	Braberg 2020 (point-mutant E-MAP)	no verdict	
101	Breker 2013 (GFP library under stress)	no verdict	
102	Tkach 2012 (protein relocalization under DNA damage)	no verdict	
103	Ehrenreich 2010 (X-QTL bulk segregant mapping)	no verdict	
104	Denervaud 2013 (microfluidic GFP dynamics)	no verdict	
105	Brogaard 2012 (chemical nucleosome map)	no verdict	
106	Xia 2022 (proteome allocation across conditions)	no verdict	
107	Melamed 2013 (PAB1 RRM domain scan)	no verdict	
108	Yvert 2013 (single-cell trait variation across isolates)	no verdict	
109	Lahtvee 2017 (absolute proteome across conditions)	no verdict	
110	Weill 2018 (SWAT libraries)	no verdict	
111	Kinsler 2020 (barcoded pleiotropy across environments)	no verdict	
112	Pelechano 2013 (transcript isoform landscape)	no verdict	
113	Deutschbauer 2005 (haploinsufficiency, genome-wide)	no verdict	
114	Valenti 2025 (SWAT degron + GFP collection)	no verdict	
115	Sousa 2013 (acetate resistance overexpression screen)	no verdict	
116	Sadhu 2016 (CRISPR-directed causal variant mapping)	no verdict	
117	Weinberg 2016 (ribosome profiling, improved)	no verdict	
118	Ghaemmaghami 2003 (absolute protein abundance)	no verdict	
119	Huh 2003 (GFP localization atlas)	no verdict	
120	Michaelis 2023 (protein interactome)	no verdict	
121	Christiano 2014 (protein turnover)	no verdict	
122	Kitzman 2015 (GAL4 DMS)	no verdict	
123	Sharon 2012 (designed promoter library)	no verdict	
124	Sadhu 2018 (CRISPR-directed mitotic recombination)	no verdict	
125	Duveau 2021 (TDH3 promoter allele series)	no verdict	
126	Strope 2015 (100 clinical and natural genomes)	no verdict	
127	Schulte 2023 (mitochondrial complexome)	no verdict	
128	Metzger 2015 (promoter mutation effects on expression and noise)	no verdict	
129	Venkataram 2016 (barcoded adaptive lineages, sequenced)	no verdict	
130	Jakociunas 2021 (degron-tuned terpene flux)	no verdict	
131	Shen 2016 (SCRaMbLE genome minimization)	no verdict	
132	Blount 2018 (SCRaMbLE for phenotype improvement)	no verdict	
133	Liti 2009 (population genomics of S. cerevisiae and S. paradoxus)	no verdict	
134	Yue 2017 (wild vs domesticated assemblies)	no verdict	
135	Zackrisson 2016 (Scan-o-matic growth curves)	blocked:wall	2026-10-11
136	Spellman 1998 (cell-cycle transcriptome)	no verdict	

Rank	Dataset	Verdict	Decided
137	Levy 2008 (morphology and phenotypic capacitance)	no verdict	
138	Causton 2001 (environmental change transcriptome)	no verdict	
139	Gasch 2001 (DNA damage response transcriptome)	no verdict	
140	Handfield 2013 (image-based localization change)	no verdict	
141	Kaplan 2009 (nucleosome sequence preferences)	no verdict	
142	Giaever 2004 (chemogenomic drug-target profiling)	no verdict	
143	Paulo 2016 (TMT proteome across carbon sources)	no verdict	
144	Gibney 2013 (osmotic and glucose stress fitness)	no verdict	
145	Meurer 2018 (genome-wide GFP library, rebuilt)	no verdict	
146	Bennett 2001 (ionizing-radiation resistance)	no verdict	
147	Sugiyama (lactic-acid tolerance)	no verdict	
148	Sousa 2013 (acetic-acid programmed cell death screen)	no verdict	
149	Nagalakshmi 2008 (RNA-seq transcriptome annotation)	no verdict	
150	Yofe 2016 (SWAT N-terminal library)	no verdict	
151	Hénault 2023 (hybrid and introgression panel)	no verdict	
152	Cubillos 2013 (F12 recombinant panel construction)	no verdict	
153	Kessi-Perez 2019 (nitrogen-consumption QTL panel)	no verdict	
154	Lian 2016 (combinatorial xylose pathway optimization)	no verdict	